# The generalized Petlyuk arrangement minimizes total vaporization: a proof of the Halvorsen–Skogestad conjecture

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**Abstract.** Halvorsen and Skogestad (Ind. Eng. Chem. Res. 2003) conjectured that, for an ideal feed (constant relative volatilities, constant molar flows, constant pressure) separated into its pure components, with infinitely many stages and no internal heat exchange, the generalized Petlyuk arrangement needs less vaporization than any other adiabatic distillation arrangement, its minimum being the highest peak $\max_j H_j$ of the $V_{\min}$ diagram. We prove the conjecture in a class $\mathcal{A}$ that contains every arrangement within their assumptions: all finite networks of constant-molar-overflow equilibrium stages, single-phase splitters and mixers, total condensers and total vaporizers, with one feed that may be split and conditioned, heat duties at any stage, recycles, and products of either phase. Infinitely many stages and pure products enter as a limit of finite-stage steady states. The proof has two halves. (i) At every steady state of every arrangement of $\mathcal{A}$, with any numbers of stages, the total heat supplied satisfies $R \geq \sum_P \max(0, \varphi_P(\theta_j))$ for every root $\theta_j$ of Underwood's feed equation, where $\varphi_P$ is the Underwood flux of the product stream $P$; with product purities $1-\varepsilon$ this gives $R \geq H_j - c_j\varepsilon$. The argument is elementary: all streams leaving an equilibrium stage have Underwood fluxes of one sign, so the units split into a light and a heavy side, and the energy balance of the heavy side, in Underwood's weights, gives the bound, with the excess as a sum of three nonnegative losses. (ii) For every $\delta > 0$ and $\varepsilon > 0$, the generalized Petlyuk arrangement with enough stages in every section has a steady state with all purities at least $1-\varepsilon$ and total vaporization $\max_j H_j + \delta$. Its operating specification keeps every column with three or more components close to its preferred split and forces a small flow of one key component into each side product; in the infinite-stage limit every section is then pinched at one end only, the limit equations have an invertible Jacobian (block triangular, with one triangular M-matrix per trace component), and the finite-stage equations are an exponentially small perturbation of them. Hence the Petlyuk value of Halvorsen and Skogestad is the exact minimum of the total vaporization over the whole class. All mathematics, code and text were produced by AI agents, and every part was checked by AI verifiers.

## 1 Introduction

### 1.1 The conjecture

The Petlyuk arrangement [1] and its realization as a dividing-wall column [2] are the best-known energy-saving devices of multicomponent distillation. Halvorsen and Skogestad analysed its minimum energy for any number of components with Underwood's equations, introduced the $V_{\min}$ diagram, and identified the minimum vapor of the *generalized* Petlyuk arrangement (one column for every set of consecutive components,

directly coupled; Figure 20) as the highest peak of that diagram [3] [4] [5] [6]. They then stated, in the abstract of [5]:

> *"We also conjecture that the minimum energy requirement for the generalized extended Petlyuk arrangement is lower than the minimum-energy requirement for any distillation configuration when we consider conventional adiabatic sections and no internal heat exchange."*

Section 6.3 of [5] (p. 626) makes the assumptions explicit:

> *"We assume constant relative volatilities, constant molar flows, constant pressure, and no internal heat integration. Then the generalized adiabatic Petlyuk arrangement has the lowest need for vaporization compared to any other adiabatic distillation arrangement for separation of an arbitrary feed mixture into its pure components. We have not presented a complete proof, so the above conclusion is a conjecture."*

The number of stages is infinite throughout [5]. Even the minimum vapor of the Petlyuk arrangement itself was established only in part (Section 5.3, p. 624):

> *"We have not carried out a detailed proof for the general N-component M-product case for other than the far left and right peaks."*

The conjecture was restated at Distillation & Absorption 2002 [7], where infinitely many stages are assumed "because this provides a lower bound on the energy requirement" for ideal mixtures and the energy is measured by the total vaporization. In this paper we prove it, together with the formula for the Petlyuk minimum, for every number of components.

## 1.2 What was known

For three components, Fidkowski and Królikowski [8] compared the Petlyuk column in closed form with side strippers, side rectifiers and the direct and indirect sequences; Halvorsen reports Petlyuk's statement that "no proof has been found in the literature, except for the ternary case" §6 [5]. For four or more components the property has been tested, not proved: Underwood-model enumerations and global optimization of the space of regular-column configurations [9] [10] [11] find the fully thermally coupled (FTC, generalized Petlyuk) value optimal on the instances solved, Fidkowski and Agrawal [12] and Fidkowski [13] note that no rigorous proof is available, and a 2021 thesis from the same group states that "for four and more components, the conjecture is yet to be proved or disproved" p. 125 [14], after proving it for one four-component family by a vapor balance on a suitably chosen boundary. Some later papers describe the property as established [15] [16] [17]; they refer back to [5], which calls it a conjecture.

The ingredients of our lower bound are classical when used one section at a time. Underwood's identity between consecutive stages holds for a free parameter [18], eqs. (2.45)–(2.49) [6]; Franklin and Forsyth used the signs of Underwood coordinates inside one column [19]; stage-level sign comparisons of an Underwood-type function inside one section of a multi-feed column prove convergence to a pinch in Prop. 5.4.1 [20] and Prop. 1 [21]; and cut or vapor-balance arguments at the common roots were used for one arrangement (the equality case of the Petlyuk arrangement §3 [5]), one junction §6.2.1 [6] and one four-component family §6.3 [14]. We found no work that combines them over a whole network. For attainment, the finite-stage behaviour of Petlyuk columns has been studied by simulation [22] [23], §5.2.7 [6], which confirms numerically that finite-stage operation approaches the Underwood values, and the theory of single sections and columns with constant volatilities is classical [18] [19] [24] [25];

we found no proof that steady states of a closed, thermally coupled network converge to the Underwood configuration.

Two difficulties stand in the way of a proof. First, "any distillation configuration" is not a finite list: besides the regular-column configurations [26] [9] it includes columns with several feed stages, stream splitters, remixing, recycles, bypasses, and operation with reverse net flows. Second, the infinite-stage Underwood model in which minimum vapor is computed is itself an idealization, and the formulations used for optimization are relaxations whose feasible points need not be limits of physical operation; we show in Remark 15 that one such formulation admits a five-component point below the Petlyuk value. We therefore work in the physical model: networks of equilibrium stages with finitely many stages, of which "infinitely many stages and pure products" is a limit.

## 1.3 Results

Let $\alpha_1 > \cdots > \alpha_N > 0$ be the relative volatilities, $z$ the feed composition ($z_p > 0$, $\sum_p z_p = 1$), $q \in \mathbb{R}$ the liquid fraction of the feed (feed flow 1), $\theta_j \in (\alpha_{j+1}, \alpha_j)$ the roots of Underwood's feed equation $\sum_p \alpha_p z_p/(\alpha_p - \theta) = 1 - q$, and

$$H_j = \sum_{p>j} \frac{\alpha_p z_p}{\theta_j - \alpha_p} \qquad (j = 1, \dots, N-1), \qquad H^* = \max_j H_j,$$

the peaks of the $V_{\min}$ diagram referred to the reboiler (Section 2.1). For a family $\mathcal{S}$ of arrangements, $R^*(\mathcal{S})$ denotes the limit, as $\varepsilon \to 0$, of the infimum of the total vaporization $R$ over the steady states of members of $\mathcal{S}$ whose products all have purity at least $1 - \varepsilon$ (Definition 5); $\mathcal{S}_{\mathrm{Pet}}$ is the generalized Petlyuk arrangement with all possible numbers of stages.

**Theorem 1** (main theorem). Let $\mathcal{A}$ be the class of Definition 3.

**(1)** (Lower bound.) At every steady state of every arrangement of $\mathcal{A}$ whose products all have purity at least $1 - \varepsilon$, $R \geq H_j - c_j\varepsilon$ for every $j$, with $c_j = \max(\alpha_j/(\alpha_j - \theta_j),\ \alpha_{j+1}/(\theta_j - \alpha_{j+1}))$. Hence $R^*(\mathcal{S}) \geq H^*$ for every family $\mathcal{S} \subseteq \mathcal{A}$.

**(2)** (Attainment.) For every $\delta > 0$ and $\varepsilon > 0$ there is $n_0$ such that the generalized Petlyuk arrangement with at least $n_0$ stages in every section has an operating specification with total heat supply $R = H^* + \delta$ and a steady state at which every mole fraction is positive and every product has purity at least $1 - \varepsilon$. Hence $R^*(\mathcal{S}_{\mathrm{Pet}}) = H^*$.

**(3)** (The conjecture.) $R^*(\mathcal{S}) \geq R^*(\mathcal{S}_{\mathrm{Pet}}) = H^*$ for every family $\mathcal{S} \subseteq \mathcal{A}$: no arrangement of $\mathcal{A}$ needs less total vaporization, with infinitely many stages and pure products, than the generalized Petlyuk arrangement, whose minimum is the highest peak of the $V_{\min}$ diagram.

Part 1 is Theorem 10 and Corollary 13; part 2 is Theorem 16 (Proposition 18 for $N = 2$); part 3 is Corollary 17. "Lowest" in the conjecture must be read as "lowest or equal": many arrangements tie with the Petlyuk value §7 [5],[27].

*The lower bound* rests on one observation. Define the *Underwood flux* of a stream with vapor flow $v$ and component flows $n_i$ as $\varphi(\theta) = v - \sum_i \alpha_i n_i/(\alpha_i - \theta)$. Under constant

molar overflow, the fluxes are conserved at every unit up to its heat duty; at $\theta = \theta_j$ the feed carries no flux; and every stream leaving an equilibrium stage, liquid or vapor, has a flux of the sign of $-\sum_i \alpha_i x_i/(\alpha_i - \theta_j)$, where $x$ is the stage liquid. The units therefore split into a light and a heavy side, and the energy balance of the heavy side gives the bound, whatever the topology, the number of stages, the signs of the net flows or the purities (Section 3). The excess of $R$ over the bound is an explicit sum of three nonnegative "pinch losses", which yields necessary conditions for an arrangement to tie with the Petlyuk value.

*Attainment* is harder. In Underwood's infinite-stage model, Halvorsen and Skogestad's construction (every internal column at its preferred split) reaches $H^*$ [5]. The difficulty is to show that finite-stage steady states of the closed, thermally coupled network converge to it. Two obstacles appear. If the binary columns are operated with both products sharp, the limit equations have a kernel: the composition of the feed stage of every binary column is not determined in the limit (a "neutral mode"), and the finite-stage solutions are then fixed only by exponentially small terms. And at the exact Petlyuk minimum several sections are pinched at both ends. We avoid both by an operating specification (Section 4.5) that keeps a margin $\delta$ in the reboiler, deforms every column with three or more components slightly away from its preferred split in a direction that preserves its common Underwood roots, and *forces* every binary column by sending a small prescribed flow of one key component into the adjacent side product. In the infinite-stage limit every section is then pinched at one end only, and the limit equations in the unknowns "junction compositions and column splits" have an invertible Jacobian, with a block-triangular structure: one top-down block for the major components (Underwood interpolation and a Cauchy-type matrix in every column) and one triangular M-matrix for every component present only in traces (Theorem 52). The finite-stage equations differ from the limit equations by terms of size $\rho^n$ in $C^1$, so a quantitative inverse-function argument gives a finite-stage steady state near the limit for all section lengths $\geq n_0$, and an M-matrix argument shows that all its mole fractions are positive (Section 4.8). The limits are taken in the order $n \to \infty$, then forcing $\to 0$, then $\delta \to 0$.

### 1.4 Outline

Section 2 defines the class $\mathcal{A}$, the limit $R^*$, and their relation to Underwood's model. Section 3 proves the lower bound and discusses its consequences. Section 4 states finite-stage attainment, proves it for two components, and gives the construction and the statements of the proof for $N \geq 3$; the proofs are in Appendices B–E. Section 5 illustrates the attainment theorem numerically, and Section 6 discusses scope, limitations and open questions. Appendix F lists the computations that check the statements (none is used in a proof).

## 2 Model

### 2.1 Data, feed roots and peaks

There are $N \geq 2$ components with constant relative volatilities $\alpha_1 > \alpha_2 > \cdots > \alpha_N > 0$, so that a vapor in equilibrium with a liquid of composition $x$ has the composition

$$y(x) = \frac{\operatorname{diag}(\alpha)\, x}{S(x)}, \qquad S(x) = \sum_k \alpha_k x_k.$$

The pressure is constant, molar flows are constant (constant molar overflow, CMO), and the molar heat of vaporization is the unit of heat, so that heat duties are measured in moles of vapor generated. The process feed has flow 1, composition $z$ (all $z_p > 0$) and liquid fraction $q \in \mathbb{R}$ ($q > 1$ subcooled, $q < 0$ superheated): it brings the vapor equivalent $1 - q$. The function $t \mapsto \sum_p \alpha_p z_p/(\alpha_p - t)$ increases from $-\infty$ to $+\infty$ on every interval $(\alpha_{j+1}, \alpha_j)$, so Underwood's feed equation

$$\sum_{p=1}^{N} \frac{\alpha_p z_p}{\alpha_p - \theta} = 1 - q \tag{2}$$

has exactly one root $\theta_j \in (\alpha_{j+1}, \alpha_j)$ for each $j = 1, \dots, N-1$, for every $q$. The *peaks* $H_j$ and $H^*$ are defined above; we also write $g_j := H^* - H_j \geq 0$. $H_j$ is the minimum boilup of a two-product column that splits the feed sharply between components $j$ and $j+1$ (Underwood's formula at the active root $\theta_j$): the peak $P_{j,j+1}$ of the $V_{\min}$ diagram [3], referred to the reboiler. Referred to the top, the peaks are $H_j + 1 - q = \sum_{p \leq j} \alpha_p z_p/(\alpha_p - \theta_j)$. We write $G_x(t) = \sum_i \alpha_i x_i/(\alpha_i - t)$ for Underwood's function of a composition (or of any vector) $x$, $\mathbf{1}$ for the all-ones vector, $\mathbf{e}_p$ for the $p$-th unit vector and $\mathcal{X} = \{x \in \mathbb{R}^N : x \geq 0,\ \mathbf{1}^\top x = 1\}$ for the simplex of compositions.

## 2.2 The class of arrangements

Halvorsen's thesis (§6.1.1 of [6]) describes an arrangement as sections of connected equilibrium stages, reboilers and condensers, directly coupled by two-way liquid and vapor streams or coupled through condensers or reboilers, divided at stages with feed or side-draw streams; under CMO the energy balance of every stage reduces to "vapor out minus vapor in equals heat input" (*ibid.*, §2.6.1). The following definition keeps exactly these physical ingredients and drops every restriction on how they are connected.

**Definition 3** (the class $\mathcal{A}$)**.** An *arrangement* is a finite directed multigraph whose vertices are *units* and whose arcs are *streams*, together with a feed source and product outlets. A stream $e$ has a phase (saturated liquid or saturated vapor), component flows $n_{e,i} \geq 0$ and vapor flow $v_e$ ($= \sum_i n_{e,i}$ for a vapor, 0 for a liquid). The units are of four types:

(E) *equilibrium stage*: any inflows; an external heat duty $Q_u \in \mathbb{R}$; its contents leave as a liquid of composition $x_u$ and a vapor of composition $y(x_u)$, each divided into any number of outflow streams (either phase may have zero flow);

(M) *single-phase mixer or splitter*: inflows of one phase, outflows of that phase and of the mixed composition, $Q_u = 0$;

(C) *total condenser*: vapor inflows, liquid outflows of the mixed composition;

(R) *total vaporizer*: liquid inflows, vapor outflows of the mixed composition.

The process feed is divided into portions of sizes $f_u \geq 0$, $\sum_u f_u = 1$, each entering an (E) unit with the feed's composition and thermal state (feed conditioning is an (E) unit with a duty). Every stream leaving the arrangement is a *product stream* and is assigned to one component. A *steady state* is an assignment of flows and

compositions such that at every unit $u$

$$\sum_{e \text{ out of } u} n_{e,i} - \sum_{e \text{ into } u} n_{e,i} = f_u z_i \quad (\text{all } i), \qquad \sum_{e \text{ out of } u} v_e - \sum_{e \text{ into } u} v_e = Q_u + (1-q) f_u, \tag{4}$$

product streams counting as outflows, and the composition rule of its type holds (this defines $Q_u$ for (C) and (R) units). Its *total vaporization* is $R = \sum_u \max(Q_u, 0)$, the total heat supplied.

Nothing is assumed on the topology (recycles are cycles of the graph), on the signs of the net component flows, on which components are present where, on the numbers of stages, or on the stability or uniqueness of steady states. An *adiabatic section* is a chain of (E) units with $Q = 0$, liquid flowing down and vapor flowing up with constant flows; a column is a stack of sections joined at junction stages.

*What $\mathcal{A}$ contains.* Every device of the arrangements considered in [5] [6] [26] [13] [9] [11] [28] is a sub-network of units of $\mathcal{A}$, with the same balances and the same compositions of every stream that leaves it: conventional columns with total or partial condensers and reboilers; thermal couplings; side draws of either phase; side strippers and rectifiers; Petlyuk, generalized Petlyuk and Kaibel arrangements and dividing-wall columns (a liquid or vapor split is a stage outflow divided into several streams, a merge is several inflows into one stage); several feed stages per column and intermediate sections; stream splitters dividing a section product among several columns, remixing, vapor or liquid bypasses (tray entrainment and weeping included), recycles; feed splitting, preheating, precooling and pre-flashing; intermediate reboilers and condensers, heaters and coolers at any stage, that is, non-adiabatic sections with external utilities (which the conjecture does not require us to admit); and trays with a Murphree vapor efficiency $E \in [0, 1]$ common to all components, which are exactly an (E) unit receiving the fraction $E$ of the vapor from below, a vapor bypass of the fraction $1 - E$ and an (M) mixer; cross-flow trays (whose Murphree efficiency may exceed one) are liquid cells in series, each contacted by a portion of the vapor. Partial condensation or vaporization giving two phases of the same composition is an (M) splitter followed by a (C) or (R) unit. Differential condensation or vaporization, and rate-based contacting with component-independent film coefficients, are limits of members of $\mathcal{A}$, to which the bound below passes by continuity.

*What $\mathcal{A}$ excludes, and why the conjecture excludes it.* Heat exchange between process streams that is not counted in $R$ ("internal heat exchange", excluded in §6.3, §8 [5]; it lowers the external heat below the Petlyuk value Table 6.1 [6]); a well-mixed tray with a Murphree efficiency above one (unphysical); several pressures (excluded by constant pressure; they act through heat exchange between columns or compression and change $\alpha$); non-ideal mixtures and non-CMO enthalpies; devices that release streams of different compositions not in phase equilibrium (decanters, membranes, component-dependent stage efficiencies or mass-transfer coefficients); heat pumps; several process feeds of different compositions or thermal states; and non-steady (batch or cyclic) operation. Finite stages and impure products are not excluded: they are the objects of the model.

### 2.3 Infinitely many stages and pure products

**Definition 5** (the limit of infinitely many stages and pure products)**.** A steady state has *purity at least* $1-\varepsilon$ if every product stream assigned to a component $p$ has mole fraction of $p$ at least $1-\varepsilon$. For a family $\mathcal{S} \subseteq \mathcal{A}$ (for example all stage numbers of one configuration, or all of $\mathcal{A}$) put

$$R^*(\mathcal{S};\varepsilon) = \inf\{R(s) : s \text{ a steady state of a member of } \mathcal{S}, \text{ purity at least } 1-\varepsilon\},$$
$$R^*(\mathcal{S}) = \lim_{\varepsilon \to 0^+} R^*(\mathcal{S};\varepsilon).$$

The limit exists in $[0,+\infty]$ because $R^*(\mathcal{S};\varepsilon)$ is nonincreasing in $\varepsilon$. For a configuration $X$ (a topology with unspecified numbers of stages), $R^*(\mathcal{S}_X)$, with $\mathcal{S}_X$ the arrangements of topology $X$ and any numbers of stages, is the faithful meaning of "the minimum vapor of $X$ with infinitely many stages and pure products": every real arrangement has finitely many stages and impure products, and infinite-stage values are used, in the words of [7], because they provide a lower bound on the energy requirement. In this language:

**Definition 6** (the Halvorsen–Skogestad conjecture in $\mathcal{A}$)**.** $R^*(\mathcal{S}_{\mathrm{Pet}}) \le R^*(\mathcal{S})$ for every family $\mathcal{S} \subseteq \mathcal{A}$, and $R^*(\mathcal{S}_{\mathrm{Pet}}) = H^*$ (the formula of [5] for the Petlyuk minimum).

*Relation to Underwood's model.* In Underwood's infinite-stage model each section is described by its flows and net component flows, and its compositions at the ends are its pinch compositions; the minimum vapor of a configuration is then a finite-dimensional optimization problem [5] [10] [11]. Definition 5 does not use this model. Theorem 1 shows that for the generalized Petlyuk arrangement the two notions agree ($R^*(\mathcal{S}_{\mathrm{Pet}})$ equals Halvorsen and Skogestad's Underwood value), and that no arrangement can do better in the physical sense. Underwood-model formulations used for global optimization are best seen as relaxations: their points need not be limits of physical steady states, and one of them admits points below $H^*$ (Remark 15).

## 3 The lower bound

### 3.1 The Underwood flux and the cut theorem

For $\theta > 0$, $\theta \notin \{\alpha_i\}$, the *Underwood flux* of a stream $e$ and of a unit of feed are

$$\varphi_e(\theta) = v_e - \sum_i \frac{\alpha_i n_{e,i}}{\alpha_i - \theta}, \qquad \varphi_F(\theta) = (1-q) - \sum_i \frac{\alpha_i z_i}{\alpha_i - \theta}, \tag{7}$$

so that $\varphi_F(\theta_j) = 0$ for every $j$ by (2). For the same component flows, a vapor stream has the flux of the liquid stream plus its total flow.

**Lemma 8** (node balance)**.** At every unit $u$ of a steady state and for every $\theta > 0$, $\theta \notin \{\alpha_i\}$,

$$\sum_{e \text{ out of } u} \varphi_e(\theta) - \sum_{e \text{ into } u} \varphi_e(\theta) = Q_u + f_u \varphi_F(\theta).$$

**Proof.** $\varphi_e = v_e - \sum_i c_i n_{e,i}$ with $c_i = \alpha_i/(\alpha_i - \theta)$ is linear in $(v_e, n_e)$. Subtract $\sum_i c_i$ times the component balances from the energy balance in (4). □

**Lemma 9** (sign uniformity). Let $\theta > 0$, $\theta \notin \{\alpha_i\}$. A liquid stream of flow $l$ and composition $x$ has $\varphi = -l\,G_x(\theta)$; a vapor stream of flow $v$ in equilibrium with a liquid of composition $x$ has $\varphi = -(\theta v/S(x))\,G_x(\theta)$. Hence all streams leaving an (E) unit have fluxes of one sign, that of $-G_{x_u}(\theta)$, and so do all streams leaving an (M), (C) or (R) unit.

**Proof.** For the vapor, with $\sum_i y_i = 1$, $y_i = \alpha_i x_i/S(x)$ and $\alpha_i/(\alpha_i - \theta) = 1 + \theta/(\alpha_i - \theta)$: $\varphi = v\sum_i y_i(1 - \frac{\alpha_i}{\alpha_i - \theta}) = -\theta v \sum_i \frac{y_i}{\alpha_i - \theta} = -\frac{\theta v}{S(x)}\,G_x(\theta)$, and $\theta > 0$, $S(x) > 0$. The outflows of an (M), (C) or (R) unit have one phase and one composition, so their fluxes are nonnegative multiples of one number. □

Fix $j$ and $\theta = \theta_j$. Call a unit *heavy-ish* if all its outflows have $\varphi(\theta_j) \geq 0$ and *light-ish* otherwise; by Lemma 9 all outflows of a light-ish unit have $\varphi(\theta_j) \leq 0$. Let $\mathcal{U}_+, \mathcal{U}_-$ be the two sets of units, $R(\mathcal{W}) = \sum_{u\in\mathcal{W}} \max(Q_u, 0)$ and $Q(\mathcal{W}) = \sum_{u\in\mathcal{W}} \max(-Q_u, 0)$.

**Theorem 10** (stage-level cut). At every steady state of every arrangement of $\mathcal{A}$, with any numbers of stages and any purities, and for every $j$,

$$R - \sum_P \max(0, \varphi_P(\theta_j)) = R(\mathcal{U}_-) + Q(\mathcal{U}_+) + \sum_{e:\,\mathcal{U}_+\to\mathcal{U}_-} \varphi_e(\theta_j) - \sum_{e:\,\mathcal{U}_-\to\mathcal{U}_+} \varphi_e(\theta_j), \quad (11)$$

the first sum running over the product streams and the last two over the streams between the two sets; every term on the right is nonnegative. In particular $R \geq \sum_P \max(0, \varphi_P(\theta_j))$.

**Proof.** Sum Lemma 8 over $u \in \mathcal{U}_+$ at $\theta_j$, where the feed terms vanish. Streams with both ends in $\mathcal{U}_+$ cancel, so $R(\mathcal{U}_+) - Q(\mathcal{U}_+) = \sum_{P \text{ from } \mathcal{U}_+} \varphi_P + \sum_{\mathcal{U}_+\to\mathcal{U}_-} \varphi_e - \sum_{\mathcal{U}_-\to\mathcal{U}_+} \varphi_e$. Product streams leaving $\mathcal{U}_+$ have $\varphi_P \geq 0$ and those leaving $\mathcal{U}_-$ have $\varphi_P \leq 0$, so the first sum is $\sum_P \max(0, \varphi_P)$; streams leaving $\mathcal{U}_+$ have $\varphi \geq 0$ and streams leaving $\mathcal{U}_-$ have $\varphi \leq 0$, so the crossing terms are nonnegative. Add $R(\mathcal{U}_-) + Q(\mathcal{U}_+)$ to both sides and use $R = R(\mathcal{U}_+) + R(\mathcal{U}_-)$. □

**Corollary 12** (pure products). If every product stream is pure, then $R \geq H_j$ for every $j$, that is, $R \geq H^*$.

**Proof.** A pure liquid product of component $p$ and flow $P$ has $\varphi = \alpha_p P/(\theta_j - \alpha_p)$, a pure vapor product $\varphi = \theta_j P/(\theta_j - \alpha_p)$; both are positive exactly when $p > j$, the vapor value being the larger. The products of component $p$ carry $z_p$ in total. □

**Corollary 13** (finite purity). If every product stream has purity at least $1 - \varepsilon$, then $R \geq H_j - c_j\varepsilon$ for every $j$, with $c_j$ as in Theorem 1. Consequently $R^*(\mathcal{S}) \geq H^*$ for every family $\mathcal{S} \subseteq \mathcal{A}$.

**Proof.** Since a vapor stream has the flux of the liquid stream with the same component flows plus its total flow, it suffices to bound liquid fluxes from below. Keeping only the products assigned to components $p > j$, $\sum_P \max(0, \varphi_P) \geq \sum_{i>j} \alpha_i n_i^h/(\theta_j - \alpha_i) - \sum_{i\leq j} \alpha_i n_i^h/(\alpha_i - \theta_j)$, where $n_i^h$ is the flow of $i$ in these products. By the overall balance,

$n_i^h = z_i - m_i$ for $i > j$, where $m_i$ is the flow of $i$ in the products assigned to components $\le j$. The flows $m_i$ $(i > j)$ and $n_i^h$ $(i \le j)$ are disjoint parts of the total impurity, which is at most $\varepsilon$ times the total product flow 1; the coefficients $\alpha_i/(\theta_j - \alpha_i)$ $(i > j$, increasing in $\alpha_i)$ and $\alpha_i/(\alpha_i - \theta_j)$ $(i \le j$, decreasing in $\alpha_i)$ are at most $c_j$. Hence $\sum_P \max(0, \varphi_P) \ge H_j - c_j\varepsilon$, and Theorem 10 applies. Taking the infimum over steady states with purity at least $1 - \varepsilon$ and letting $\varepsilon \to 0$ gives $R^*(\mathcal{S}) \ge H_j$ for every $j$. □

*What the proof uses.* The CMO energy balance at every unit (Lemma 8); that every unit releases streams of a single composition per phase, the two phases being in equilibrium with constant relative volatilities when both occur (Lemma 9); $\theta_j > 0$; one feed composition and thermal state, so that every feed portion has zero flux at $\theta_j$; and the counting of every positive duty in $R$. Nothing is used about topology, numbers of stages, signs of net flows (reverse flows), which components are present where (degenerate distributions), stability, adiabaticity of sections, or product phases. In particular the bound holds for the Petlyuk arrangement itself: $R^*(\mathcal{S}_{\mathrm{Pet}}) \ge H^*$ for every peak, which is the half of the formula of [5] that was proved there only for the extreme peaks.

## 3.2 Consequences

*The pinch hyperplane and three losses.* For $0 \le q \le 1$ the equilibrium liquid $x^F$ of the feed flash satisfies $G_{x^F}(\theta_j) = 0$ for every $j$: inserting $z = (1-q)y(x^F) + qx^F$ in (2) gives $G_{x^F}(\theta_j)\,[(1-q)\theta_j/S(x^F) + q] = 0$. So an (E) unit is light-ish or heavy-ish according to the side of the hyperplane $\{x : G_x(\theta_j) = 0\}$ through the feed-flash liquid on which its liquid lies (for an (R) unit, the liquid in equilibrium with its vapor). Theorem 10 says that the heat supplied to the heavy side is at least $H_j$ for pure products, and that the excess of $R$ over the bound is the sum of three nonnegative losses: heat supplied on the light side, heat removed on the heavy side, and flux across the hyperplane, much as in pinch analysis of heat-exchanger networks.

*Sections.* In an adiabatic section with vapor $V$, liquid $L$ and net upward component flows $w = Vy_k - Lx_{k+1}$ (constant along the section), the difference "flux of the vapor going up minus flux of the liquid coming down" across any cross-section equals the *section flux*

$$\Psi(\theta) = V - \sum_i \frac{\alpha_i w_i}{\alpha_i - \theta} = L\,G_{x_{k+1}}(\theta) - \frac{\theta V}{S(x_k)}\,G_{x_k}(\theta) \tag{14}$$

(Lemma 9); at the roots of $\Psi$ this is Underwood's stage identity eq. (2.49) [6], here for every $\theta$. Hence, going up a section, the side changes at most once: from heavy-ish to light-ish if $\Psi(\theta_j) > 0$, the reverse if $\Psi(\theta_j) < 0$, never if $\Psi(\theta_j) = 0$; a section whose stages change side contributes $|\Psi(\theta_j)|$ to the crossing terms of (11). A pinch composition $\hat{x}$ of the section, $\hat{x}_i \propto w_i/(\alpha_i - \psi)$ with $\Psi(\psi) = 0$, has $G_{\hat{x}}(\theta) = -(S(\hat{x})/V)\,\Psi(\theta)/(\theta - \psi)$.

*Ties.* By (11), a family of steady states with purity tending to 1 and $R \to H_j$ must have, in the limit, no heat supplied to light-ish units, no heat removed from heavy-ish units, vanishing flux on every stream between the two sides, and $\theta_j$ as a root of $\Psi$ for every section whose stages change side, the change taking place in a zone where the liquid is close to the hyperplane: Underwood's pinch at an active root, now a necessary condition for every arrangement. In particular, if the highest peak $H_{j^*}$ is strict, then along steady states that attain $H^*$ the duty of every reboiler or heater whose outflow is (in the limit) a pure light product $p \le j^*$, and of every condenser or cooler whose outflow is a pure heavy product $p > j^*$, tends to zero. With finitely many stages the inequality is strict as soon as some section changes side.

*Internal heat exchange.* The theorem bounds the heat delivered to the heavy-ish units, whatever its origin. If heat exchanged between process streams is counted in $R$, the bound still holds; the external heat can fall below $H^*$ only by moving heat into heavy-ish units from light-ish ones, which is what heat-integrated arrangements do §6.5 [6],[29]. This is why the conjecture must exclude internal heat exchange.

*Several feeds and non-sharp products.* With feeds $F_k$ of sizes $f_k$, compositions $z^{(k)}$ and thermal states $q_k$, the same proof gives

$$R \geq \sup_{\theta} \Big[ \sum_{P} \max(0, \varphi_P(\theta)) - \sum_{k} f_k \max(0, \varphi_{F_k}(\theta)) \Big],$$

and for arbitrary product specifications $R \geq \max_j \sum_P \max(0, \varphi_P(\theta_j))$ as it stands; for a two-product column the latter is Underwood's classical bound $V_B \geq \sum_i \alpha_i b_i/(\theta_j - \alpha_i)$, here for any number of stages. Whether these bounds are attained is not studied here.

**Remark 15** (a relaxation of Underwood's model can go below the Petlyuk value)**.** In the Underwood model of [11] (constraints on the section vapors at the column feed roots, with equalities at the interior roots and the option of carrying vapor above the Underwood threshold), the five-component regular-column configuration with splits ABCDE→ABCD/BCDE, ABCD→ABC/BCD, BCDE→BCD/CDE, ABC→AB/BC, BCD→BC/D, CDE→C/DE and binary finishing columns, all transfers thermally coupled, has, for a saturated-vapor feed and suitable rational $(\alpha, z)$, a feasible point with $R/H^* \approx 0.99946$, certified by a Krawczyk test in interval arithmetic (Appendix F). By Corollary 13 no family of steady states of any arrangement of $\mathcal{A}$ with purity tending to 1 approaches it. Concretely, orient every section from its bottom end to its top end if its flux at $\theta_3$ is positive, and the other way if it is negative. At a steady state the side can only change once along a section, from heavy-ish below to light-ish above when $\Psi > 0$ and the reverse when $\Psi < 0$, so along an oriented section a light-ish end is followed by a light-ish end. The section fluxes of the permissive point contain an oriented path from the light product C through the columns CDE, BCDE and BCD to the heavy product D, which would therefore have to be light-ish: impossible. The violation is global, a chain of four sections, which is why no column-by-column condition excludes it. For global optimization this suggests adding, for every $j$, binary "side" variables at the junctions with the sign constraints above: every point of the repaired model satisfies $R \geq H^*$, and physical limits satisfy its constraints when each junction is one equilibrium stage. ⌟

# 4 Finite-stage attainment

## 4.1 Statement

The generalized Petlyuk arrangement is defined precisely in Section 4.3 (Definition 19, Figure 20); $\mathcal{S}_{\mathrm{Pet}}$ denotes this arrangement with all possible numbers of stages in its sections.

**Theorem 16** (finite-stage attainment)**.** Let $N \geq 2$, $\alpha_1 > \cdots > \alpha_N > 0$, $z > 0$ with $\sum_p z_p = 1$, and $q \in \mathbb{R}$. For every $\delta > 0$ and $\varepsilon > 0$ there is $n_0$ such that, for every choice of section lengths $n_\sigma \geq n_0$ (equal or not), the generalized Petlyuk arrangement with these section lengths has an operating specification (all stream flows prescribed, the same for all such $n_\sigma$) with total heat supply $R = H^* + \delta$, and a steady state at which every mole fraction of every stream is positive and every

product has purity at least $1-\varepsilon$. Hence $R^*(\mathcal{S}_{\mathrm{Pet}}) \le H^*$.

**Corollary 17** (the Halvorsen–Skogestad conjecture in $\mathcal{A}$)**.** $R^*(\mathcal{S}_{\mathrm{Pet}}) = H^*$, and $R^*(\mathcal{S}) \ge R^*(\mathcal{S}_{\mathrm{Pet}})$ for every family $\mathcal{S} \subseteq \mathcal{A}$.

**Proof.** $\mathcal{S}_{\mathrm{Pet}} \subseteq \mathcal{A}$, so Corollary 13 gives $R^*(\mathcal{S}) \ge H^*$ for every $\mathcal{S} \subseteq \mathcal{A}$, in particular $R^*(\mathcal{S}_{\mathrm{Pet}}) \ge H^*$; Theorem 16 gives $R^*(\mathcal{S}_{\mathrm{Pet}}) \le H^*$. □

Attainment is sufficient for the conjecture, not logically necessary: without it the inequality $R^*(\mathcal{S}_{\mathrm{Pet}}) \le R^*(\mathcal{S})$ could still hold, but the formula $R^*(\mathcal{S}_{\mathrm{Pet}}) = H^*$ of [5] would fail. The operating specification of Theorem 16 does not depend on the numbers of stages; only the purity does. The limits are taken in the order $n \to \infty$ at fixed specification, then the forcing parameters $\to 0$, then $\delta \to 0$; no uniformity is needed because $R^*(\mathcal{S};\varepsilon)$ is an infimum over all numbers of stages.

## 4.2 Two components

**Proposition 18** (two components)**.** Let $N = 2$, $\alpha_1 = \alpha > 1 = \alpha_2$, light feed fraction $z \in (0,1)$ and any $q$. For every $s > H_1$ and $\varepsilon > 0$ there is $n_0$ such that a column with a total condenser, $n_T \ge n_0$ stages above the feed stage, $n_B \ge n_0$ stages below it and a partial reboiler of duty $s$, operated with distillate flow $z$, has a steady state with both products of purity at least $1-\varepsilon$, at which every mole fraction lies in $(0,1)$. Hence Theorem 16 holds for $N = 2$.

The proof (Appendix A) is a shooting argument in the McCabe–Thiele plane: for $s > H_1$ the pinch of the top section lies below the pinch of the bottom section, and the intermediate value theorem matches the two profiles at the feed stage. This is classical in substance; we include it for completeness. From now on $N \ge 3$.

## 4.3 The arrangement, its specification and the unknowns

**Definition 19** (the generalized Petlyuk arrangement)**.** The *columns* are $c = [i,m]$, $1 \le i < m \le N$ (the set $\mathcal{C}$, $|\mathcal{C}| = N(N-1)/2$); $c$ is *binary* if $m = i+1$ and *non-binary* otherwise. The *nodes* are $J[i,m]$, $1 \le i \le m \le N$. Column $[i,m]$ consists of a top section $\mathrm{T}[i,m]$, with bottom end $J[i,m]$ and top end $J[i,m-1]$, and a bottom section $\mathrm{B}[i,m]$, with top end $J[i,m]$ and bottom end $J[i+1,m]$. The node $J_c := J[i,m]$, $i < m$, is the *junction* (feed stage) of $c$, an (E) stage without duty; the process feed enters $J[1,N]$. The node $J[p,p]$ is the *product node* of component $p$: $J[1,1]$ is a total condenser (reflux and liquid product 1), $J[N,N]$ a partial reboiler (an (E) stage delivering liquid product $N$, the only heat input), and $J[p,p]$, $2 \le p \le N-1$, an (E) side-draw stage without duty (liquid product $p$), where $\mathrm{B}[p-1,p]$ ends from above and $\mathrm{T}[p,p+1]$ from below. A *section* $\sigma$ consists of $n_\sigma \ge 1$ (E) stages without duty, numbered $1,\dots,n_\sigma$ upward, with constant liquid and vapor flows $L_\sigma, V_\sigma > 0$; it receives vapor from its bottom-end node and liquid from its top-end node and returns liquid and vapor to them (thermal couplings). We write $\mathrm{bot}(\sigma)$, $\mathrm{top}(\sigma)$ for the end nodes of $\sigma$, $\mathrm{T}_c, \mathrm{B}_c$ for the sections of $c$, $\Delta_\sigma := V_\sigma - L_\sigma$, and $h(J[i,m]) := i+m$.

A junction $J[i,m]$ is thus attached to four sections: $\mathrm{T}[i,m]$ (as its bottom end), $\mathrm{B}[i,m]$ (top end), $\mathrm{T}[i,m+1]$ (top end, if $m < N$) and $\mathrm{B}[i-1,m]$ (bottom end, if $i > 1$). This

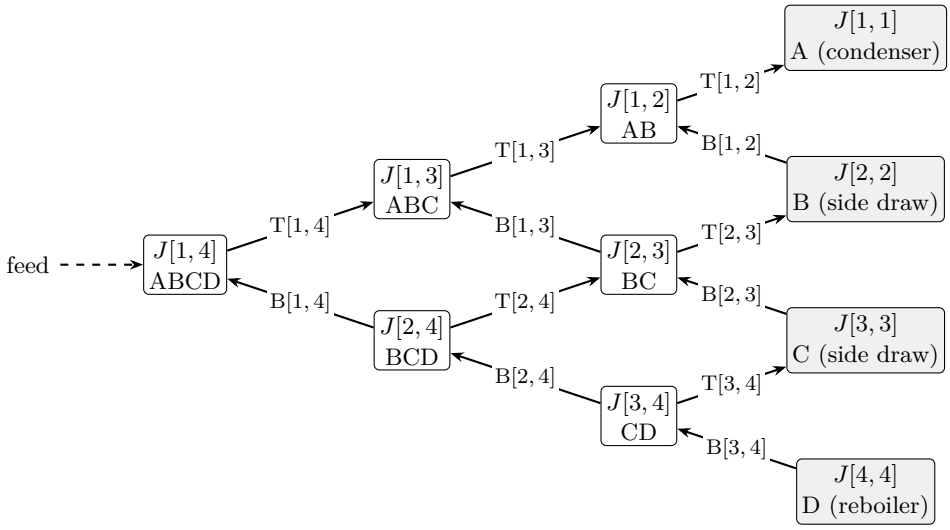



**Figure 20 /** The generalized Petlyuk arrangement for $N = 4$ (components A, B, C, D) as a network of nodes and sections. Arrows point upward (direction of the vapor); every section also carries liquid downward, so every arrow is a two-way thermal coupling. Each column $[i, m]$ is the pair of sections $\mathrm{T}[i, m]$ (from $J[i, m]$ up to $J[i, m-1]$) and $\mathrm{B}[i, m]$ (from $J[i+1, m]$ up to $J[i, m]$). The right-hand nodes form the main column, with one condenser, one reboiler and liquid side draws. The vertical position of $J[i, m]$ is $-h = -(i + m)$.

is the arrangement of Figure 1 of [5], with the main column written as the stack of the binary columns $[p, p+1]$ and their side draws; for $N = 3$ it is the Petlyuk column. In [5] the side products are drawn between consecutive columns of the last row; the side-draw stage $J[p, p]$ can be counted as a stage of $\mathrm{B}[p-1, p]$, so the family $\mathcal{S}_{\mathrm{Pet}}$ is the same. Halvorsen and Skogestad also allow some heat to be supplied or removed at the product outlets; we do not use this freedom. The arrangement belongs to $\mathcal{A}$.

**Definition 21** (operating specification)**.** An *operating specification* prescribes the flows $L_\sigma, V_\sigma > 0$ of every section and the product flows $P_1, \dots, P_N > 0$, such that at every unit the total material balance and the CMO balance hold, with zero duty at every junction and every side-draw stage. The condenser duty is then $-V_{\mathrm{T}[1,2]}$ and the reboiler duty is $R = V_{\mathrm{B}[N-1,N]}$.

*Unknowns.* For every column $c$ let $d_c \in \mathbb{R}^N$ be its *top split* (the net upward component flow in $\mathrm{T}_c$), with $\mathbf{1}^\top d_c = \Delta_{\mathrm{T}_c}$, and $X_c \in \mathbb{R}^N$ its *junction composition*, with $\mathbf{1}^\top X_c = 1$. The set of $v = (d_c, X_c)_{c \in \mathcal{C}}$ is an affine space $\mathcal{V}$ of dimension $2(N-1)|\mathcal{C}| = N(N-1)^2$. The following quantities are affine in $v$, computed parents first (labels shrink from parent to child):

$$f_{[1,N]} := z, \qquad f_{[i,m]} := d_{[i,m+1]}\,[m < N] + b_{[i-1,m]}\,[i > 1] \quad \text{(the net feed of } [i, m]),$$
$$b_c := f_c - d_c, \qquad w_{\mathrm{T}_c} := d_c, \qquad w_{\mathrm{B}_c} := -b_c \quad \text{(net upward flows)},$$
$$\bar{x}_p := \big(d_{[p,p+1]}\,[p < N] + b_{[p-1,p]}\,[p > 1]\big)/P_p \quad \text{(product compositions)},$$

where $[\cdot]$ is 1 if the condition holds and 0 otherwise. The composition of a node is $X_c$ at $J_c$ and $\bar{x}_p$ at $J[p, p]$; $x_{\mathrm{bot}}(\sigma)$, $x_{\mathrm{top}}(\sigma)$ are those of $\mathrm{bot}(\sigma)$, $\mathrm{top}(\sigma)$. The total balances of a specification give $\mathbf{1}^\top w_\sigma = \Delta_\sigma$ and $\mathbf{1}^\top \bar{x}_p = 1$.

*Sections.* For $w \in \mathbb{R}^N$ put $M_\sigma(w) := V_\sigma \operatorname{diag}(\alpha) - w\,\alpha^\top$. If $\mathbf{1}^\top w = \Delta_\sigma$ then $\mathbf{1}^\top M_\sigma(w) =$

$L_\sigma \alpha^\top$, and the balance $V_\sigma y(x_k) - L_\sigma x_{k+1} = w$ between consecutive stages is equivalent to $L_\sigma S(x_k)\, x_{k+1} = M_\sigma(w)\, x_k$: the stage-to-stage map of a section is projective-linear.

**Proposition 22** (exact reformulation). Fix an operating specification and section lengths $n_\sigma$.

(a) Every steady state defines, through its column splits and junction compositions, a point $v \in \mathcal{V}$ at which, for every section, $x_{\mathrm{top}}(\sigma)$ is a positive multiple of $M_\sigma(w_\sigma)^{n_\sigma+1} x_{\mathrm{bot}}(\sigma)$; its stage compositions are the normalized iterates $x_{k+1} = M_\sigma x_k / (\mathbf{1}^\top M_\sigma x_k)$, $x_0 = x_{\mathrm{bot}}(\sigma)$.

(b) Conversely, let $v \in \mathcal{V}$ be such that, for every section, $x_{\mathrm{top}}(\sigma)$ is a real multiple of $M_\sigma^{n_\sigma+1} x_{\mathrm{bot}}(\sigma)$, the iterates $x_k^\sigma := M_\sigma^k x_{\mathrm{bot}}(\sigma)/(\mathbf{1}^\top M_\sigma^k x_{\mathrm{bot}}(\sigma))$ are defined and satisfy $S(x_k^\sigma) > 0$ for $0 \le k \le n_\sigma$, and $S > 0$ at every node composition. Assigning $x_k^\sigma$ and $y(x_k^\sigma)$ to stage $k$ of $\sigma$, $X_c$ to $J_c$ and $\bar{x}_p$ to $J[p,p]$ defines a steady state of the arrangement with the prescribed flows, at which *every mole fraction of every unit and stream is strictly positive* and $R = V_{\mathrm{B}[N-1,N]}$.

The positivity in (b) needs no sign information on net flows or on trace contents, only $S > 0$: at fixed $K$-values $\alpha_i / S(x_u)$ the component balances form, for each component, a linear system whose matrix is a nonsingular M-matrix with a positive inverse between any two units joined by a path of positive flows (Appendix B).

### 4.4 One section

Fix a section with flows $L, V > 0$ and $w \in \mathbb{R}^N$, and put

$$\Psi(t; w) := V - \sum_i \frac{\alpha_i w_i}{\alpha_i - t}, \qquad P_w(t) := \prod_i (\alpha_i - t)\, \Psi(t; w),$$

terms with $w_i = 0$ being omitted from $\Psi$; $\Psi^{(k)}$ and $G_x^{(k)}$ denote the sums without the term $i = k$. $\Psi$ is the section flux (14). The following lemmas are proved in Appendix B.

**Lemma 23** (spectrum, eigenvectors and the pinch identity). (i) $\det(Vt - M(w)) = (-1)^N V^{N-1} P_w(t)$: the eigenvalues of $M(w)$ are $Vt$ for the roots $t$ of $P_w$, which are the roots of $\Psi$ (*major modes*) and the $\alpha_k$ with $w_k = 0$ (*trace modes*).

(ii) For a root $\psi$ of $\Psi$ put $\xi_\psi := (w_i/(\alpha_i - \psi))_i$ and $\ell_\psi := (\alpha_i/(\alpha_i - \psi))_i$, so that $\ell_\psi \cdot x = G_x(\psi)$; for a trace mode $\alpha_k$ put $(\xi_{\alpha_k})_i := w_i/(\alpha_i - \alpha_k)$ $(i \ne k)$, $(\xi_{\alpha_k})_k := \Psi^{(k)}(\alpha_k)/\alpha_k$ and $\ell_{\alpha_k} := \mathbf{e}_k$. Then $M\xi_\psi = V\psi\,\xi_\psi$, $\ell_\psi^\top M = V\psi\,\ell_\psi^\top$, $\mathbf{1}^\top \xi_\psi = L/\psi$, and $\ell_\psi \cdot \xi_\chi = 0$ for distinct modes $\psi \ne \chi$; $\ell_\psi \cdot \xi_\psi = -\Psi'(\psi)$ for a root and $\Psi^{(k)}(\alpha_k)/\alpha_k$ for a trace mode. The vector $\hat{\xi}_\psi := (\psi/L)\,\xi_\psi$, with $\mathbf{1}^\top \hat{\xi}_\psi = 1$, is the *pinch composition* of the mode $\psi$.

(iii) (Pinch identity.) For every mode $\psi$, $G_{\xi_\psi}(t) = -\Psi(t)/(t - \psi)$ for every $t \notin \{\alpha_i : w_i \ne 0\} \cup \{\psi\}$ (for a trace mode $\alpha_k$, $\Psi = \Psi^{(k)}$ because $w_k = 0$). In particular every eigenvector of a mode $\psi$ is annihilated by $\ell_\chi$ for every other mode $\chi$.

(iv) If the $N$ modes are distinct, the $\hat{\xi}_\psi$ form a basis, and $x = \sum_\psi [x]_\psi\, \hat{\xi}_\psi$ with $[x]_\psi = \ell_\psi \cdot x / \ell_\psi \cdot \hat{\xi}_\psi$.

**Lemma 24** (one-signed net flows)**.** Let $w \neq 0$ have all its nonzero entries of one sign, with support $\Sigma$, and let $a_1 > \cdots > a_s$ be the $\alpha_i$, $i \in \Sigma$.

(a) (Top type: $w \geq 0$, $L > 0$.) $\Psi$ is strictly decreasing on each interval of $\mathbb{R}\backslash\{a_j\}$; $\Psi > V$ on $(a_1, \infty)$; $\Psi$ has exactly one root in each $(a_{j+1}, a_j)$, one root $\psi_{\text{low}} \in (0, a_s)$, and no other; $\Psi > 0$ on $(-\infty, \psi_{\text{low}})$.

(b) (Bottom type: $w \leq 0$.) $\Psi$ is strictly increasing on each interval of $\mathbb{R}\backslash\{a_j\}$; $\Psi > V$ on $(-\infty, a_s)$; $\Psi$ has exactly one root in each $(a_{j+1}, a_j)$, one root $\psi_{\text{high}} \in (a_1, \infty)$, and no other; $\Psi > 0$ on $(\psi_{\text{high}}, \infty)$.

All these roots are simple and positive.

**Lemma 25** (positive directions)**.** (a) If $w \leq 0$, then $M \geq 0$ entrywise. (b) If $w \geq 0$ and $L > 0$, then $M^{-1} = V^{-1}\operatorname{diag}(\alpha)^{-1} + (VL)^{-1}\operatorname{diag}(\alpha)^{-1} w\mathbf{1}^\top \geq 0$. (c) If all modes are positive and $x = \sum_\psi [x]_\psi \hat{\xi}_\psi$ with $[x]_\psi \geq 0$, and $\hat{\xi}_\psi \geq 0$ whenever $[x]_\psi \neq 0$, then $M^k x \geq 0$ for every $k \in \mathbb{Z}$.

**Lemma 26** (analytic eigen-data through zero trace flows)**.** Let $M(w^0)$ have $N$ simple positive eigenvalues $Vt_1^0 > \cdots > Vt_N^0$. On a neighbourhood $O$ of $w^0$ the roots $t_1(w) > \cdots > t_N(w) > 0$ of $P_w$ are simple and real-analytic, with analytic left and right eigenvectors $\ell_r(w)$, $\xi_r(w)$; any other analytic choice differs by nonvanishing analytic factors. If $t_r^0 = \alpha_k$ is a trace mode at $w^0$ (so $w_k^0 = 0$), then $\Psi^{(k)}(\alpha_k; w^0) \neq 0$,

$$(\alpha_k - t_r(w))\, \Psi^{(k)}(t_r(w); w) = \alpha_k w_k \qquad (w \in O), \tag{27}$$

and the left eigenvector normalized by its $k$-th entry, $\tilde{\ell}_r(w)_i := \frac{w_k}{\Psi^{(k)}(t_r;w)} \frac{\alpha_i}{\alpha_i - t_r}$ $(i \neq k)$, $\tilde{\ell}_r(w)_k := 1$, is analytic on $O$ for either sign of $w_k$ and satisfies

$$\tilde{\ell}_r(w) \cdot x = x_k + \frac{w_k}{\Psi^{(k)}(t_r(w); w)}\, G_x^{(k)}(t_r(w)). \tag{28}$$

In particular $\tilde{\ell}_r = \mathbf{e}_k$ where $w_k = 0$, and at a point with $w_k = 0$, $x_k = 0$ the derivative of $(w, x) \mapsto \tilde{\ell}_r(w) \cdot x$ is $(\delta w, \delta x) \mapsto \delta x_k + \delta w_k\, G_x^{(k)}(\alpha_k)/\Psi^{(k)}(\alpha_k; w)$.

*Product form.* Let $\psi_* = t_{r_*}$ be a designated major mode (the *pinch*), $U := \{r < r_*\}$ the modes above it and $D := \{s > r_*\}$ those below, $\lambda_r := Vt_r$, $\mu_r := \lambda_r/\lambda_{r_*}$ $(r \in U)$, $\mu_s := \lambda_s/\lambda_{r_*}$ $(s \in D)$, all $< 1$, and $\varpi_{\text{bot}} := \ell_{r_*} \cdot x_{\text{bot}}$, $\varpi_{\text{top}} := \ell_{r_*} \cdot x_{\text{top}}$ (the *pivots*). For $n \geq 0$ define

$$\begin{aligned} E_r^{(n)} &:= (\ell_r \cdot x_{\text{bot}})\, \varpi_{\text{top}} - \mu_r^{n+1} (\ell_r \cdot x_{\text{top}})\, \varpi_{\text{bot}} \quad (r \in U), \\ E_s^{(n)} &:= (\ell_s \cdot x_{\text{top}})\, \varpi_{\text{bot}} - \mu_s^{n+1} (\ell_s \cdot x_{\text{bot}})\, \varpi_{\text{top}} \quad (s \in D), \end{aligned} \tag{29}$$

and $E^{(\infty)}$ by the same formulas with $\mu^{n+1}$ replaced by 0.

**Lemma 30** (product form of the section relation)**.** On $O$, where $\varpi_{\text{bot}}\varpi_{\text{top}} \neq 0$:

(a) $E^{(n)}(x_{\text{bot}}, x_{\text{top}}, w) = 0$ if and only if $x_{\text{top}} = c\, M(w)^{n+1} x_{\text{bot}}$ for some $c \neq 0$, and then $c = \varpi_{\text{top}}/(\lambda_{r_*}^{n+1} \varpi_{\text{bot}})$;

(b) $E^{(\infty)} = 0$ if and only if $\ell_r \cdot x_{\text{bot}} = 0$ for $r \in U$ and $\ell_s \cdot x_{\text{top}} = 0$ for $s \in D$: every

mode above the pinch is *closed* at the bottom end and every mode below it at the top end (the limit relation of a long section);

(c) on compact sets of $(x_{\mathrm{bot}}, x_{\mathrm{top}})$ and a compact neighbourhood $O' \subset O$ of $w^0$ with $\bar\rho := \sup_{O'} \max_r \lambda_{r+1}/\lambda_r < 1$, $|E^{(n)} - E^{(\infty)}| \le C\bar\rho^{\,n+1}$ and $\|D(E^{(n)} - E^{(\infty)})\| \le C(n+2)\bar\rho^{\,n+1}$; at $w = w^0$, $|E^{(n)} - E^{(\infty)}| \le C\rho_0^{n+1}$ with $\rho_0 := \max(\lambda^0_{r_*+1}/\lambda^0_{r_*}, \lambda^0_{r_*}/\lambda^0_{r_*-1})$;

(d) changing the analytic normalization of the $\ell_r$ multiplies the components of $E^{(n)}$ and $E^{(\infty)}$ by nonvanishing analytic factors, and at a zero of $E^{(\infty)}$, $DE^{(\infty)} = \mathrm{diag}(\varpi_{\mathrm{top}}\ (r \in U),\ \varpi_{\mathrm{bot}}\ (s \in D))\, D\Lambda$ with $\Lambda := ((\ell_r \cdot x_{\mathrm{bot}})_{r\in U}, (\ell_s \cdot x_{\mathrm{top}})_{s\in D})$. Hence the invertibility of the derivative of the network limit map below depends neither on these normalizations nor on whether its rows are divided by the pivots.

With $a_s := \ell_s \cdot x_{\mathrm{bot}}/\varpi_{\mathrm{bot}}$ and $b_r := \ell_r \cdot x_{\mathrm{top}}/\varpi_{\mathrm{top}}$ (eigenvectors normalized by $\ell_r \cdot \hat\xi_r = 1$), (a) says that $x_{\mathrm{bot}} \propto \hat\xi_{r_*} + \sum_D a_s\hat\xi_s + \sum_U \mu_r^{n+1} b_r \hat\xi_r$ and $x_{\mathrm{top}} \propto \hat\xi_{r_*} + \sum_U b_r\hat\xi_r + \sum_D \mu_s^{n+1} a_s \hat\xi_s$: a long section is a bottom "half-profile" and a top half-profile glued at the pinch. The coefficients $a, b$ are not unknowns; they stay bounded as long as both pivots stay away from 0: the section is then *singly pinched*. In a *doubly pinched* section a pivot vanishes in the limit and the section relation degenerates into a product of two small factors equal to an exponentially small number; the design below avoids this.

## 4.5 The forced design

*Non-binary columns.* Fix a non-binary column $c = [i, m]$ with feed $f$ ($f_p > 0$ for $i \le p \le m$, 0 otherwise) and net vapor $U \in \mathbb{R}$ entering its junction. Its *feed function* $\Phi_c(t) := U - \sum_{p=i}^m \alpha_p f_p/(\alpha_p - t)$ decreases from $+\infty$ to $-\infty$ on each $(\alpha_{p+1}, \alpha_p)$, so it has exactly one root $\theta_p^c$ there, $i \le p \le m-1$ (the *feed roots* of $c$), simple and analytic in $(f, U)$. With $U_{[i,m]} := V_{\mathrm{T}[i,m+1]} - V_{\mathrm{B}[i-1,m]}$ (absent terms 0) and $U_{[1,N]} := 1 - q$,

$$\Phi_{[i,m]} = \Psi_{\mathrm{T}[i,m+1]} - \Psi_{\mathrm{B}[i-1,m]}, \qquad \Phi_c = \Psi_{\mathrm{T}_c} - \Psi_{\mathrm{B}_c}, \tag{31}$$

where $\Psi_\sigma = \Psi_\sigma(\cdot\,; w_\sigma)$ (for $[1, N]$ the feed function is $\varphi_F$).

**Lemma 32** (preferred split relative to the actual feed)**.** Put $\omega_p := \prod_{r=i}^{m-1}(\alpha_p - \theta_r^c)/\prod_{p'\in[i,m-1],\,p'\neq p}(\alpha_p - \alpha_{p'})$, $V_{\mathrm{T}}^0 := \alpha_i f_i/\omega_i$, $d_p^0 := V_{\mathrm{T}}^0\omega_p/\alpha_p$ ($i \le p \le m-1$), $d_p^0 := 0$ otherwise, $b^0 := f - d^0$ and $V_{\mathrm{B}}^0 := V_{\mathrm{T}}^0 - U$. Then $d_i^0 = f_i$, $d_p^0 > 0$ ($i \le p \le m-1$), $b_i^0 = 0$, $b_p^0 > 0$ ($i+1 \le p \le m$), $V_{\mathrm{T}}^0, V_{\mathrm{B}}^0, L_{\mathrm{T}}^0, L_{\mathrm{B}}^0 > 0$, and

$$\Psi_{\mathrm{T}}^0(t) = V_{\mathrm{T}}^0 \frac{\prod_r (t - \theta_r^c)}{\prod_{p=i}^{m-1}(t - \alpha_p)}, \qquad \Psi_{\mathrm{B}}^0(t) = V_{\mathrm{B}}^0 \frac{\prod_r (t - \theta_r^c)}{\prod_{p=i+1}^{m}(t - \alpha_p)}:$$

every feed root is a common (active) root of both sections.

This is Halvorsen and Skogestad's preferred split [4] [5], computed for the actual feed of the column. At the preferred split the junction composition is the pinch of both sections: the feed stage is doubly pinched. To move away from it without losing the common middle roots, which are needed by the children, we use a deformation that keeps them exactly.

**Lemma 33** (slack direction). If $m-i=2$ put $K\equiv 1$ and $h:=0$. If $m-i\geq 3$ put $\theta_c^\star:=\prod_{p=i+1}^{m-1}\alpha_p/\prod_{r=i+1}^{m-2}\theta_r^c$, $K(t):=(t-\theta_c^\star)\prod_{r=i+1}^{m-2}(t-\theta_r^c)/\prod_{p=i+1}^{m-1}(t-\alpha_p)$, and

$$h_p:=\frac{(\alpha_p-\theta_c^\star)\prod_{r=i+1}^{m-2}(\alpha_p-\theta_r^c)}{\alpha_p\prod_{p'\in[i+1,m-1],\,p'\neq p}(\alpha_p-\alpha_{p'})}\qquad (i+1\leq p\leq m-1),\qquad h_p:=0 \text{ otherwise.}$$

Then $K(t)=1-\sum_p \alpha_p h_p/(\alpha_p-t)$, $\alpha_{m-1}<\theta_c^\star<\alpha_{i+1}$, $\sum_p h_p=0$, $K(\theta_r^c)=0$ for $i+1\leq r\leq m-2$, and $K(\theta_i^c),K(\theta_{m-1}^c),K(\alpha_i),K(\alpha_m)>0$.

**Definition 34** (deformed column). For $\eta\geq 0$ put $d:=d^0+\eta h$, $b:=f-d$, $V_{\mathrm{T}}:=V_{\mathrm{T}}^0+\eta$, $V_{\mathrm{B}}:=V_{\mathrm{B}}^0+\eta$, $L_{\mathrm{T}}:=V_{\mathrm{T}}-\mathbf{1}^\top d$, $L_{\mathrm{B}}:=V_{\mathrm{B}}+\mathbf{1}^\top b$. Then $\Psi_{\mathrm{T}_c}=\Psi_{\mathrm{T}}^0+\eta K$ and $\Psi_{\mathrm{B}_c}=\Psi_{\mathrm{B}}^0+\eta K$. The column is *valid* if (V) $d_p>0$ for $i\leq p\leq m-1$ and $b_p>0$ for $i+1\leq p\leq m$; this holds for all $\eta\geq 0$ if $m-i=2$, and for $\eta<\min(\{d_p^0/|h_p|:h_p<0\}\cup\{b_p^0/h_p:h_p>0\})$ in general.

**Proposition 35** (a valid deformed column). Let $\eta>0$ satisfy (V).

(a) $V_{\mathrm{T}},L_{\mathrm{T}},V_{\mathrm{B}},L_{\mathrm{B}}>0$.

(b) $\Psi_{\mathrm{T}_c}$ has exactly the simple roots $r_c,\theta_{i+1}^c,\dots,\theta_{m-2}^c,u_c$, and $\Psi_{\mathrm{B}_c}$ has exactly the simple roots $v_c,\theta_{i+1}^c,\dots,\theta_{m-2}^c,s_c$, with $\alpha_i>r_c>\theta_i^c>v_c>\alpha_{i+1}$ and $\alpha_{m-1}>u_c>\theta_{m-1}^c>s_c>\alpha_m$; the middle roots are common; $\Psi_{\mathrm{T}_c}(\alpha_m)>0$ and $\Psi_{\mathrm{B}_c}(\alpha_i)>0$.

(c) $\mathrm{T}_c$ has the trace modes $\alpha_k$, $k\notin[i,m-1]$, and $\mathrm{B}_c$ the trace modes $\alpha_k$, $k\notin[i+1,m]$; the trace modes of $\mathrm{T}_c$ are $>r_c$ ($k<i$) or $<u_c$ ($k\geq m$), those of $\mathrm{B}_c$ are $>v_c$ ($k\leq i$) or $<s_c$ ($k>m$); all modes are positive and distinct.

(d) The pinch compositions $\hat\xi_{(m)}^{\mathrm{T}}:=\hat\xi_{\alpha_m}^{\mathrm{T}_c}$, $\hat\xi_u^{\mathrm{T}}:=\hat\xi_{u_c}^{\mathrm{T}_c}$, $\hat\xi_v^{\mathrm{B}}:=\hat\xi_{v_c}^{\mathrm{B}_c}$, $\hat\xi_{(i)}^{\mathrm{B}}:=\hat\xi_{\alpha_i}^{\mathrm{B}_c}$ satisfy $\hat\xi_{(m)}^{\mathrm{T}},\hat\xi_{(i)}^{\mathrm{B}}>0$ on $[i,m]$, $\hat\xi_u^{\mathrm{T}}>0$ on $[i,m-1]$, $\hat\xi_v^{\mathrm{B}}>0$ on $[i+1,m]$, and vanish outside these supports.

(e) (Interpolation.) The vector $X_c$ with

$$X_{c,p}\propto\frac{\prod_\chi(\alpha_p-\chi)}{\alpha_p\prod_{p'\in[i,m],\,p'\neq p}(\alpha_p-\alpha_{p'})}\qquad (p\in[i,m]),\qquad X_{c,p}=0 \text{ otherwise,}$$

$\chi$ ranging over the interlacing points $r_c,\theta_{i+1}^c,\dots,\theta_{m-2}^c,s_c$, is the unique normalized vector supported on $[i,m]$ with $G_{X_c}(\chi)=0$ at these $m-i$ points; $X_c>0$ on $[i,m]$; the matrix with rows $\mathbf{1}^\top$ and $(\alpha_p/(\alpha_p-\chi))_{p\in[i,m]}$ is nonsingular. Moreover

$$X_c=(1-\beta_c^{\mathrm{T}})\,\hat\xi_{(m)}^{\mathrm{T}}+\beta_c^{\mathrm{T}}\,\hat\xi_u^{\mathrm{T}}=\beta_c^{\mathrm{B}}\,\hat\xi_v^{\mathrm{B}}+(1-\beta_c^{\mathrm{B}})\,\hat\xi_{(i)}^{\mathrm{B}},\qquad \beta_c^{\mathrm{T}},\beta_c^{\mathrm{B}}\in(0,1).$$

At $\eta=0$, $\beta_c^{\mathrm{T}}=\beta_c^{\mathrm{B}}=0$: $X_c$ is the pinch of both sections, which is why every non-binary column needs $\eta_c>0$. The interpolation property and the slack direction first appeared in the analysis of an isolated prefractionator with prescribed returns (Section 4.10).

*Binary columns.* Fix a binary column $c = [p, p+1]$ with feed $f = f_p\mathbf{e}_p + f_{p+1}\mathbf{e}_{p+1}$ ($f_p, f_{p+1} > 0$), net vapor $U$, vapors $V_{\mathrm{T}}$ and $V_{\mathrm{B}} = V_{\mathrm{T}} - U$; let $\theta' \in (\alpha_{p+1}, \alpha_p)$ be its feed root and

$$\Psi^0_{\mathrm{T}}(t) := V_{\mathrm{T}} - \frac{\alpha_p f_p}{\alpha_p - t}, \qquad \Psi^0_{\mathrm{B}}(t) := V_{\mathrm{B}} + \frac{\alpha_{p+1} f_{p+1}}{\alpha_{p+1} - t}, \qquad \kappa := \Psi^0_{\mathrm{T}}(\theta') = \Psi^0_{\mathrm{B}}(\theta'), \tag{36}$$

the section functions of the sharp split and the *binary margin*: the vapor of the column above its Underwood minimum, for its actual feed.

**Proposition 37** (sharp binary column)**.** If $\kappa > 0$, then $V_{\mathrm{T}} > f_p$, $V_{\mathrm{B}} > 0$, the roots $\psi^0_{\mathrm{T}} := \alpha_p(V_{\mathrm{T}} - f_p)/V_{\mathrm{T}}$ of $\Psi^0_{\mathrm{T}}$ and $\psi^0_{\mathrm{B}} := \alpha_{p+1}(V_{\mathrm{B}} + f_{p+1})/V_{\mathrm{B}}$ of $\Psi^0_{\mathrm{B}}$ satisfy $\alpha_{p+1} < \psi^0_{\mathrm{B}} < \theta' < \psi^0_{\mathrm{T}} < \alpha_p$, and $\Psi^0_{\mathrm{B}} > \kappa$ on $(\theta', \infty)$, $\Psi^0_{\mathrm{T}} > \kappa$ on $(-\infty, \theta')$ (outside the poles).

If both products of a binary column are sharp, the junction composition is not determined in the infinite-stage limit (Remark 55). We therefore *force* the column: a small prescribed net flow $\eta > 0$ of one key is sent into the adjacent side product. *B-forcing* ($\pi_p = \mathrm{B}$): $d = (f_p - \eta)\mathbf{e}_p$, $b = \eta\mathbf{e}_p + f_{p+1}\mathbf{e}_{p+1}$, $L_{\mathrm{T}} = V_{\mathrm{T}} - f_p + \eta$, $L_{\mathrm{B}} = V_{\mathrm{B}} + f_{p+1} + \eta$; component $p$ is sent down the bottom section into product $p+1$. *T-forcing* ($\pi_p = \mathrm{T}$): $d = f_p\mathbf{e}_p + \eta\mathbf{e}_{p+1}$, $b = (f_{p+1} - \eta)\mathbf{e}_{p+1}$, $L_{\mathrm{T}} = V_{\mathrm{T}} - f_p - \eta$, $L_{\mathrm{B}} = V_{\mathrm{B}} + f_{p+1} - \eta$; component $p+1$ is sent up the top section into product $p$.

**Proposition 38** (B-forced binary column)**.** Let $\kappa > 0$ and $0 < \eta < \hat{\eta}_p$, with $\hat{\eta}_1 := f_1$ and $\hat{\eta}_p := \min(f_p, (\alpha_{p-1} - \alpha_p)\Psi^0_{\mathrm{B}}(\alpha_{p-1})/\alpha_p)$ for $p \geq 2$. Then:

(a) all phase flows are positive;

(b) $\mathrm{T}_c$ has the single major mode $\psi^{\mathrm{T}} = \alpha_p L_{\mathrm{T}}/V_{\mathrm{T}} \in [\psi^0_{\mathrm{T}}, \alpha_p)$ and the trace modes $\alpha_k$, $k \neq p$;

(c) $\mathrm{B}_c$ has exactly the major modes $\psi'' \in (\alpha_{p+1}, \psi^0_{\mathrm{B}})$ and $\psi' \in (\alpha_p, \alpha_{p-1})$ ($\psi' > \alpha_1$ if $p = 1$), and the trace modes $\alpha_k$ ($k < p$), which exceed $\psi'$, and $\alpha_k$ ($k > p+1$), which are below $\psi''$;

(d) (margin) $\Psi_{\mathrm{B}_c}(\psi^{\mathrm{T}}) > \kappa > 0$, equivalently $\psi'' < \psi^{\mathrm{T}}$;

(e) the *forced pinch* $X := \hat{\xi}^{\mathrm{B}_c}_{\psi'} \propto (\eta/(\psi' - \alpha_p))\mathbf{e}_p + (f_{p+1}/(\psi' - \alpha_{p+1}))\mathbf{e}_{p+1}$ is positive on $\{p, p+1\}$ and $X = (1 - \beta^{\mathrm{T}}_c)\hat{\xi}^{\mathrm{T}}_{(p+1)} + \beta^{\mathrm{T}}_c\mathbf{e}_p$ with $\beta^{\mathrm{T}}_c \in (0, 1)$, where $\mathbf{e}_p = \hat{\xi}^{\mathrm{T}_c}_{\psi^{\mathrm{T}}}$ and $\hat{\xi}^{\mathrm{T}}_{(p+1)} := \hat{\xi}^{\mathrm{T}_c}_{\alpha_{p+1}} > 0$ on $\{p, p+1\}$;

(f) as $\eta \to 0$, $X$ tends to the pinch of the unforced bottom section at its trace mode $\alpha_p$.

**Proposition 39** (T-forced binary column)**.** Let $\kappa > 0$ and $0 < \eta < \hat{\eta}_p$, with $\hat{\eta}_{N-1} := \min(f_N, V_{\mathrm{T}} - f_{N-1})$ and $\hat{\eta}_p := \min(f_{p+1}, (\alpha_{p+1} - \alpha_{p+2})\Psi^0_{\mathrm{T}}(\alpha_{p+2})/\alpha_{p+1})$ for $p + 1 < N$. Then all phase flows are positive; $\mathrm{B}_c$ has the single major mode $\psi^{\mathrm{B}} = \alpha_{p+1}L_{\mathrm{B}}/V_{\mathrm{B}} \in (\alpha_{p+1}, \psi^0_{\mathrm{B}}]$; $\mathrm{T}_c$ has exactly the major modes $\psi^{\mathrm{mid}} \in (\psi^0_{\mathrm{T}}, \alpha_p)$ and $\psi^{\mathrm{low}} \in (\alpha_{p+2}, \alpha_{p+1})$ ($\psi^{\mathrm{low}} \in (0, \alpha_N)$ if $p + 1 = N$); (margin) $\Psi_{\mathrm{T}_c}(\psi^{\mathrm{B}}) > \kappa > 0$, equivalently $\psi^{\mathrm{B}} < \psi^{\mathrm{mid}}$; the forced pinch $X := \hat{\xi}^{\mathrm{T}_c}_{\psi^{\mathrm{low}}}$ is positive on $\{p, p+1\}$ and

$X = \beta_c^{\mathrm{B}}\mathbf{e}_{p+1} + (1-\beta_c^{\mathrm{B}})\hat{\xi}_{(p)}^{\mathrm{B}}$ with $\beta_c^{\mathrm{B}} \in (0,1)$, $\mathbf{e}_{p+1} = \hat{\xi}_{\psi^{\mathrm{B}}}^{\mathrm{B}_c}$, $\hat{\xi}_{(p)}^{\mathrm{B}} := \hat{\xi}_{\alpha_p}^{\mathrm{B}_c} > 0$ on $\{p, p+1\}$; as $\eta \to 0$, $X$ tends to the pinch of the unforced top section at its trace mode $\alpha_{p+1}$.

*The specification.* Let $\eta = ((\eta_c)_{c \text{ non-binary}}, (\eta_p)_{1\le p\le N-1})$ (we write $\eta_{\mathrm{nb}}$ and $\eta_{\mathrm{bin}}$ for the two parts) and let $\pi \in \{\mathrm{B},\mathrm{T}\}^{N-1}$ be a *forcing pattern*.

**Definition 40** (the forced specification $\mathcal{F}(\delta,\eta,\pi)$)**.**

(F1) Non-binary columns, by decreasing length: the feed $(f_c, U_c)$ (for $[1,N]$: $z$ and $1-q$); the preferred split relative to $(f_c, U_c)$ (Lemma 32), deformed with $\eta_c$ (Definition 34).

(F2) Binary columns: feeds and $U$'s likewise; vapors from the reboiler up,

$$V_{\mathrm{B}[N-1,N]} := R := H^* + \delta, \qquad V_{\mathrm{T}[p,p+1]} := V_{\mathrm{B}[p,p+1]} + U_{[p,p+1]},$$
$$V_{\mathrm{B}[p-1,p]} := V_{\mathrm{T}[p,p+1]} \quad \text{(zero-duty side draws)};$$

then $d$, $b$, $L_{\mathrm{T}}$, $L_{\mathrm{B}}$ by B- or T-forcing with $\eta_p$.

(F3) Products: $P_1 := \mathbf{1}^\top d_{[1,2]}$, $P_p := \mathbf{1}^\top b_{[p-1,p]} + \mathbf{1}^\top d_{[p,p+1]}$ $(2 \le p \le N-1)$, $P_N := \mathbf{1}^\top b_{[N-1,N]}$.

The *operating specification* consists of the stream flows only ($V_\sigma$, $L_\sigma$, $P_p$, and the feed); the net component flows $d$, $b$ belong to the limit configuration below.

**Lemma 41** (balances and flux bookkeeping)**.** If every non-binary column is valid: every feed $f_c$ is positive on the range of $c$ and 0 outside; every junction and side-draw stage satisfies the total and CMO balances with zero duty; the condenser removes $V_{\mathrm{T}[1,2]}$ and the reboiler receives $R = H^* + \delta$, the only heat input; $V_{\mathrm{T}[p,p+1]} = R + \sum_{r\ge p} U_{[r,r+1]}$ and $V_{\mathrm{B}[p,p+1]} = R + \sum_{r>p} U_{[r,r+1]}$; $\sum_p P_p = 1$, and the net flows satisfy every component balance: $d_c + b_c = f_c$, $P_p\bar{x}_p = \bar{n}_p := b_{[p-1,p]} + d_{[p,p+1]}$, $\sum_p \bar{n}_p = z$. Moreover, with $\Psi_\sigma$ computed from these net flows and $\varphi_{P_p}(t) := -\sum_k \alpha_k \bar{n}_{p,k}/(\alpha_k - t)$ (the Underwood flux of the liquid product $p$), at every node $u$ and every $t \notin \{\alpha_i\}$

$$\sum_{\sigma:\,\mathrm{bot}(\sigma)=u} \Psi_\sigma(t) - \sum_{\sigma:\,\mathrm{top}(\sigma)=u} \Psi_\sigma(t) + \varphi_{P_p}(t)\,[u = J[p,p]] = Q_u + \varphi_F(t)\,[u = J[1,N]], \tag{42}$$

$Q_u$ being the duty (0 at junctions and side draws, $-V_{\mathrm{T}[1,2]}$ at the condenser, $R$ at the reboiler).

**Lemma 43** (the middle roots are global)**.** If every non-binary column is valid, then $\theta_r^c = \theta_r$ for every non-binary $c = [i,m]$ and $i+1 \le r \le m-2$; at $\eta_{\mathrm{nb}} = 0$ every feed root of every column, binary columns included, is global.

The second statement is the carry-over of the common roots [30] [4]. The binary margins are then computed by a cut identity of the same kind as Theorem 10:

| type | section | modes (decreasing) |
|---|---|---|
| 1 | $\mathrm{T}_c$, non-binary | $\alpha_k\,(k<i) > r_c > \theta_{i+1} > \cdots > \theta_{m-2} > u_c^* > \alpha_m > \alpha_k\,(k>m)$ |
| 2 | $\mathrm{B}_c$, non-binary | $\alpha_k\,(k<i) > \alpha_i > v_c^* > \theta_{i+1} > \cdots > \theta_{m-2} > s_c > \alpha_k\,(k>m)$ |
| 3 | $\mathrm{T}_c$, binary, B-forced | $\alpha_k\,(k<p) > \psi^{\mathrm{T}*} > \alpha_k\,(k\ge p+1)$ |
| 4 | $\mathrm{B}_c$, binary, B-forced | $\alpha_k\,(k<p) > \psi'^{*} > \psi'' > \alpha_k\,(k>p+1)$ |
| 5 | $\mathrm{T}_c$, binary, T-forced | $\alpha_k\,(k<p) > \psi^{\mathrm{mid}} > \psi^{\mathrm{low}*} > \alpha_k\,(k>p+1)$ |
| 6 | $\mathrm{B}_c$, binary, T-forced | $\alpha_k\,(k\le p) > \psi^{\mathrm{B}*} > \alpha_k\,(k>p+1)$ |

**Table 48 /** Modes, pinch and closed ends of the six section types of the limit configuration (modes in decreasing order; the pinch is marked $*$; modes above the pinch are closed at the bottom end, modes below it at the top end).

**Proposition 44** (binary margins)**.** Let $\kappa_p(\eta_{\mathrm{nb}};\delta)$ be the margin (36) of the binary column $[p,p+1]$ in $\mathcal{F}(\delta,\eta,\pi)$.

(a) The non-binary part of $\mathcal{F}$ and the binary feeds and $U$'s do not depend on $(\delta,\eta_{\mathrm{bin}},\pi)$.

(b) $\kappa_p(\eta_{\mathrm{nb}};\delta)=\kappa_p(\eta_{\mathrm{nb}};0)+\delta$.

(c) $\kappa_p(0;0)=g_p=H^*-H_p$.

(d) There are $\eta_1>0$ and $C_1\ge 0$, depending only on $(\alpha,z,q)$, such that for $\eta_{\mathrm{nb}}\in[0,\eta_1]^{(N-1)(N-2)/2}$ every non-binary column is valid and $|\kappa_p(\eta_{\mathrm{nb}};0)-g_p|\le C_1\max_c\eta_c$.

**Definition 45** (admissible parameters)**.** $(\delta,\eta,\pi)$ with $\delta>0$ and all $\eta$'s positive is *admissible* if (i) every non-binary column is valid along (F1); (ii) $\kappa_p>0$ for every binary column; (iii) $0<\eta_p<\hat\eta_p$ for every binary column (the bounds of Propositions 38–39, computed from $\mathcal{F}$).

**Theorem 46** (the admissible set is nonempty for every $\delta$)**.** Let $\delta>0$ and $\bar\eta(\delta):=\min(\eta_1,\delta/(2C_1))$ ($\bar\eta=\eta_1$ if $C_1=0$). If $0<\eta_c\le\bar\eta(\delta)$ for every non-binary column, then (i) holds and $\kappa_p\ge g_p+\delta/2>0$ for every binary column; any $\eta_p\in(0,\hat\eta_p)$ then gives an admissible parameter. The admissible set is open in $\eta$, and every quantity below is real-analytic on it.

## 4.6 The limit configuration

**Definition 47** (limit configuration $\mathcal{L}(\delta,\eta,\pi)$)**.** For admissible parameters: the net flows $d_c$, $b_c$ of $\mathcal{F}$, every component outside a column's range having zero net flow in its sections; the junction compositions $X_c$ of Proposition 35(e) (non-binary), $X_c:=\hat\xi^{\mathrm{B}_c}_{\psi'}$ (B-forced) or $X_c:=\hat\xi^{\mathrm{T}_c}_{\psi^{\mathrm{low}}}$ (T-forced); the product compositions $\bar x_p:=\bar n_p/P_p$. Then $\operatorname{supp}X_c=[i,m]$ and $\operatorname{supp}\bar x_p=\mathrm{Maj}(p):=\{p\}\cup\{p-1:\pi_{p-1}=\mathrm{B}\}\cup\{p+1:\pi_p=\mathrm{T}\}$, with positive entries on the supports. The point $v_\infty:=(d_c,X_c)_c\in\mathcal{V}$.

Every top section pinches at its smallest major mode and every bottom section at its largest; every trace mode lies above all major modes or below all of them.

**Theorem 49** (properties of the limit configuration)**.** For admissible parameters:

(H1) every $V_\sigma$, $L_\sigma$, $\mathbf{1}^\top d_c$, $\mathbf{1}^\top b_c$, $P_p$ is positive and every feed is positive on its column's range;

(H2) top sections have $w \geq 0$ and bottom sections $w \leq 0$, with the supports of Definition 47; the forced flows are nonzero; the middle roots are global (Lemma 43);

(H3) the conclusions of Proposition 35 hold for every non-binary column;

(H4) the conclusions of Propositions 38–39 hold for every binary column, with $\kappa_p > 0$;

(H5) Table 48 holds: in every section all $N$ modes are positive and distinct, the pinch is a major mode, and $\rho_\sigma :=$ max over consecutive modes of (smaller)/(larger) $< 1$; *limit relations*: in every section, $\ell_\psi \cdot x_{\mathrm{bot}} = 0$ for every mode $\psi$ above the pinch and $\ell_\psi \cdot x_{\mathrm{top}} = 0$ for every mode below it;

(H6) (half-profiles) in every section the vectors $M^k x_{\mathrm{bot}}$ and $M^{-k} x_{\mathrm{top}}$ ($k \geq 0$) are $\geq 0$ and $\neq 0$, and the vectors

$$v_k^{\mathrm{bot}} := \frac{M^k x_{\mathrm{bot}}}{(V\psi_*)^k [x_{\mathrm{bot}}]_{\psi_*}} = \hat{\xi}_{\psi_*} + \sum_{\psi<\psi_*} \frac{[x_{\mathrm{bot}}]_\psi}{[x_{\mathrm{bot}}]_{\psi_*}} \left(\frac{\psi}{\psi_*}\right)^k \hat{\xi}_\psi,$$
$$v_k^{\mathrm{top}} := \frac{(V\psi_*)^k M^{-k} x_{\mathrm{top}}}{[x_{\mathrm{top}}]_{\psi_*}} = \hat{\xi}_{\psi_*} + \sum_{\psi>\psi_*} \frac{[x_{\mathrm{top}}]_\psi}{[x_{\mathrm{top}}]_{\psi_*}} \left(\frac{\psi_*}{\psi}\right)^k \hat{\xi}_\psi$$

are $\geq 0$ with $\inf_k \mathbf{1}^\top v_k > 0$;

(H7) (traces) for every trace mode $\alpha_k$ of a section $\sigma$, closed at the node $J$: $\Psi_\sigma^{(k)}(\alpha_k) > 0$; $k$ is lighter (if $J$ is the bottom end) or heavier (top end) than every component of $\operatorname{supp} x_J$, so that $G_{x_J}^{(k)}(\alpha_k) < 0$, respectively $> 0$; and $x_{J,k} = 0$. Every junction $J_c$ at which $k$ is a trace is the closed end, for $k$, of $\mathrm{T}_c$ ($k$ lighter than the range of $c$) or of $\mathrm{B}_c$ ($k$ heavier). Every component with nonzero net flow in $\sigma$ belongs to the supports of both end compositions of $\sigma$;

(H8) $R = H^* + \delta$ is the only heat input; the impurity of product $p$ is $1 - \bar{x}_{p,p} = (\eta_{p-1}[\pi_{p-1} = \mathrm{B}] + \eta_p[\pi_p = \mathrm{T}])/P_p$; the pivots $[x_{\mathrm{bot}}]_{\psi_*}$, $[x_{\mathrm{top}}]_{\psi_*}$ are positive in every section (at a junction they are $\beta_c^{\mathrm{T}}$, $\beta_c^{\mathrm{B}}$, or 1 for the forced section).

(H5) says that in the limit every section is pinched at exactly one end (singly pinched), and (H6) that the two half-profiles of every section are nonnegative. (H7) is the sign information that makes the trace blocks below M-matrices. Proofs: Appendix C.

## 4.7 The limit system and its nondegeneracy

For section lengths $(n_\sigma)$ and the pinches of Table 48, define on a neighbourhood of $v_\infty$

$$\begin{aligned} G_n(v) &:= \left(E_\sigma^{(n_\sigma)}(x_{\mathrm{bot}}(\sigma)(v), x_{\mathrm{top}}(\sigma)(v), w_\sigma(v))\right)_\sigma \in \mathbb{R}^{N(N-1)^2}, \\ G_\infty(v) &:= \left(E_\sigma^{(\infty)}(x_{\mathrm{bot}}(\sigma)(v), x_{\mathrm{top}}(\sigma)(v), w_\sigma(v))\right)_\sigma, \end{aligned} \tag{50}$$

with $N-1$ rows per section and $N(N-1)$ sections: a square system on the $N(N-1)^2$-dimensional space $\mathcal{V}$. The left eigenvectors are Underwood's $\ell_\psi$ for major modes and the regularized $\tilde{\ell}$ of Lemma 26 for trace modes. By Lemma 30(a) and Proposition 22, a zero of $G_n$ near $v_\infty$ whose reconstructed profiles have $S > 0$ is a steady state, and every steady state near $v_\infty$ is a zero of $G_n$.

**Lemma 51** (domain). There are a compact neighbourhood $\mathcal{W}$ of $v_\infty$ and $\bar{\rho} < 1$, $\varpi_0 > 0$ such that on $\mathcal{W}$, for every section, the modes of $M_\sigma(w_\sigma(v))$ are simple, positive, ordered as at $v_\infty$, analytic, with consecutive ratios at most $\bar{\rho}$, and both pivots have modulus at least $\varpi_0$. $G_\infty$ and all $G_n$ are analytic on $\mathcal{W}$, $G_\infty(v_\infty) = 0$, and with $n := \min_\sigma n_\sigma$, $|G_n - G_\infty| \le C\bar{\rho}^{n+1}$ and $\|D(G_n - G_\infty)\| \le C(n+2)\bar{\rho}^{n+1}$ on $\mathcal{W}$.

The rows of $G_\infty$ are of two kinds. A *major row* is the limit relation of a non-pinch major mode; a *trace row of component $k$* is that of the trace mode $\alpha_k$ of a section in which $k$ has zero net flow. The coordinates of $\mathcal{V}$ split accordingly into *trace coordinates* of each component (net flows and junction contents that vanish at $v_\infty$) and *major coordinates* (Appendix D, Table 66).

**Theorem 52** (nondegeneracy of the limit system). For every admissible $(\delta, \eta, \pi)$, $DG_\infty(v_\infty)$ is invertible. In the coordinates of Table 66, with rows ordered (major; trace of component $1, \dots, N$) and coordinates likewise,

$$DG_\infty(v_\infty) = \begin{pmatrix} \mathbb{M} & * \\ 0 & \mathbb{T} \end{pmatrix}, \qquad \mathbb{T} = \operatorname{diag}(\mathbb{T}_1, \dots, \mathbb{T}_N),$$

$$\det DG_\infty(v_\infty) = \det \mathbb{M} \prod_k \det \mathbb{T}_k \neq 0,$$

where $\mathbb{M}$ is block lower triangular over the columns ordered by decreasing length, with nonsingular diagonal blocks, and every $\mathbb{T}_k$ is nonsingular because a vector in its kernel solves $A_k\zeta = 0$ for a triangular nonsingular M-matrix $A_k$.

*Idea of the proof* (Appendix D). (1) By (28), a trace row of component $k$ closed at a node $J$ has the form $x_{J,k} + w_{\sigma,k} \cdot$ (analytic), and both $x_{J,k}$ and $w_{\sigma,k}$ vanish at $v_\infty$ and are linear in the trace coordinates of $k$ alone. So at $v_\infty$ a trace row of $k$ depends only on the trace coordinates of $k$: this gives the zero block and the block-diagonal $\mathbb{T}$. (2) Its derivative is $(\delta x_{J,k} - \gamma_\sigma^{(k)}\, \delta\nu^k_{J\to\sigma})/\varpi$, where $\nu^k_{J\to\sigma}$ is the net flow of $k$ from $J$ into $\sigma$ and $\gamma_\sigma^{(k)} = |G^{(k)}_{x_J}(\alpha_k)|/\Psi^{(k)}_\sigma(\alpha_k) > 0$ by (H7). Combined with the component balance of $k$ at every node where $k$ is absent, a kernel vector of $\mathbb{T}_k$ gives contents $\zeta_J$ with $A_k\zeta = 0$, where $A_k$ has diagonal entries $P_J + \sum 1/\gamma$ and off-diagonal entries $-1/\gamma$ along the moves from the closed end to the open end of a section. Every such move changes $h = i + m$ by one unit in a fixed direction for each component, so $A_k$ is triangular with positive diagonal. (3) The major rows of a column involve only the coordinates of the column and of its ancestors; for a non-binary column, the two rows at a common middle root give a Cauchy-type system $\sum_p \alpha_p\, \delta d_p/(\alpha_p - \theta_j) = 0$ at the $m - i - 2$ middle roots, which with $\sum_p \delta d_p = 0$ forces $\delta d = 0$, and then Underwood interpolation (Proposition 35(e)) forces $\delta X_c = 0$; for a binary column the single major row is a nonzero multiple of $\delta X_{c,p+1}$.

### 4.8 From the limit system to finite-stage steady states

**Theorem 53** (network perturbation). For admissible parameters there are $r_1 > 0$, $n_0$ and $C$ such that, for all section lengths with $n := \min_\sigma n_\sigma \geq n_0$, $G_n$ has exactly one zero $v_n$ in the closed ball $\bar{B}(v_\infty, r_1)$, and

$$|v_n - v_\infty| \leq C\rho_\infty^{n+1}, \qquad \rho_\infty := \max_\sigma \max\Big(\frac{\lambda^\infty_{\sigma,r_*+1}}{\lambda^\infty_{\sigma,r_*}}, \frac{\lambda^\infty_{\sigma,r_*}}{\lambda^\infty_{\sigma,r_*-1}}\Big) < 1.$$

**Theorem 54** (finite-stage steady states near the limit configuration). For every admissible $(\delta, \eta, \pi)$ there are $n_1$ and $C$ such that, for all section lengths with $\min_\sigma n_\sigma = n \geq n_1$, the arrangement operated with $\mathcal{F}(\delta, \eta, \pi)$ has a steady state with

(i) every mole fraction of every unit and stream strictly positive;

(ii) every node composition and column split within $C\rho_\infty^{n+1}$ of its value in $\mathcal{L}$; in particular product $p$ has purity at least $1 - (\eta_{p-1}[\pi_{p-1} = \mathrm{B}] + \eta_p[\pi_p = \mathrm{T}])/P_p - C\rho_\infty^{n+1}$;

(iii) stage compositions within $C(\rho_\infty^{n+1} + \bar{\rho}^{\min(k, n_\sigma+1-k)})$ of the limit half-profiles;

(iv) heat supply $R = H^* + \delta$;

(v) local uniqueness: no other steady state has its $(d_c, X_c)$ in $\bar{B}(v_\infty, r_1)$.

*Idea of the proof* (Appendix E). Theorem 53 is a quantitative inverse function theorem applied to $G_n$, which differs from $G_\infty$ by $O(n\bar{\rho}^n)$ in $C^1$ (Lemma 51) while $DG_\infty(v_\infty)$ is invertible (Theorem 52). At $v_n$ every stage profile is reconstructed exactly: in the eigenbasis it is the sum of a bottom half-profile decaying upward and a top half-profile decaying downward, and it is exponentially close, stage by stage, to one of the two nonnegative limit half-profiles of (H6). This gives $S > 0$ at every stage, and Proposition 22(b) turns $v_n$ into a steady state with all mole fractions positive.

### 4.9 Proof of Theorem 16

Let $N \geq 3$ ($N = 2$ is Proposition 18), $\delta > 0$, $\varepsilon > 0$, and fix a pattern $\pi$ (for example all B). Choose $\eta_c := \bar{\eta}(\delta)$ for every non-binary column (Theorem 46). The bounds $\hat{\eta}_p > 0$ of Propositions 38–39 involve only the binary feeds and vapors, which depend on $(\delta, \eta_{\mathrm{nb}})$ but not on $\eta_{\mathrm{bin}}$ (Proposition 44(a), Lemma 41); the product flows $P_p$ are affine in $\eta_{\mathrm{bin}}$, positive on the admissible set, and tend to $\bar{n}_{p,p} > 0$ as $\eta_{\mathrm{bin}} \to 0$. So we can choose $\eta_p \in (0, \hat{\eta}_p)$ so small that $(\eta_{p-1}[\pi_{p-1} = \mathrm{B}] + \eta_p[\pi_p = \mathrm{T}])/P_p \leq \varepsilon/2$ for every $p$. The parameters are admissible, so Theorem 54 applies: take $n_0 \geq n_1(\delta, \eta, \pi)$ with $C(\delta, \eta, \pi)\rho_\infty^{n_0+1} \leq \varepsilon/2$. For all section lengths $n_\sigma \geq n_0$, the arrangement operated with $\mathcal{F}(\delta, \eta, \pi)$, a specification independent of the $n_\sigma$, has a steady state with all mole fractions positive, every product of purity at least $1 - \varepsilon$, and $R = H^* + \delta$. Hence $R^*(\mathcal{S}_{\mathrm{Pet}}; \varepsilon) \leq H^* + \delta$ for all $\delta, \varepsilon > 0$, that is, $R^*(\mathcal{S}_{\mathrm{Pet}}) \leq H^*$. □

## 4.10 Remarks

**Remark 55** (why forcing is needed)**.** If a binary column $[p, p+1]$ is not forced ($\eta_p = 0$), both of its sections pinch at modes whose eigenvectors do not fix the ratio $X_{c,p} : X_{c,p+1}$, every row closed at $J_c$ is a trace row, and $v_\infty + t(\mathbf{e}_{p+1} - \mathbf{e}_p)$ (in the coordinates $X_c$) is a curve of zeros of $G_\infty$: $DG_\infty(v_\infty)$ is singular. Physically, the position of the composition front inside the column is then fixed only by exponentially small terms, and the steady-state Jacobian of the finite-stage equations has a smallest singular value decaying geometrically with $n$ (Section 5). Forcing turns $p$ (or $p+1$) into a major component of one section, whose pinch fixes $X_c$, while its second major mode supplies the missing major row. ⌟

**Remark 56** (why $\delta > 0$ and $\eta_c > 0$)**.** At $\delta = 0$ the binary column at the highest peak has margin $\kappa_p = 0$, and its junction composition becomes the pinch of both of its sections; at $\eta_c = 0$ the junction composition of a non-binary column is the pinch of both of its sections ($\beta_c^{\mathrm{T}} = \beta_c^{\mathrm{B}} = 0$). These are the doubly pinched "knots" of the Underwood picture; the design stays at a positive distance from them for fixed $(\delta, \eta)$. ⌟

**Remark 57** (order of limits and constants)**.** The constants $n_0, C, r_1$ depend on $(\delta, \eta)$ and are not uniform as $\eta \to 0$: the non-binary pivots $\beta_c^{\mathrm{T}}, \beta_c^{\mathrm{B}}$ are $O(\eta_c)$, and in a forced section the ratio of the two pivots is $O(\eta_p)$, so the exponentially small terms carry a factor $1/\eta$ and $n_0 \gtrsim \log(1/\eta)/\log(1/\rho_\infty)$, in agreement with the computations of Section 5. No uniformity is needed, since $R^*(\mathcal{S}; \varepsilon)$ is an infimum over all numbers of stages. ⌟

**Remark 58** (section lengths, uniqueness, and non-physical zeros)**.** Theorem 16 holds for any family of section lengths with $\min_\sigma n_\sigma \geq n_0$: the limit system does not involve the lengths because every section of $\mathcal{L}$ is singly pinched. The steady state is unique near $\mathcal{L}$; nothing is claimed far from it. The exact equations $G_n = 0$ do have other zeros, some with negative stage mole fractions, so the condition $S > 0$ in Proposition 22(b) cannot be dropped; near $\mathcal{L}$ it is delivered by the nonnegativity of the limit half-profiles (H6) and the positivity of the pivots. These are not consequences of the algebraic structure: in a three-component case deliberately chosen outside the admissible region (a negative binary margin), $G_\infty(v_\infty) = 0$ and $DG_\infty(v_\infty)$ is invertible with the block structure of Theorem 52, but one half-profile has a negative entry and the zeros of $G_n$ converging to $v_\infty$ have negative stage mole fractions (Section 5). ⌟

**Remark 59** (signs of trace flows)**.** The trace net flows and trace contents of the finite-stage steady state are $O(\rho_\infty^{n+1})$; they may have either sign (for $N \geq 4$ some trace flows go against the "natural" direction and change sign with $n$), and no step of the proof uses their signs: positivity of all contents comes from Proposition 22(b). ⌟

**Remark 60** (relation with the Underwood construction and with an isolated prefractionator)**.** At $\delta = \eta = 0$ the specification is Halvorsen and Skogestad's operating point: every non-binary column at its preferred split, the binary columns with excess vapor $g_p$, and $R = H^*$ [5]; Proposition 44(c) re-proves that this point closes the side-draw balances. The single-column case of Proposition 35, with the returns prescribed instead of computed by the rest of the network, is a theorem on an isolated prefractionator: it admits exact positive finite-stage profiles whose unwanted extreme components at the two returns are $O(\eta^{-1}\rho^n)$. The forced design closes the network around such columns. ⌟

| $n$ | impurity of products 1, 2, 3 | $\lvert v_n - v_\infty\rvert$ | $\sigma_{\min}(DG_n)$ | min. mole fraction | digits |
|---|---|---|---|---|---|
| 16 | $2.12\cdot10^{-4}$, $2.11\cdot10^{-2}$, $3.25\cdot10^{-3}$ | $4.67\cdot10^{-2}$ | 0.0049615 | $1.8\cdot10^{-11}$ | 91 |
| 24 | $6.58\cdot10^{-6}$, $1.77\cdot10^{-2}$, $2.54\cdot10^{-5}$ | $8.22\cdot10^{-3}$ | 0.0069240 | $2.3\cdot10^{-16}$ | 96 |
| 32 | $2.15\cdot10^{-7}$, $1.77\cdot10^{-2}$, $4.68\cdot10^{-7}$ | $9.87\cdot10^{-4}$ | 0.0072625 | $8.0\cdot10^{-22}$ | 101 |
| 48 | $2.31\cdot10^{-10}$, $1.77\cdot10^{-2}$, $2.01\cdot10^{-10}$ | $1.48\cdot10^{-5}$ | 0.0072911 | $1.2\cdot10^{-32}$ | 112 |
| 64 | $2.47\cdot10^{-13}$, $1.77\cdot10^{-2}$, $8.80\cdot10^{-14}$ | $2.25\cdot10^{-7}$ | 0.0072915 | $1.7\cdot10^{-43}$ | 123 |
| 96 | $2.83\cdot10^{-19}$, $1.77\cdot10^{-2}$, $1.68\cdot10^{-20}$ | $5.48\cdot10^{-11}$ | 0.0072915 | $3.6\cdot10^{-65}$ | 144 |
| 128 | $3.25\cdot10^{-25}$, $1.77\cdot10^{-2}$, $3.22\cdot10^{-27}$ | $1.44\cdot10^{-14}$ | 0.0072915 | $7.9\cdot10^{-87}$ | 166 |
| 192 | $4.28\cdot10^{-37}$, $1.77\cdot10^{-2}$, $1.18\cdot10^{-40}$ | $1.18\cdot10^{-21}$ | 0.0072915 | $3.6\cdot10^{-130}$ | 209 |
| 256 | $5.63\cdot10^{-49}$, $1.77\cdot10^{-2}$, $4.30\cdot10^{-54}$ | $1.10\cdot10^{-28}$ | 0.0072915 | $1.7\cdot10^{-173}$ | 253 |

**Table 61 /** $N = 3$, $\alpha = (4, 2, 1)$, $z_p = 1/3$, $q = 1$, $\delta = 0.05$, prefractionator slack $\eta_{[1,3]} = 0.025$, pattern (B, T), $\eta_1 = \eta_2 = 0.003$: $R = H^* + \delta = 1.415722514$ ($H^* = 1.365722514$), $n$ stages in every section. The limit impurities are 0, $1.768\cdot10^{-2}$ (designed, $(\eta_1 + \eta_2)/P_2$) and 0; $\sigma_{\min}(DG_\infty(v_\infty)) = 0.0072915$. The observed rate of $\lvert v_n - v_\infty\rvert$ is $0.7765^n$, the predicted $\rho_\infty = 0.7773$.

| $n$ | impurity of products 1, 2, 3, 4 | $\lvert v_n - v_\infty\rvert$ | $\sigma_{\min}(DG_n)$ | min. mole fraction |
|---|---|---|---|---|
| 16 | $5.10\cdot10^{-5}$, $1.55\cdot10^{-2}$, $2.31\cdot10^{-2}$, $1.94\cdot10^{-2}$ | $1.50\cdot10^{-2}$ | 0.00447990 | $1.3\cdot10^{-23}$ |
| 32 | $2.04\cdot10^{-8}$, $1.22\cdot10^{-2}$, $1.26\cdot10^{-2}$, $1.23\cdot10^{-2}$ | $1.07\cdot10^{-3}$ | 0.00095545 | $5.9\cdot10^{-44}$ |
| 64 | $3.55\cdot10^{-15}$, $1.20\cdot10^{-2}$, $1.20\cdot10^{-2}$, $1.19\cdot10^{-2}$ | $7.37\cdot10^{-7}$ | 0.00099035 | $7.5\cdot10^{-86}$ |
| 96 | $6.16\cdot10^{-22}$, $1.20\cdot10^{-2}$, $1.20\cdot10^{-2}$, $1.19\cdot10^{-2}$ | $1.00\cdot10^{-9}$ | 0.00099042 | $7.7\cdot10^{-128}$ |
| 128 | $1.07\cdot10^{-28}$, $1.20\cdot10^{-2}$, $1.20\cdot10^{-2}$, $1.19\cdot10^{-2}$ | $1.53\cdot10^{-12}$ | 0.00099042 | $7.8\cdot10^{-170}$ |
| 192 | $3.22\cdot10^{-42}$, $1.20\cdot10^{-2}$, $1.20\cdot10^{-2}$, $1.19\cdot10^{-2}$ | $3.63\cdot10^{-18}$ | 0.00099042 | $8.1\cdot10^{-254}$ |

**Table 62 /** $N = 4$, $\alpha = (8, 4, 2, 1)$, $z_p = 1/4$, $q = 1$, $\delta = 0.05$, $\eta_c = 0.005$ for the three non-binary columns, pattern (B, B, B), $\eta_p = 0.003$: $R = 1.322854785$ ($H^* = 1.272854785$). Limit impurities 0, $1.200\cdot10^{-2}$, $1.200\cdot10^{-2}$, $1.186\cdot10^{-2}$ (designed); $\sigma_{\min}(DG_\infty(v_\infty)) = 0.00099042$. Observed rate $0.8168^n$, predicted $\rho_\infty = 0.8169$.

## 5 Numerical illustration

*Everything in this section is numerical evidence (high-precision floating point), not part of any proof.* The finite-stage steady states below were computed by a solver written separately from the proof, in the unknowns of Section 4.3 with the exact product-form section relations, Newton's method in `mpmath` at 90 to about 1500 significant digits, and two precision audits; every reported steady state was then re-checked stage by stage (every stage, junction, side-draw, condenser and reboiler balance with the actual stream compositions; all mole fractions positive; $S > 0$ everywhere). The limit configuration was computed separately from Definition 47 and agrees with a second implementation to $10^{-54}$.

*Two illustrative cases* (Tables 61–62, Figure 64). The distance to the limit configuration decays like $\rho_\infty^n$ with the predicted rate, the products 1 and $N$ purify geometrically, the side products tend to their designed impurities $O(\eta)$, the smallest singular value of the finite-stage Jacobian tends to that of the limit Jacobian, and $R - H^* = \delta$ exactly. Table 63 follows the order of limits of the proof: as $(\delta, \eta) \to 0$, the side-product impurity follows its designed value and $R \to H^*$, with stage numbers growing like $\log(1/\eta)$, since $\sigma_{\min}(DG_\infty)$ is proportional to $\eta$ in the product-form normalization.

*A broader campaign.* The same solver was run on 62 cases: $N = 3$ with all four forcing

| $\delta$ | $\eta_{[1,3]}$ | $\eta_1 = \eta_2$ | $n$ | impurity of products 1, 2, 3 | $\sigma_{\min}(DG_\infty)$ |
|---|---|---|---|---|---|
| 0.05 | 0.025 | 0.003 | 128 | $3.25 \cdot 10^{-25}$, $1.77 \cdot 10^{-2}$, $3.22 \cdot 10^{-27}$ | 0.007291 |
| 0.02 | 0.01 | 0.001 | 192 | $2.29 \cdot 10^{-36}$, $5.96 \cdot 10^{-3}$, $8.21 \cdot 10^{-40}$ | 0.002692 |
| 0.005 | 0.0025 | 0.0002 | 256 | $1.57 \cdot 10^{-47}$, $1.20 \cdot 10^{-3}$, $3.38 \cdot 10^{-52}$ | 0.0005757 |
| 0.001 | 0.0005 | 0.00003 | 384 | $2.14 \cdot 10^{-70}$, $1.80 \cdot 10^{-4}$, $8.43 \cdot 10^{-78}$ | 0.00009037 |

**Table 63 /** Approach to the Petlyuk minimum (data of Table 61): $(\delta, \eta) \to 0$ jointly, with $R - H^* = \delta$ exactly and a number of stages large enough for the limit regime. The designed impurities $(\eta_1 + \eta_2)/P_2$ of the side product are $1.768 \cdot 10^{-2}$, $5.964 \cdot 10^{-3}$, $1.199 \cdot 10^{-3}$ and $1.800 \cdot 10^{-4}$.

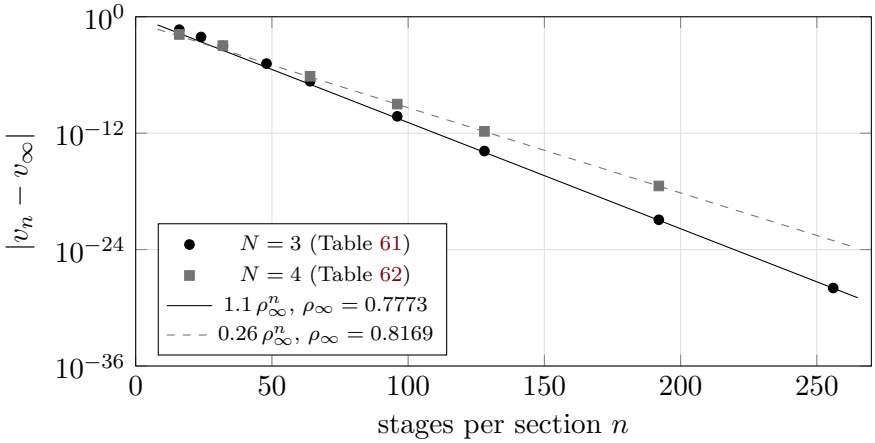



**Figure 64 /** Distance between the finite-stage steady state and the limit configuration (junction compositions and column splits) against the number of stages per section, with the rate $\rho_\infty$ predicted by Theorem 53 (largest ratio of consecutive eigenvalues at the pinches of the limit configuration).

patterns, $q$ from $-10$ to 10, $\alpha$ from $(1.05, 1.02, 1)$ to $(100, 10, 1)$, feed fractions down to $10^{-4}$, two equal highest peaks, $\delta$ down to 0.001 and forcing up to $0.95 f_p$; $N = 4$ with all eight patterns, $q$ from $-5$ to 6, three equal peaks and close volatilities; random instances for $N = 5$ to 8; $n$ up to 768 (4000 for nearly equal volatilities); unequal section lengths with ratios up to 8. In all 61 cases inside the admissible region, $DG_\infty(v_\infty)$ is invertible with the block structure of Theorem 52, every trace block reduces to a nonsingular M-matrix, and the finite-stage steady states are physical and converge to $\mathcal{L}$ at the predicted rate. Without forcing, $\sigma_{\min}(DG_n)$ decays geometrically ($8.2 \cdot 10^{-5}$, $9.8 \cdot 10^{-7}$, $1.4 \cdot 10^{-8}$ at $n = 16, 24, 32$ for $N = 3$), while with forcing it stays near 0.007: the neutral mode of Remark 55. The one failing case was chosen outside the admissible region on purpose (a three-component case with $\delta = 0.002$ and a prefractionator slack $0.05 > \bar{\eta}(\delta)$, so that a binary margin is negative): all algebraic hypotheses and the structure of Theorem 52 hold, but a half-profile has a negative entry and the zeros of $G_n$ converging to $v_\infty$ ($n = 32, 64, 128$) have negative stage mole fractions (Remark 58). The exact equations also have non-physical zeros far from $\mathcal{L}$ at admissible parameters. Separately, the limit Jacobian was checked for 16 random instances with $N = 3$ to 6 (block structure exact, every trace row as predicted by Lemma 69), and the combinatorial case analysis of Theorem 52 (rows, coordinates, closed-to-open moves) was checked exhaustively for all $N \le 12$ and all $2^{N-1}$ forcing patterns.

*Separate re-implementations.* The two verifications of the attainment proof wrote their own codes from the statements. The first built the design by solving its defining conditions, and the limit system from a general eigen-solver instead of Underwood's formulas; on 23 cases ($N = 3$ to 6, all kinds of forcing patterns, forcing from 5% to 99.9% of the bound $\hat{\eta}_p$, nearly equal volatilities) it confirmed every property of Theorem 49, the block structure of Theorem 52 to finite-difference accuracy, and physical finite-stage steady states with every unit balance satisfied to $10^{-85}$. The second built the limit system from the explicit Underwood and trace formulas, measured the constants of Lemma 51 and checked the inequalities of Theorem 53 where they apply; it also observed that below $n_0$ the zero of $G_n$ closest to $\mathcal{L}$ need not be physical (a three-component case with $q = -0.5$ at $n = 16$), while it is physical from $n = 32$ on, as the theorem predicts. A deliberately broken version of the combinatorial check of Theorem 52 (for example, unforced binary columns) was detected in every mutation.

*Earlier evidence.* The lower bound was checked on finite-stage steady states computed by four separate solvers (random networks of up to twelve units with recycles, bypasses, split feeds, duties anywhere, products of either phase and reverse net flows; columns and Petlyuk arrangements with up to 120 stages per section; Murphree trays), and its identities in exact rational arithmetic; no violation was found. At fixed flows with a margin spread over all sections, three separate solvers found the impurities of the three-component Petlyuk arrangement decreasing geometrically with $n$, which motivated the proof; with sharp binary columns their Jacobians showed the neutral mode.

## 6 Discussion

*What is proved.* Within constant relative volatilities, constant molar flows, constant pressure, one feed, no heat exchange between process streams and steady-state operation, every arrangement of equilibrium stages, single-phase splitters and mixers, total condensers and vaporizers needs a total vaporization of at least $\max_j H_j$ in the limit of infinitely many stages and pure products (in fact $R \geq H_j - c_j\varepsilon$ at every finite number of stages and purity $1 - \varepsilon$), and the generalized Petlyuk arrangement attains this value. This is the Halvorsen–Skogestad conjecture, read as in Definition 6, together with their formula for the Petlyuk minimum, for every number of components. Both halves were checked by verifiers who re-derived the proofs and re-implemented the constructions. The lower bound is the stronger statement: it is uniform in the numbers of stages, holds at every steady state of every arrangement (including non-adiabatic sections with external utilities and Murphree trays), and yields necessary conditions for ties. The attainment is the more delicate one: it requires the order of limits $n \to \infty$, then forcing $\to 0$, then $\delta \to 0$, and gives no explicit stage numbers.

*Scope and limitations.* (1) *Constant relative volatilities* enter the lower bound only through Lemma 9: the vapor flux $-(\theta v/S)G_x(\theta)$ has the sign of the liquid flux because $y_i \propto \alpha_i x_i$. For non-ideal mixtures this sign uniformity can fail, and the conjecture is open there (and not stated by its authors). (2) *Constant molar overflow* enters only through the energy balance. Under the constant-heat-transport model of [31] (constant but component-dependent molar latent heats $\lambda_i$, no sensible heat), the latent-heat weighted flows $\tilde{n}_{e,i} = \lambda_i n_{e,i}$ satisfy the same balances, and the weighted compositions satisfy $\tilde{y}_i = \alpha_i \tilde{x}_i / \sum_k \alpha_k \tilde{x}_k$ again, so the lower bound extends verbatim, with the peaks of the correspondingly weighted $V_{\min}$ diagram; we have not examined attainment in

that model. (3) *One feed*: with several feeds the bound of Section 3.2 holds, but it is not known whether it is attained, nor by which arrangement. (4) *Internal heat exchange* is excluded by the conjecture, and Section 3.2 shows why it must be: the external heat can then fall below $H^*$, as in Halvorsen's examples Table 6.1 [6]; the minimum external heat with heat integration (called unknown in [7]) is not determined here. (5) *Steady state*: periodic or batch operation is outside the model; with holdups, time-averaged outflows of a tray need not be sign-uniform. (6) *Equilibrium stages*: devices that release streams of different compositions not in phase equilibrium (decanters, membranes, component-specific efficiencies or mass-transfer rates) are outside $\mathcal{A}$ and outside the conjecture's assumptions.

*Relation to Underwood's model and to optimization.* Underwood's infinite-stage model remains the right tool to *compute* minimum vapor, and our results justify its value for the generalized Petlyuk arrangement. For global optimization over configurations, Remark 15 shows that relaxations of the model can return values below $H^*$; the sign-consistency constraints derived from Lemma 9 remove such points without removing physically admissible excess vapor. The tie conditions of Section 3.2 explain why many configurations reach the Petlyuk value [27] and give necessary conditions for a configuration to do so.

*Open questions.* (i) Explicit stage numbers: the proof gives $n_0 \gtrsim \log(1/\eta)/\log(1/\rho_\infty)$ at fixed margins, but no explicit constants; a quantitative version would turn the attainment into a design rule. (ii) Uniqueness: we prove local uniqueness near the limit configuration; whether the finite-stage Petlyuk network at the flows of $\mathcal{F}(\delta, \eta, \pi)$ has a unique physical steady state is open (multiplicity is known for single columns under other specifications [32]; uniqueness for binary columns with molar flow specifications goes back to Rosenbrock, see [33]). (iii) Which other arrangements attain $H^*$: the necessary conditions are given here, sufficient conditions are not. (iv) Non-ideal mixtures, several feeds, non-sharp products, and arrangements with heat integration. (v) A formal (machine-checked) proof: the lower bound is short enough for this; the attainment proof uses analytic perturbation theory and would be a substantial formalization.

## Data and code availability

All scripts, logs and certificates are in the repository `chemistry-open-problems`, folder `problems/halvorsen-skogestad-petlyuk-minimum-vapor/`; Appendix F lists them with what each checks. No result of this paper depends on a computation.

## Use of artificial intelligence

All mathematics, code and text of this work, including this manuscript and the literature review behind it, were produced by AI agents (Claude Opus 5.5 and GPT-6-Astra), working under human direction in a structured workflow with separate roles (solvers, red team, computation, verifiers). Every result was meant to be checked by an agent that had not seen the solver's reasoning: the Underwood-model results were verified, the lower bound received an early separate check (confirmed, with editorial corrections that are incorporated here), and the attainment proof was verified separately by two further agents, one for the design and the limit system, one for the analysis and the integrated proof (both confirmed it; the corrections were minor and editorial, and are incorporated here). Each verifier re-derived the proofs by hand and re-implemented the construction

in its own code. No human has checked the proofs line by line. Responsibility for any submission of this work rests with the humans who submit it. The contributions are listed by role under the title.

## A Two components: proof of Proposition 18

Take $\alpha_1 = \alpha > 1 = \alpha_2$, so $\theta = \theta_1 \in (1, \alpha)$ and $H_1 = (1-z)/(\theta-1)$. Write $x, y$ for light mole fractions and $e(x) = \alpha x/(1+(\alpha-1)x)$ (increasing, strictly concave, $e(0) = 0$, $e(1) = 1$). The column has a partial reboiler (an (E) unit with duty $s$, liquid product of composition $x_B$), $n_B$ stages, a feed stage, $n_T$ stages and a total condenser (reflux and distillate of composition $x_D$); the flows are $D = z$, $B = 1-z$, $V_B = s$, $V_T = s+1-q$, $L_T = V_T - D$, $L_B = s+1-z$.

*Flows.* $H_1 + 1 - q = \alpha z/(\alpha-\theta) > z$ by (2), so $V_T > 0$ and $L_T > 0$ for $s \geq H_1$; and $H_1 > (1-z)/(\alpha-1)$ since $\theta < \alpha$.

*Pinches.* The bottom map for a pure bottom product, $\beta_0(x) = (s/L_B)e(x)$, has the fixed points 0 and $\hat{x}_B = (\alpha s/L_B - 1)/(\alpha-1)$, which lies in $(0,1)$ because $s(\alpha-1) > 1-z$; and $\beta_0(x) > x$ on $(0, \hat{x}_B)$ because $e(x)/x$ decreases. The downward top map for a pure top product, $\tau_0(x) = e^{-1}((L_T x + D)/V_T)$, has the fixed points 1 and $\hat{x}_T = (V_T/L_T - 1)/(\alpha-1)$; between them the line $(L_T x + D)/V_T$ lies below the concave curve $e$, so $\tau_0(x) < x$ on $(\hat{x}_T, 1)$. Now $\hat{x}_T < \hat{x}_B$ if and only if $f(s) := \alpha s L_T - V_T L_B > 0$. Eliminating $q$ with the feed equation $1-q = \alpha z/(\alpha-\theta) + (1-z)/(1-\theta)$, $f$ is a quadratic in $s$ with leading coefficient $\alpha - 1 > 0$ and roots $H_1$ and $s_- = 1/(\alpha-1) - z\theta/(\alpha-\theta)$, and

$$H_1 - s_- = \frac{z\alpha(\theta-1)^2 + (1-z)(\alpha-\theta)^2}{(\alpha-1)(\alpha-\theta)(\theta-1)} > 0$$

(a two-line computation, checked symbolically). Hence $s > H_1$ gives $\hat{x}_T < \hat{x}_B < 1$.

*Shooting.* Fix $\eta_1 \in (0,1]$ with $\eta_1 \max(1, (1-z)/z) \leq \varepsilon$. For $\eta \in [0, \eta_1]$ put $x_B = \eta$ and $x_D(\eta) = 1 - (1-z)\eta/z \in [1-\varepsilon, 1]$ (light balance). Upward from the reboiler liquid, $x_0 = \eta$, $x_{k+1} = \beta_\eta(x_k) := (s\,e(x_k) + B\eta)/L_B$ (the balance $s y_k - L_B x_{k+1} = -B x_B$); after $n_B + 1$ steps this gives a feed-stage liquid $x_f^{\mathrm{bot}}(\eta)$. Downward from the condenser: the top-stage vapor has composition $x_D$, each stage has the liquid $e^{-1}(y)$, and the vapor from the stage below is $y_{\mathrm{next}} = (L_T x + D x_D)/V_T$ (the balance $V_T y - L_T x = D x_D$); after $n_T + 1$ stages this gives $x_f^{\mathrm{top}}(\eta)$. All iterates stay in $[0,1]$, and both functions are continuous in $\eta$. If $x_f^{\mathrm{bot}}(\eta) = x_f^{\mathrm{top}}(\eta)$, the two profiles form a steady state: every stage balance holds by construction, the feed-stage balance is the difference of the two cross-section balances adjacent to it plus the overall light balance $D x_D + B x_B = z$, and the reboiler and condenser balances hold. At $\eta = 0$, $x_f^{\mathrm{bot}} = 0 < 1 = x_f^{\mathrm{top}}$. Since $\beta_{\eta_1} \geq \beta_0$ and both are increasing, $x_f^{\mathrm{bot}}(\eta_1) \geq \beta_0^{n_B+1}(\eta_1)$, which tends to $\hat{x}_B$ (or is $\geq \hat{x}_B$ if $\eta_1 \geq \hat{x}_B$); since the downward step with $x_D(\eta_1) \leq 1$ is at most $\tau_0$ and $\tau_0$ is increasing, $x_f^{\mathrm{top}}(\eta_1) \leq \max(\hat{x}_T, \tau_0^{n_T}(e^{-1}(x_D(\eta_1))))$, which tends to $\hat{x}_T$. So for $n_T, n_B \geq n_0$, $x_f^{\mathrm{bot}}(\eta_1) > (\hat{x}_T + \hat{x}_B)/2 > x_f^{\mathrm{top}}(\eta_1)$, and the intermediate value theorem gives $\eta^* \in (0, \eta_1)$ with a steady state of purities $1-\eta^*$ and $1-(1-z)\eta^*/z$, both at least $1-\varepsilon$, and $R = s$. Every mole fraction of this steady state lies in $(0,1)$: $x_B = \eta^* > 0$ and $x_D \in (0,1)$; the upward step $x \mapsto (s\,e(x) + B\eta^*)/L_B$ maps $(0,1)$ into $(0,1)$ because $s\,e(x) + B\eta^* < s + B = L_B$; and the downward step is a convex combination of points of $(0,1)$ followed by $e^{-1}$, which maps $(0,1)$ onto itself. □

## B One section and the exact reformulation

**Proof** (Proof of Lemma 23). (i) By the rank-one determinant formula, $\det(\lambda - V\operatorname{diag}\alpha + w\alpha^\top) = \prod_i(\lambda - V\alpha_i)(1 + \sum_i \alpha_i w_i/(\lambda - V\alpha_i))$; put $\lambda = Vt$. At a root $t \notin \{\alpha_i\}$ of $P_w$, $\Psi(t) = 0$; $P_w(\alpha_k) = -\alpha_k w_k \prod_{j\neq k}(\alpha_j - \alpha_k)$ vanishes if and only if $w_k = 0$. (ii) For a root, $\alpha\cdot\xi_\psi = V - \Psi(\psi) = V$, so $(M\xi_\psi)_i = V\alpha_i(\xi_\psi)_i - Vw_i = V\psi(\xi_\psi)_i$; for a trace mode $\alpha\cdot\xi_{\alpha_k} = (V - \Psi^{(k)}(\alpha_k)) + \Psi^{(k)}(\alpha_k) = V$ and the same computation works (the $k$-th row uses $w_k = 0$). $(\ell^\top M)_j = V\alpha_j\ell_j - (\ell\cdot w)\alpha_j$ with $\ell_\psi\cdot w = V$ for a root and $\ell_{\alpha_k}\cdot w = w_k = 0$ for a trace mode. For $\mathbf{1}^\top\xi_\psi$: $\sum_i w_i/(\alpha_i - \psi) = \psi^{-1}\sum_i w_i(\alpha_i/(\alpha_i - \psi) - 1) = \psi^{-1}(V - \mathbf{1}^\top w) = L/\psi$, and similarly for a trace mode. Orthogonality and the values of $\ell_\psi\cdot\xi_\psi$ follow from (iii). (iii) With $1/((\alpha - \psi)(\alpha - t)) = (1/(\alpha - t) - 1/(\alpha - \psi))/(t - \psi)$, $G_{\xi_\psi}(t) = ((V - \Psi(t)) - (V - \Psi(\psi)))/(t - \psi)$ for a root; for a trace mode the sum over $i \neq k$ gives $-(\Psi^{(k)}(t) - \Psi^{(k)}(\alpha_k))/(t - \alpha_k)$ and the $k$-th term adds $\Psi^{(k)}(\alpha_k)/(\alpha_k - t)$. Then $\ell_\chi\cdot\xi_\psi = G_{\xi_\psi}(\chi) = 0$ for a root $\chi \neq \psi$, and for a trace mode $\chi = \alpha_k \neq \psi$, $\ell_\chi\cdot\xi_\psi = (\xi_\psi)_k = w_k/(\alpha_k - \psi) = 0$. (iv) Distinct eigenvalues; $\ell_\psi\cdot\hat\xi_\psi \neq 0$ because the modes are simple. □

**Proof** (Proof of Lemma 24). $\Psi'(t) = -\sum_{i\in\Sigma}\alpha_i w_i/(\alpha_i - t)^2$ has constant sign. (a) Near $a_j$ the term $-\alpha_j w_j/(a_j - t)$ tends to $-\infty$ from the left and to $+\infty$ from the right; $\Psi(-\infty) = V$, $\Psi(0) = V - \mathbf{1}^\top w = L > 0$, and on $(a_1, \infty)$ every term $-\alpha_i w_i/(\alpha_i - t)$ is positive; monotonicity gives the roots. (b) The mirror image, with $\Psi(+\infty) = V$. □

**Proof** (Proof of Lemma 25). (a) $M = V\operatorname{diag}(\alpha) + |w|\alpha^\top$. (b) Sherman–Morrison, with $1 - \alpha^\top(V\operatorname{diag}\alpha)^{-1}w = 1 - \mathbf{1}^\top w/V = L/V$ and $\alpha^\top\operatorname{diag}(\alpha)^{-1} = \mathbf{1}^\top$. (c) $M^k\hat\xi_\psi = (V\psi)^k\hat\xi_\psi$. □

**Proof** (Proof of Lemma 26). Simple roots of a polynomial whose coefficients are analytic in $w$ are analytic (implicit function theorem); simple real roots stay real and keep their order; the spectral projectors $\prod_{s\neq r}(M - \lambda_s)/(\lambda_r - \lambda_s)$ are analytic and of rank one, which gives analytic $\xi_r$, $\ell_r$. For a trace mode write $P_w(t) = (\alpha_k - t)Q_w(t) - \alpha_k w_k\prod_{j\neq k}(\alpha_j - t)$ with $Q_w(t) := \prod_{j\neq k}(\alpha_j - t)\Psi^{(k)}(t; w)$. At $w^0$, $P_{w^0} = (\alpha_k - t)Q_{w^0}$, and simplicity of $\alpha_k$ means $Q_{w^0}(\alpha_k) \neq 0$, that is $\Psi^{(k)}(\alpha_k; w^0) \neq 0$. Shrinking $O$ so that $t_r(w)$ stays away from the $\alpha_j$, $j \neq k$, and $\Psi^{(k)}(t_r; w) \neq 0$, the equation $P_w(t_r) = 0$ divided by $\prod_{j\neq k}(\alpha_j - t_r)$ is (27). If $w_k \neq 0$, then $t_r \neq \alpha_k$ and $\tilde\ell_r = ((\alpha_k - t_r)/\alpha_k)\,\ell_{t_r}$ by (27), a left eigenvector by Lemma 23; if $w_k = 0$, $\tilde\ell_r = \mathbf{e}_k$, a left eigenvector by Lemma 23(ii). Formula (28) is the definition written out, and the derivative follows because $t_r = \alpha_k$ where $w_k = 0$. □

**Proof** (Proof of Lemma 30). (a) $x_{\text{top}} = cM^{n+1}x_{\text{bot}}$ if and only if $\ell_r\cdot x_{\text{top}} = c\lambda_r^{n+1}\ell_r\cdot x_{\text{bot}}$ for all $r$. Given this with $c \neq 0$, the index $r_*$ fixes $c$; multiplying the relation of $r \in U$ by $\varpi_{\text{bot}}\lambda_{r_*}^{n+1}/\lambda_r^{n+1}$ and that of $s \in D$ by $\varpi_{\text{bot}}$ gives $E_r = 0$, $E_s = 0$. Conversely, with $c := \varpi_{\text{top}}/(\lambda_{r_*}^{n+1}\varpi_{\text{bot}})$, $E = 0$ gives back every relation. (b) Immediate. (c) The difference consists of terms $\mu^{n+1} \times$ (analytic), and $\nabla(\mu^{n+1}) = (n+1)\mu^{n+1}\nabla\log\mu$. (d) Product rule at a zero. □

**Lemma 65** (weakly chained column-dominant Z-matrices). Let $\mathfrak{A} = D - F$ be a square matrix with $D = \operatorname{diag}(d_u)$, $d_u > 0$, $F \geq 0$ with zero diagonal, and nonnegative column sums ($d_u \geq \sum_{u'} F_{u'u}$ for every $u$). Call $u$ *leaky* if its column sum is positive, and suppose that from every index $u$ there is a chain $u = u_0, u_1, \ldots, u_\ell$

with $F_{u_{t+1}u_t} > 0$ ending at a leaky index. Then $\mathfrak{A}$ is nonsingular, $\mathfrak{A}^{-1} \geq 0$, and $(\mathfrak{A}^{-1})_{us} > 0$ whenever there is a chain from $s$ to $u$ (in particular $u = s$).

**Proof.** Put $P := FD^{-1} \geq 0$, so $\mathfrak{A} = (I - P)D$ and the column sums of $P$ are at most 1, with strict inequality exactly at the leaky indices. The column sum of $P^K$ at $u$ is the mass left after $K$ steps of the substochastic chain started at $u$; it is $< 1$ as soon as the chain can reach a leaky index within $K - 1$ steps. With $K$ the number of indices, $\|P^K\|_1 < 1$, so $\rho(P) < 1$, $(I - P)^{-1} = \sum_k P^k \geq 0$, and its $(u, s)$ entry is positive if a chain leads from $s$ to $u$. Finally $\mathfrak{A}^{-1} = D^{-1}(I - P)^{-1}$. (These are the weakly chained diagonally dominant matrices of [34]; see also [35].) □

**Proof** (Proof of Proposition 22)**.** (a) At a steady state all compositions lie in $\mathcal{X}$, so $S \geq \alpha_N > 0$. The component balances of the stages between a cross-section and the bottom end of $\sigma$ give $V_\sigma y(x_k) - L_\sigma x_{k+1} = w_\sigma$ at every cross-section ($0 \leq k \leq n_\sigma$, $x_0 := x_{\mathrm{bot}}$, $x_{n_\sigma+1} := x_{\mathrm{top}}$), that is $L_\sigma S(x_k) x_{k+1} = M_\sigma x_k$; iterating, $x_{\mathrm{top}} = M_\sigma^{n_\sigma+1} x_{\mathrm{bot}} / \prod_k (L_\sigma S(x_k))$. The junction and product balances are $b_c = f_c - d_c$ and the formula for $\bar{x}_p$.

(b) Fix $\sigma$ and drop the index. Since $\mathbf{1}^\top M = L\alpha^\top$, $\mathbf{1}^\top M^{k+1} x_{\mathrm{bot}} = L\, S(x_k)\, \mathbf{1}^\top M^k x_{\mathrm{bot}}$, so the iterates are defined as long as $S(x_k) \neq 0$, $x_{k+1} = M x_k / (L S(x_k))$, and the cross-section balances $V y(x_k) - L x_{k+1} = w$ hold for $0 \leq k \leq n$. The hypothesis on $x_{\mathrm{top}}$ and $\mathbf{1}^\top x_{\mathrm{top}} = 1$ give $x_{\mathrm{top}} = M^{n+1} x_{\mathrm{bot}} / (\mathbf{1}^\top M^{n+1} x_{\mathrm{bot}}) = x_{n+1}$, and the positive factor is $1/(L^{n+1} \prod_{k=0}^{n} S(x_k))$. The balance of stage $k$ ($1 \leq k \leq n$) is the difference of the cross-section balances $k - 1$ and $k$. The vapor entering stage 1 is that of the bottom-end node, of composition $y(x_0)$; the liquid entering stage $n$ is that of the top-end node, of composition $x_{n+1}$. At a node $J$, by the end cross-section balances, the component balance (out minus in) is $\sum_{\mathrm{bot}(\sigma)=J} w_\sigma - \sum_{\mathrm{top}(\sigma)=J} w_\sigma - (\text{feed}) + (\text{product})$. At $J_c$, $c = [i, m]$, this is $d_c - b_{[i-1,m]} + b_c - d_{[i,m+1]}$ (minus $z$ if $c = [1, N]$), that is $d_c + b_c - f_c = 0$. At $J[p, p]$ it is $-d_{[p,p+1]} - b_{[p-1,p]} + P_p \bar{x}_p = 0$. At the condenser, the top cross-section of $\mathrm{T}[1, 2]$ gives $V y(x_n) = L x_1 + d_{[1,2]} = (L + P_1)\bar{x}_1 = V \bar{x}_1$, so the condensed vapor has the composition $\bar{x}_1$ of the reflux and of the distillate, and the composition rule and the balance hold. Total and CMO balances hold by the specification, so the duties are those of the specification.

*Positivity.* Put $K_{u,i} := \alpha_i / S(x_u) > 0$ at every unit $u$ with a vapor outflow. For each component $i$ the balances just verified form a linear system $\mathfrak{A}_i\, X_{\cdot,i} = z_i\, \mathbf{e}_{J[1,N]}$ in the vector of $i$-contents of all units, where $\mathfrak{A}_i$ has diagonal entries $L_u + V_u K_{u,i} > 0$ (total liquid and vapor outflows of $u$, products included) and off-diagonal entries $-(\ell_{u'u} + v_{u'u} K_{u',i}) \leq 0$ in column $u'$, $\ell_{u'u}$ and $v_{u'u}$ being the liquid and vapor flows from $u'$ to $u$. The column sum at $u'$ equals the flow that $u'$ sends to product outlets, and every unit is joined to a product node along positive flows (liquid flows down every section and reaches the reboiler). By Lemma 65, $\mathfrak{A}_i$ is nonsingular with a nonnegative inverse that is positive between units joined by a path of positive flows. Every unit is reached from $J[1, N]$ along positive flows, so $x_{u,i} > 0$ for every unit and every component, and $y = Kx > 0$ too. □

## C The forced design: proofs

**Proof** (Proof of (31)). Insert the definitions: $\Psi_{\mathrm{T}[i,m+1]} - \Psi_{\mathrm{B}[i-1,m]} = V_{\mathrm{T}[i,m+1]} - V_{\mathrm{B}[i-1,m]} - \sum_k \alpha_k(d_{[i,m+1],k} + b_{[i-1,m],k})/(\alpha_k - t) = \Phi_{[i,m]}$, and $\Psi_{\mathrm{T}_c} - \Psi_{\mathrm{B}_c} = V_{\mathrm{T}} - V_{\mathrm{B}} - \sum_k \alpha_k(d_k + b_k)/(\alpha_k - t) = \Phi_c$ because $d + b = f$ and $V_{\mathrm{T}} - V_{\mathrm{B}} = U$. □

**Proof** (Proof of Lemma 32). Partial fractions of $\prod_r (t-\theta_r^c)/\prod_{p=i}^{m-1}(t-\alpha_p) = 1+\sum_p \omega_p/(t-\alpha_p)$ give the formula for $\Psi_{\mathrm{T}}^0$. Numerator and denominator of $\omega_p$ have $p-i$ negative factors each, so $\omega_p > 0$, $V_{\mathrm{T}}^0 > 0$ and $d^0 > 0$. Since $\Psi_{\mathrm{T}}^0 - \Psi_{\mathrm{B}}^0 = \Phi_c$, $\Psi_{\mathrm{B}}^0$ vanishes at every $\theta_r^c$; since $b_i^0 = 0$, $\Psi_{\mathrm{B}}^0(t)\prod_{p=i+1}^m (t-\alpha_p)$ is a polynomial of degree at most $m-i$ with leading coefficient $V_{\mathrm{B}}^0$ and the $m-i$ roots $\theta_r^c$; and $V_{\mathrm{B}}^0 = 0$ would give $b^0 = 0$, contradicting $b_m^0 = f_m > 0$. The residues give $-\alpha_p b_p^0 = V_{\mathrm{B}}^0 \prod_r(\alpha_p - \theta_r^c)/\prod_{p'\neq p}(\alpha_p - \alpha_{p'})$, of the sign of $-V_{\mathrm{B}}^0$, and $b_m^0 > 0$ forces $V_{\mathrm{B}}^0 > 0$. Finally $L_{\mathrm{T}}^0 = \Psi_{\mathrm{T}}^0(0)$ and $L_{\mathrm{B}}^0 = \Psi_{\mathrm{B}}^0(0)$ are positive by the product forms. □

**Proof** (Proof of Lemma 33). $\theta_c^\star < \alpha_{i+1}$ and $\theta_c^\star > \alpha_{m-1}$ by comparing the factors $\theta_r^c$ with $\alpha_{r+1}$ and $\alpha_r$. $K$ has numerator and denominator of equal degree, leading coefficient 1 and simple poles at $\alpha_{i+1}, \dots, \alpha_{m-1}$ with residues $\alpha_p h_p$; $K(0) = 1$ by the choice of $\theta_c^\star$, which gives $\sum_p h_p = 0$. At $\theta_i^c$ and $\alpha_i$ all factors are positive; at $\theta_{m-1}^c$ and $\alpha_m$ numerator and denominator each have $m-i-1$ negative factors. □

**Proof** (Proof of Definition 34 (the section functions)). $\Psi_{\mathrm{T}_c} = V_{\mathrm{T}}^0 + \eta - \sum \alpha_p(d_p^0 + \eta h_p)/(\alpha_p - t) = \Psi_{\mathrm{T}}^0 + \eta K$, and with $b = b^0 - \eta h$, $\Psi_{\mathrm{B}_c} = V_{\mathrm{B}}^0 + \eta + \sum \alpha_p(b_p^0 - \eta h_p)/(\alpha_p - t) = \Psi_{\mathrm{B}}^0 + \eta K$. Validity for $m - i = 2$ is immediate ($h = 0$). □

**Proof** (Proof of Proposition 35). (a) Lemma 32 and $\sum h = 0$: $L_{\mathrm{T}} = L_{\mathrm{T}}^0 + \eta$, $L_{\mathrm{B}} = L_{\mathrm{B}}^0 + \eta$. (b) $\mathrm{T}_c$ has $w = d \geq 0$ with support $[i, m-1]$ and $L_{\mathrm{T}} > 0$: by Lemma 24(a) it has one root in each $(\alpha_{p+1}, \alpha_p)$, $i \leq p \leq m-2$, and one in $(0, \alpha_{m-1})$. $\Psi_{\mathrm{T}_c}(\theta_r^c) = \eta K(\theta_r^c) = 0$ for the middle $r$; $\Psi_{\mathrm{T}_c}(\theta_i^c) = \eta K(\theta_i^c) > 0$, with $\Psi_{\mathrm{T}_c}$ decreasing on $(\alpha_{i+1}, \alpha_i)$, gives $r_c > \theta_i^c$; $\Psi_{\mathrm{T}_c}(\theta_{m-1}^c) > 0$, with $\Psi_{\mathrm{T}_c}$ decreasing on $(-\infty, \alpha_{m-1})$, gives $u_c > \theta_{m-1}^c > \alpha_m$ and $\Psi_{\mathrm{T}_c}(\alpha_m) > 0$. $\mathrm{B}_c$ is symmetric (Lemma 24(b), $\Psi_{\mathrm{B}_c}(\theta_i^c) > 0$ and increasing). (c) Lemma 23 and (b). (d) Lemma 23(ii): $(\hat{\xi}_u^{\mathrm{T}})_p = (u_c/L_{\mathrm{T}})d_p/(\alpha_p - u_c) > 0$ for $p \leq m-1$; $(\hat{\xi}_{(m)}^{\mathrm{T}})_p = (\alpha_m/L_{\mathrm{T}})d_p/(\alpha_p - \alpha_m) > 0$ ($p < m$) and $(\hat{\xi}_{(m)}^{\mathrm{T}})_m = \Psi_{\mathrm{T}_c}(\alpha_m)/L_{\mathrm{T}} > 0$; symmetrically for $\mathrm{B}_c$. (e) $G_X(t)\prod_{p\in[i,m]}(\alpha_p - t)$ is a polynomial of degree at most $m-i$ (exactly $m-i$, with leading coefficient $\pm S(X)$); prescribing its $m-i$ zeros fixes it up to a scalar, the residues give the formula, and numerator and denominator of $X_{c,p}$ have $p-i$ negative factors each; the homogeneous version gives the nonsingularity. In the eigenbasis of $\mathrm{T}_c$, the coefficient of $X_c$ on a major mode $\chi \in \{r_c, \theta_{\mathrm{middle}}^c\}$ is proportional to $G_{X_c}(\chi) = 0$, and on a trace mode $\alpha_k$ ($k \notin [i,m]$) to $X_{c,k} = 0$; so $X_c = (1-\beta)\hat{\xi}_{(m)}^{\mathrm{T}} + \beta\hat{\xi}_u^{\mathrm{T}}$. By the pinch identity, $G_{\hat{\xi}_{(m)}^{\mathrm{T}}}(s_c) = -(\alpha_m/L_{\mathrm{T}})\Psi_{\mathrm{T}_c}(s_c)/(s_c - \alpha_m) < 0$ ($\Psi_{\mathrm{T}_c}(s_c) > 0$ since $s_c < \theta_{m-1}^c < u_c$) and $G_{\hat{\xi}_u^{\mathrm{T}}}(s_c) = -(u_c/L_{\mathrm{T}})\Psi_{\mathrm{T}_c}(s_c)/(s_c - u_c) > 0$; $G_{X_c}(s_c) = 0$ then forces $0 < \beta < 1$. The bottom decomposition is the same at $r_c$. □

**Proof** (Proof of Proposition 37). $V_{\mathrm{T}} = \kappa + \alpha_p f_p/(\alpha_p - \theta') > \alpha_p f_p/(\alpha_p - \theta') > f_p$ and $V_{\mathrm{B}} = \kappa + \alpha_{p+1} f_{p+1}/(\theta' - \alpha_{p+1}) > 0$. $\Psi_{\mathrm{T}}^0$ decreases on $(-\infty, \alpha_p)$ with $\Psi_{\mathrm{T}}^0(\theta') = \kappa > 0$, and $\Psi_{\mathrm{B}}^0$ increases on $(\alpha_{p+1}, \infty)$ with $\Psi_{\mathrm{B}}^0(\theta') = \kappa > 0$; their roots solve $\alpha_p - t = \alpha_p f_p/V_{\mathrm{T}}$ and $t - \alpha_{p+1} = \alpha_{p+1} f_{p+1}/V_{\mathrm{B}}$. □

**Proof** (Proof of Proposition 38). $\hat{\eta}_p > 0$ because $\Psi_{\mathrm{B}}^0(\alpha_{p-1}) > \kappa$ (Proposition 37). (a) $\mathbf{1}^\top d = f_p - \eta > 0$, $\mathbf{1}^\top b = f_{p+1} + \eta$, $L_{\mathrm{T}} = (V_{\mathrm{T}} - f_p) + \eta > 0$, $L_{\mathrm{B}} > 0$. (b) $\Psi_{\mathrm{T}_c}(t) = V_{\mathrm{T}} -$ ⤵

$\alpha_p(f_p-\eta)/(\alpha_p-t)$ vanishes at $t=\alpha_p L_{\mathrm{T}}/V_{\mathrm{T}}$, which increases with $\eta$. (c) Lemma 24(b) with poles $\alpha_p,\alpha_{p+1}$: one root in $(\alpha_{p+1},\alpha_p)$ and one in $(\alpha_p,\infty)$. On $(\alpha_{p+1},\alpha_p)$ the added term $\alpha_p\eta/(\alpha_p-t)$ is positive, so $\Psi_{\mathrm{B}_c}(\psi^0_{\mathrm{B}})>0$ and $\psi''<\psi^0_{\mathrm{B}}$. Next, $\psi'<\alpha_{p-1}$ if and only if $\Psi_{\mathrm{B}_c}(\alpha_{p-1})>0$ ($\Psi_{\mathrm{B}_c}$ increases on $(\alpha_p,\infty)$), that is $\Psi^0_{\mathrm{B}}(\alpha_{p-1})-\alpha_p\eta/(\alpha_{p-1}-\alpha_p)>0$, which is the bound on $\eta$. The trace modes are placed by Lemma 24(b). (d) $\psi^{\mathrm{T}}\ge\psi^0_{\mathrm{T}}>\theta'$ and $\psi^{\mathrm{T}}<\alpha_p$, so $\Psi_{\mathrm{B}_c}(\psi^{\mathrm{T}})=\Psi^0_{\mathrm{B}}(\psi^{\mathrm{T}})+\alpha_p\eta/(\alpha_p-\psi^{\mathrm{T}})>\Psi^0_{\mathrm{B}}(\psi^{\mathrm{T}})>\kappa$; as $\Psi_{\mathrm{B}_c}$ increases on $(\alpha_{p+1},\alpha_p)$, $\Psi_{\mathrm{B}_c}(\psi^{\mathrm{T}})>0$ is equivalent to $\psi^{\mathrm{T}}>\psi''$. (e) Positivity: $\psi'>\alpha_p>\alpha_{p+1}$. On the face $\{p,p+1\}$ the modes of $\mathrm{T}_c$ are $\psi^{\mathrm{T}}$ (vector $\mathbf{e}_p$, since $d$ is supported on $\{p\}$) and $\alpha_{p+1}$ (left vector $\mathbf{e}_{p+1}$), and $(\hat\xi^{\mathrm{T}}_{(p+1)})_{p+1}=\Psi_{\mathrm{T}_c}(\alpha_{p+1})/L_{\mathrm{T}}>0$ because $\alpha_{p+1}<\psi^{\mathrm{T}}$ (Lemma 24(a)). The coefficient of $\hat\xi^{\mathrm{T}}_{(p+1)}$ in $X$ is $X_{p+1}/(\hat\xi^{\mathrm{T}}_{(p+1)})_{p+1}>0$, and that of $\mathbf{e}_p$ is $G_X(\psi^{\mathrm{T}})(\alpha_p-\psi^{\mathrm{T}})/\alpha_p$, where $G_X(\psi^{\mathrm{T}})=-(\psi'/L_{\mathrm{B}})\Psi_{\mathrm{B}_c}(\psi^{\mathrm{T}})/(\psi^{\mathrm{T}}-\psi')>0$ by (d) and the pinch identity. (f) Lemma 26 with $k=p$: by (27), $\psi'-\alpha_p=\alpha_p\eta/\Psi^{(p)}_{\mathrm{B}_c}(\psi')\to 0$ with $\Psi^{(p)}_{\mathrm{B}_c}=\Psi^0_{\mathrm{B}}$, so $\eta/(\psi'-\alpha_p)\to\Psi^0_{\mathrm{B}}(\alpha_p)/\alpha_p$. □

**Proof** (Proof of Proposition 39). The mirror image of Proposition 38. $\hat\eta_p>0$ since $V_{\mathrm{T}}>f_p$ and $\Psi^0_{\mathrm{T}}(\alpha_{p+2})>\kappa$. $\mathbf{1}^\top b=f_{p+1}-\eta>0$; $L_{\mathrm{T}}>0$ by the bound (for $p+1<N$ via Lemma 24(a): $\Psi_{\mathrm{T}_c}>0$ on $(-\infty,\psi^{\mathrm{low}})\ni 0,\alpha_{p+2}$). $\psi^{\mathrm{B}}$ solves $t-\alpha_{p+1}=\alpha_{p+1}(f_{p+1}-\eta)/V_{\mathrm{B}}$; it exceeds $\alpha_{p+1}$ because $f_{p+1}>\eta$ and decreases with $\eta$. Lemma 24(a) gives one root of $\Psi_{\mathrm{T}_c}$ in $(\alpha_{p+1},\alpha_p)$ and one in $(0,\alpha_{p+1})$; on $(\alpha_{p+1},\alpha_p)$ the added term $\alpha_{p+1}\eta/(t-\alpha_{p+1})$ is positive, so $\psi^{\mathrm{mid}}>\psi^0_{\mathrm{T}}$; $\psi^{\mathrm{low}}>\alpha_{p+2}$ if and only if $\Psi_{\mathrm{T}_c}(\alpha_{p+2})>0$, which is the bound on $\eta$. Margin: $\psi^{\mathrm{B}}\le\psi^0_{\mathrm{B}}<\theta'$ and $\psi^{\mathrm{B}}>\alpha_{p+1}$, so $\Psi_{\mathrm{T}_c}(\psi^{\mathrm{B}})>\Psi^0_{\mathrm{T}}(\psi^{\mathrm{B}})>\kappa$, and $\Psi_{\mathrm{T}_c}$ decreases on $(\alpha_{p+1},\alpha_p)$. Cone: $G_X(\psi^{\mathrm{B}})=-(\psi^{\mathrm{low}}/L_{\mathrm{T}})\Psi_{\mathrm{T}_c}(\psi^{\mathrm{B}})/(\psi^{\mathrm{B}}-\psi^{\mathrm{low}})<0$; the coefficient of $\mathbf{e}_{p+1}=\hat\xi^{\mathrm{B}_c}_{\psi^{\mathrm{B}}}$ is $G_X(\psi^{\mathrm{B}})(\alpha_{p+1}-\psi^{\mathrm{B}})/\alpha_{p+1}>0$ and that of $\hat\xi^{\mathrm{B}}_{(p)}$ is $X_p/(\hat\xi^{\mathrm{B}}_{(p)})_p>0$, with $(\hat\xi^{\mathrm{B}}_{(p)})_p=\Psi_{\mathrm{B}_c}(\alpha_p)/L_{\mathrm{B}}>0$ since $\alpha_p>\psi^{\mathrm{B}}$. The limit $\eta\to 0$: Lemma 26 with $k=p+1$. □

**Proof** (Proof of Lemma 41). *Feeds*: induction in the order of (F1); $f_{[i,m]}=d_{[i,m+1]}+b_{[i-1,m]}$, where each parent contributes a vector positive exactly on $[i,m]$ (validity), and a binary column is never a parent. *Vapor at $J_c$*: the outflows are $V_{\mathrm{T}_c}$ and $V_{\mathrm{B}[i-1,m]}$, the inflows $V_{\mathrm{B}_c}$ and $V_{\mathrm{T}[i,m+1]}$; the difference is $V_{\mathrm{T}_c}-V_{\mathrm{B}_c}-U_c=0$ (for $[1,N]$ the feed brings $1-q$). *Liquid at $J_c$*: $L_{\mathrm{B}_c}-L_{\mathrm{T}_c}=\mathbf{1}^\top d_c+\mathbf{1}^\top b_c-U_c=\mathbf{1}^\top f_c-U_c=L_{\mathrm{B}[i-1,m]}+q\,[c=[1,N]]-L_{\mathrm{T}[i,m+1]}$. *Side draws*: $V_{\mathrm{B}[p-1,p]}=V_{\mathrm{T}[p,p+1]}$ by (F2) and $L_{\mathrm{B}[p-1,p]}-L_{\mathrm{T}[p,p+1]}=\mathbf{1}^\top b_{[p-1,p]}+\mathbf{1}^\top d_{[p,p+1]}=P_p$. Telescoping gives the vapors; summing $d_c+b_c=f_c$ over the tree of columns gives $\sum_p\bar n_p=z$. *Flux bookkeeping* (42): at $J_c$ the left side is $(V_{\mathrm{T}_c}-V_{\mathrm{B}_c}-U_c)-\sum_k\alpha_k(d_{c,k}+b_{c,k}-f_{c,k})/(\alpha_k-t)=0$ (for $J[1,N]$: $\varphi_F(t)$); at a side draw, $(V_{\mathrm{B}[p-1,p]}-V_{\mathrm{T}[p,p+1]})+\sum_k\alpha_k(b_{[p-1,p],k}+d_{[p,p+1],k}-\bar n_{p,k})/(\alpha_k-t)=0$; at the condenser, $-\Psi_{\mathrm{T}[1,2]}+\varphi_{P_1}=-V_{\mathrm{T}[1,2]}$; at the reboiler, $\Psi_{\mathrm{B}[N-1,N]}+\varphi_{P_N}=V_{\mathrm{B}[N-1,N]}=R$. (This is Lemma 8 summed over the stages of each section.) □

**Proof** (Proof of Lemma 43). Induction top-down with (31). For a valid parent deformed with $\eta\ge 0$, $\Psi_{\mathrm{T}}=\Psi^0_{\mathrm{T}}+\eta K$ and $\Psi_{\mathrm{B}}=\Psi^0_{\mathrm{B}}+\eta K$, where $\Psi^0_{\mathrm{T}}$, $\Psi^0_{\mathrm{B}}$ vanish at all the parent's feed roots (Lemma 32) and $K$ at its middle ones (Lemma 33); by induction these middle roots are global. So $\Psi_{\mathrm{T}[i,m+1]}$ vanishes at $\theta_{i+1},\dots,\theta_{m-1}$ and $\Psi_{\mathrm{B}[i-1,m]}$ at $\theta_i,\dots,\theta_{m-2}$, hence $\Phi_{[i,m]}$ vanishes at $\theta_{i+1},\dots,\theta_{m-2}$, and $\Phi_{[i,m]}$ has exactly one root per interval. At $\eta_{\mathrm{nb}}=0$ both section functions of every non-binary column vanish at all of its feed roots, and the same induction makes every feed root of every column global. □

**Proof** (Proof of Proposition 44). (a) (F1) involves neither $R$ nor $\eta_{\mathrm{bin}}$ nor $\pi$. (b) $V_{\mathrm{T}[p,p+1]}=R+\mathrm{const}$ (Lemma 41), with the constant and the binary feed independent of $R$, and

$\kappa_p = V_{\mathrm{T}[p,p+1]} - \alpha_p f_p/(\alpha_p - \theta')$.

(c) (A light-side cut.) At $\eta_{\mathrm{nb}} = 0$ every feed root of every column is global (Lemma 43); in particular $\theta_p$ is the feed root of $[p, p+1]$ and a root of $\Psi_{\mathrm{T}[i,p+1]}$ for every non-binary $[i, p+1]$ (Lemma 32). Since $\kappa_p$ does not depend on $\eta_{\mathrm{bin}}$, we may evaluate the identities (42), which are algebraic consequences of the balances, with the sharp binary net flows ($\eta_{\mathrm{bin}} = 0$), for which $\bar{n}_r = z_r \mathbf{e}_r$ for every product $r$. Sum (42) at $t = \theta_p$ over the *light* nodes: the junctions $J[i,m]$ with $m \le p$, the condenser and the side draws $J[2,2], \dots, J[p,p]$. A section with both ends light contributes $\Psi - \Psi = 0$; a bottom section $\mathrm{B}[i,m]$ with a light end has $m \le p$, so both ends are light; a top section $\mathrm{T}[i,m]$ with a light end has $m \le p+1$; so the sections with exactly one light end are the $\mathrm{T}[i,p+1]$, $1 \le i \le p$, whose top end $J[i,p]$ is light. The feed enters $J[1,N]$, which is not light. Hence $-\sum_{i=1}^{p} \Psi_{\mathrm{T}[i,p+1]}(\theta_p) + \sum_{r=1}^{p} \varphi_{P_r}(\theta_p) = -V_{\mathrm{T}[1,2]}$. Summing (42) over all nodes, every section cancels and $\sum_r \varphi_{P_r}(t) = R - V_{\mathrm{T}[1,2]} + \varphi_F(t)$ for every $t$; letting $t \to \infty$ gives $V_{\mathrm{T}[1,2]} = R + 1 - q$. With $\Psi_{\mathrm{T}[i,p+1]}(\theta_p) = 0$ for $i < p$, $\varphi_{P_r}(\theta_p) = -\alpha_r z_r/(\alpha_r - \theta_p)$ and the feed equation,

$$\kappa_p(0;\delta) = \Psi_{\mathrm{T}[p,p+1]}(\theta_p) = R + (1-q) - \sum_{r \le p} \frac{\alpha_r z_r}{\alpha_r - \theta_p} = R - \sum_{r > p} \frac{\alpha_r z_r}{\theta_p - \alpha_r} = R - H_p = g_p + \delta.$$

(d) All data of every column are real-analytic in $\eta_{\mathrm{nb}}$ near 0 (simple roots of the feed functions, rational expressions with nonvanishing denominators) as long as the feeds remain positive; (V) is strict at $\eta_{\mathrm{nb}} = 0$ (Lemma 32), and the binary feed roots are simple. So there is $\eta_1 > 0$ such that on the cube $[0, \eta_1]^{(N-1)(N-2)/2}$ all columns are valid and $\kappa_p(\cdot\,;0)$ is $C^1$; take $C_1 := \max_p \max \|\nabla \kappa_p(\cdot\,;0)\|_1$ over the cube and use the mean value theorem and (c). □

**Proof** (Proof of Theorem 46)**.** By Proposition 44(b), (d), $\kappa_p(\eta_{\mathrm{nb}};\delta) = \kappa_p(\eta_{\mathrm{nb}};0) + \delta \ge g_p - C_1 \bar{\eta}(\delta) + \delta \ge g_p + \delta/2$. Openness and analyticity: finitely many strict inequalities between analytic functions. □

**Proof** (Proof of Theorem 49)**.** (H1)–(H4) collect Lemma 41, Propositions 35, 38, 39 and Theorem 46.

*(H5) and the limit relations, type by type.* The mode lists are those of Proposition 35(c) (types 1, 2) and Propositions 38–39 (types 3–6), with distinct positive modes, so $\rho_\sigma < 1$. We use $\ell_\psi \cdot x = G_x(\psi)$ for a major mode and $\ell_\psi \cdot x = x_k$ for a trace mode $\alpha_k$. Type 1: $G_{X_c}(r_c) = G_{X_c}(\theta_r) = 0$ by Proposition 35(e), $X_{c,k} = 0$ for $k < i$, and the top end $X_{[i,m-1]}$ is supported on $[i, m-1]$. Type 2: the bottom end $X_{[i+1,m]}$ is supported on $[i+1, m]$; $G_{X_c}(\theta_r) = G_{X_c}(s_c) = 0$ and $X_{c,k} = 0$ for $k > m$. Type 3: $X_c$ is supported on $\{p, p+1\}$; the top end $\bar{x}_p$ has $\mathrm{Maj}(p) \subseteq \{p-1, p\}$ ($\pi_p = \mathrm{B}$). Type 4: $\bar{x}_{p+1}$ has $\mathrm{Maj}(p+1) \subseteq \{p, p+1, p+2\}$; $X_c = \hat{\xi}_{\psi'}$ is annihilated by $\ell_{\psi''}$ (Lemma 23(iii)) and has no component $k > p+1$. Types 5, 6: the mirror images ($\mathrm{Maj}(p) \subseteq \{p-1, p, p+1\}$, $\mathrm{Maj}(p+1) \subseteq \{p+1, p+2\}$).

*Pivots (H8).* At a junction the pivot of the section whose pinch vector appears in the cone decomposition is its coefficient: $\beta_c^{\mathrm{T}}$ (type 1), $\beta_c^{\mathrm{B}}$ (type 2), $\beta_c^{\mathrm{T}}$ (type 3), 1 (type 4), 1 (type 5), $\beta_c^{\mathrm{B}}$ (type 6), all positive. At the other end $x$ (a product, or a child's junction for types 1–2), $\operatorname{supp} x$ lies entirely on one side of the pinch: for a top section every component of $\operatorname{supp} x$ is lighter than $\psi_*$ (type 1: $\operatorname{supp} \subseteq [i, m-1]$, $u_c < \alpha_{m-1}$; type 3: $\subseteq \{p-1, p\}$, $\psi^{\mathrm{T}} < \alpha_p$; type 5: $\subseteq \{p-1, p, p+1\}$, $\psi^{\mathrm{low}} < \alpha_{p+1}$), so $G_x(\psi_*) > 0$, and

$\ell_{\psi_*} \cdot \hat{\xi}_{\psi_*} = (\psi_*/L)(-\Psi'(\psi_*)) > 0$ ($\Psi$ decreasing); for a bottom section every component of $\operatorname{supp} x$ is heavier than $\psi_*$ (type 2: $\subseteq [i+1, m]$, $v_c > \alpha_{i+1}$; type 4: $\subseteq \{p, p+1, p+2\}$, $\psi' > \alpha_p$; type 6: $\subseteq \{p+1, p+2\}$, $\psi^{\mathrm{B}} > \alpha_{p+1}$), so $G_x(\psi_*) < 0$ and $\ell_{\psi_*} \cdot \hat{\xi}_{\psi_*} < 0$. In both cases $[x]_{\psi_*} > 0$.

*(H6).* By the limit relations, $x_{\mathrm{bot}} = [x_{\mathrm{bot}}]_{\psi_*}(\hat{\xi}_{\psi_*} + \sum_{\psi<\psi_*} \dots)$ and similarly at the top, which gives the expressions of $v_k^{\mathrm{bot}}$, $v_k^{\mathrm{top}}$; they converge to $\hat{\xi}_{\psi_*}$. Nonnegativity: going up a bottom section (types 2, 4, 6), $M \geq 0$ (Lemma 25(a)); going down a top section (types 1, 3, 5), $M^{-1} \geq 0$ (Lemma 25(b)). The two other directions start at a junction and use the cone decompositions with nonnegative pinch vectors and Lemma 25(c): going up $\mathrm{T}_c$ from $X_c$, type 1 uses $X_c = (1-\beta_c^{\mathrm{T}})\hat{\xi}^{\mathrm{T}}_{(m)} + \beta_c^{\mathrm{T}}\hat{\xi}^{\mathrm{T}}_u$, type 3 uses $X_c = (1-\beta_c^{\mathrm{T}})\hat{\xi}^{\mathrm{T}}_{(p+1)} + \beta_c^{\mathrm{T}}\mathbf{e}_p$, and in type 5 $X_c = \hat{\xi}_{\psi^{\mathrm{low}}}$ is an eigenvector; going down $\mathrm{B}_c$ from $X_c$, type 2 uses $X_c = \beta_c^{\mathrm{B}}\hat{\xi}^{\mathrm{B}}_v + (1-\beta_c^{\mathrm{B}})\hat{\xi}^{\mathrm{B}}_{(i)}$, in type 4 $X_c = \hat{\xi}_{\psi'}$ is an eigenvector, and type 6 uses $X_c = \beta_c^{\mathrm{B}}\mathbf{e}_{p+1} + (1-\beta_c^{\mathrm{B}})\hat{\xi}^{\mathrm{B}}_{(p)}$. Hence $M^{\pm k}x \geq 0$, and $M^{\pm k}x \neq 0$ since $M$ is invertible; with positive pivots, $v_k \geq 0$ and $\mathbf{1}^\top v_k > 0$, and since $\mathbf{1}^\top v_k \to \mathbf{1}^\top \hat{\xi}_{\psi_*} = 1$, $\inf_k \mathbf{1}^\top v_k > 0$.

*(H7).* $x_{J,k} = 0$ is part of the limit relations. The closed-at-bottom trace modes of Table 48 are $\alpha_k$ with $k < i$ (type 1; $\operatorname{supp} X_c = [i, m]$), $k \leq i$ (type 2; $\operatorname{supp} = [i+1, m]$), $k < p$ (types 3, 5; $\operatorname{supp} X_c = \{p, p+1\}$), $k < p$ (type 4; $\operatorname{supp} \bar{x}_{p+1} \subseteq \{p, p+1, p+2\}$), $k \leq p$ (type 6; $\operatorname{supp} \subseteq \{p+1, p+2\}$); the closed-at-top ones are $k \geq m$ (type 1; $\operatorname{supp} \subseteq [i, m-1]$), $k > m$ (type 2; $\operatorname{supp} X_c = [i, m]$), $k \geq p+1$ (type 3; $\operatorname{supp} \subseteq \{p-1, p\}$), $k > p+1$ (types 4, 6; $\operatorname{supp} X_c = \{p, p+1\}$), $k > p+1$ (type 5; $\operatorname{supp} \subseteq \{p-1, p, p+1\}$). In each case $k$ is lighter (bottom) or heavier (top) than the whole support, so every term of $G^{(k)}_{x_J}(\alpha_k) = \sum_{i \in \operatorname{supp}} \alpha_i x_i/(\alpha_i - \alpha_k)$ is negative, respectively positive. The components with nonzero net flow are $[i, m-1]$ (type 1), $[i+1, m]$ (2), $\{p\}$ (3), $\{p, p+1\}$ (4, 5), $\{p+1\}$ (6); comparing with the supports of the end compositions gives the last statement of (H7). Hence a closed-at-bottom trace of a top section is lighter than every component with nonzero net flow, so $\alpha_k$ exceeds every pole and $\Psi_\sigma(\alpha_k) > V > 0$ (Lemma 24(a)); a closed-at-top trace of a top section lies below its smallest root, where $\Psi_\sigma > 0$; a closed-at-bottom trace of a bottom section lies above its largest root, where $\Psi_\sigma > 0$; a closed-at-top trace of a bottom section lies below every pole, where $\Psi_\sigma > V$ (Lemma 24(b)); and $\Psi^{(k)}_\sigma = \Psi_\sigma$ since $w_k = 0$. Finally, for a junction $X_c$, $c = [i, m]$, and $k \notin [i, m]$: if $k < i$, $\alpha_k$ is a trace mode of $\mathrm{T}_c$ above its pinch (types 1, 3, 5), closed at $X_c$; if $k > m$, $\alpha_k$ is a trace mode of $\mathrm{B}_c$ below its pinch (types 2, 4, 6), closed at $X_c$.

*(H8).* $R$: Lemma 41. Purities: $\bar{n}_p = b_{[p-1,p]} + d_{[p,p+1]}$, whose entries other than $p$ are $\eta_{p-1}$ (on $p-1$, if $\pi_{p-1} = \mathrm{B}$) and $\eta_p$ (on $p+1$, if $\pi_p = \mathrm{T}$). Pivots: shown above. □

## D The limit system: proof of Theorem 52

Throughout, the parameters are admissible and $v_\infty = \mathcal{L}$. In this appendix the rows of $G_\infty$ are divided by the nonvanishing pivot product $\varpi_{\mathrm{bot}}\varpi_{\mathrm{top}}$: the row of a mode $r$ closed at the end $J$ of $\sigma$ is $\ell_r \cdot x_J/\ell_{r_*} \cdot x_J$. By Lemma 30(d) this does not affect the invertibility of the derivative at $v_\infty$. Left eigenvectors are Underwood's $\ell_\psi$ for major modes and the regularized $\tilde{\ell}$ of Lemma 26 for trace modes.

*Coordinates.* On $\mathcal{V}$ we use the affine coordinates of Table 66; all other entries of $(d_c, X_c)$ are dependent ($\mathbf{1}^\top d_c = \Delta_{\mathrm{T}_c}$, $\mathbf{1}^\top X_c = 1$, $b_c = f_c - d_c$ given the parents). The change of

variables is affine, triangular in the order of the columns (parents first) and invertible. A trace coordinate carries the index $k$ of its component; its value at $v_\infty$ is 0.

| column | trace coordinates | major coordinates | dependent |
|---|---|---|---|
| non-binary $[i,m]$ | $d_{c,k}$ $(k \notin [i,m])$, $d_{c,m}$, $b_{c,i}$; $X_{c,k}$ $(k \notin [i,m])$ | $d_{c,p}$ $(i<p<m-1)$; $X_{c,p}$ $(i<p\le m)$ | $d_{c,i}=f_{c,i}-b_{c,i}$, $d_{c,m-1}$, $X_{c,i}$ |
| binary, $\pi_p=\mathrm{B}$ | $d_{c,k}$ $(k\neq p)$; $X_{c,k}$ $(k\notin\{p,p+1\})$ | $X_{c,p+1}$ | $d_{c,p}$, $X_{c,p}$ |
| binary, $\pi_p=\mathrm{T}$ | $d_{c,k}$ $(k\notin\{p,p+1\})$, $b_{c,p}$; $X_{c,k}$ $(k\notin\{p,p+1\})$ | $X_{c,p+1}$ | $d_{c,p}=f_{c,p}-b_{c,p}$, $d_{c,p+1}$, $X_{c,p}$ |

**Table 66 /** Coordinates on $\mathcal{V}$, column by column.

The rows are the *major rows* (non-pinch major modes) and the *trace rows of component* $k$ (the trace mode $\alpha_k$ of a section $\sigma$ with $w^\infty_{\sigma,k}=0$). For a component $k$ let $\mathcal{J}_k := \{J : x^\infty_{J,k}=0\} = \{J[i,m] : i<m,\ k\notin[i,m]\} \cup \{J[p,p] : k\notin \mathrm{Maj}(p)\}$. Every $J\in\mathcal{J}_k$ has first index $>k$ ($k$ is lighter than the node, $J\in\mathcal{J}_k^{<}$) or second index $<k$ ($J\in\mathcal{J}_k^{>}$), and by (H7) every section attached to a node of $\mathcal{J}_k$ has zero limit net flow of $k$.

**Lemma 67** (counts)**.** For every column $c$ and component $k$, the trace rows of $k$ carried by $\mathrm{T}_c$ and $\mathrm{B}_c$ are as many as the trace coordinates of $k$ of $c$; the major rows of $c$ are as many as its major coordinates ($2(m-i-1)$ for a non-binary column, 1 for a binary one).

**Proof.** For $k\notin[i,m]$: two trace rows ($\mathrm{T}_c$, $\mathrm{B}_c$) and two coordinates ($d_{c,k}$, $X_{c,k}$). Non-binary, $k=i$: one row ($\mathrm{B}_c$), one coordinate ($b_{c,i}$); $k=m$: one row ($\mathrm{T}_c$), one coordinate ($d_{c,m}$); $i<k<m$: none. Binary with $\pi_p=\mathrm{B}$, $k=p+1$: one row ($\mathrm{T}_c$), coordinate $d_{c,p+1}$; with $\pi_p=\mathrm{T}$, $k=p$: one row ($\mathrm{B}_c$), coordinate $b_{c,p}$. Major rows: $r_c$ and the $m-i-2$ middle modes in $\mathrm{T}_c$, the middle modes and $s_c$ in $\mathrm{B}_c$; binary: $\psi''$ or $\psi^{\mathrm{mid}}$. Major coordinates: $m-i$ (in $X_c$) plus $m-i-2$ (in $d_c$), or 1. □

**Lemma 68** (trace quantities are linear in the traces of their component)**.** Fix $k$. For every section $\sigma$ with $w^\infty_{\sigma,k}=0$, $w_{\sigma,k}(v)$ is a linear form in the trace coordinates of component $k$ only, and so is $x_{J,k}(v)$ for every $J\in\mathcal{J}_k$.

**Proof.** Induction over the columns, parents first: a $d_{c,k}$ with $d^\infty_{c,k}=0$ is a trace coordinate; a $b_{c,k}$ with $b^\infty_{c,k}=0$ is either a trace coordinate ($b_{c,i}$, $b_{c,p}$) or, for $k\notin[i,m]$, $d_{[i,m+1],k}+b_{[i-1,m],k}-d_{c,k}$ with both parental limit flows zero (they are supported on $[i,m]$). $x_{J_c,k}=X_{c,k}$, and $\bar{x}_{p,k}=(b_{[p-1,p],k}+d_{[p,p+1],k})/P_p$. □

**Lemma 69** (linearized trace rows)**.** Let $\alpha_k$ be a trace mode of $\sigma$ closed at $J\in\mathcal{J}_k$. With $\nu^k_{J\to\sigma}$ the net flow of $k$ from $J$ into $\sigma$ ($=w_{\sigma,k}$ if $J=\mathrm{bot}(\sigma)$, $=-w_{\sigma,k}$ if $J=\mathrm{top}(\sigma)$) and $\gamma^{(k)}_\sigma := |G^{(k)}_{x^\infty_J}(\alpha_k)|/\Psi^{(k)}_\sigma(\alpha_k)>0$ (by (H7)), the derivative at $v_\infty$ of the row $\tilde{\ell}_k\cdot x_J/\ell_{r_*}\cdot x_J$ is $(\delta x_{J,k}-\gamma^{(k)}_\sigma\,\delta\nu^k_{J\to\sigma})/\ell_{r_*}\cdot x^\infty_J$. In particular, at $v_\infty$ the trace rows of $k$ depend only on the trace coordinates of $k$.

**Proof.** By (28) the numerator is $x_{J,k}+w_{\sigma,k}\,g(v)$ with $g$ analytic; $x_{J,k}$ and $w_{\sigma,k}$ vanish at $v_\infty$, so its derivative is $\delta x_{J,k}+g(v_\infty)\,\delta w_{\sigma,k}$ with $g(v_\infty)=G^{(k)}_{x^\infty_J}(\alpha_k)/\Psi^{(k)}_\sigma(\alpha_k)$; the signs

of (H7) ($G^{(k)} < 0$ at a bottom end, $> 0$ at a top end, $\Psi^{(k)} > 0$) turn $g\,\delta w$ into $-\gamma\,\delta\nu$ at both kinds of ends. Lemma 68 gives the last statement. □

**Lemma 70** (node identities). For every component $k$ and every node $J \neq J[1,N]$, identically on $\mathcal{V}$, $\sum_{\sigma \text{ attached to } J} \nu^k_{J\to\sigma}(v) + P_J x_{J,k}(v) = 0$, with $P_J := 0$ for a junction.

**Proof.** The junction balance $d_c + b_c - f_c = 0$ and the definition of $\bar{x}_p$. □

**Proposition 71** (the trace blocks). For every $k$, the square matrix $\mathbb{T}_k$ of derivatives at $v_\infty$ of the trace rows of $k$ with respect to the trace coordinates of $k$ is nonsingular. More precisely, if $\mathbb{T}_k\,\delta t = 0$, the contents $\zeta_J := \delta x_{J,k}$ ($J \in \mathcal{J}_k$) solve $A_k\zeta = 0$, where

$$(A_k)_{JJ} = P_J + \sum_{\sigma \text{ closed at } J} \frac{1}{\gamma_\sigma^{(k)}},$$

$$(A_k)_{JJ'} = -\sum_{\sigma \text{ closed at } J', \text{ with open end } J} \frac{1}{\gamma_\sigma^{(k)}} \quad (J \neq J'),$$

and $A_k$ is, after a permutation, upper triangular with positive diagonal: a nonsingular M-matrix.

**Proof.** Extend $\delta t$ by zero to the other coordinates. By Lemma 69 every trace row of $k$, of a section $\sigma$ closed at $c(\sigma)$, gives $\zeta_{c(\sigma)} = \gamma_\sigma\,\delta\nu_\sigma$ with $\delta\nu_\sigma := \delta\nu^k_{c(\sigma)\to\sigma}$. At $J \in \mathcal{J}_k$ every attached section has zero limit net flow of $k$, hence $\alpha_k$ as a trace mode, closed at one of its ends, and that end lies in $\mathcal{J}_k$ by (H7). The node identity (Lemma 70, linear) gives $\sum_{\sigma \text{ closed at } J} \zeta_J/\gamma_\sigma - \sum_{\sigma \text{ open at } J} \zeta_{c(\sigma)}/\gamma_\sigma + P_J\zeta_J = 0$ (at an open end the flow into $\sigma$ is $-\delta\nu_\sigma$), that is $A_k\zeta = 0$. *Triangularity.* If $k$ is a lighter trace of $\sigma$, then $c(\sigma) = \mathrm{bot}(\sigma)$, the open end is $\mathrm{top}(\sigma)$, and $h(\mathrm{top}) = h(\mathrm{bot}) - 1$ (the top of $\mathrm{T}[i,m]$ is $J[i,m-1]$, the bottom of $\mathrm{B}[i,m]$ is $J[i+1,m]$); if $k$ is heavier, $c(\sigma) = \mathrm{top}(\sigma)$ and $h(\mathrm{bot}) = h(\mathrm{top}) + 1$; in both cases both ends lie in the same class $\mathcal{J}_k^<$ or $\mathcal{J}_k^>$. Ordering $\mathcal{J}_k^<$ by increasing $h$ and then $\mathcal{J}_k^>$ by decreasing $h$, every nonzero off-diagonal entry lies above the diagonal. *Diagonal.* $P_J > 0$ at products; a junction $J[i,m] \in \mathcal{J}_k^<$ is the closed bottom end of $\mathrm{T}[i,m]$ for $k$ ($k < i$), and $J[i,m] \in \mathcal{J}_k^>$ the closed top end of $\mathrm{B}[i,m]$ ($k > m$), by (H7); and $\gamma > 0$. A triangular Z-matrix with positive diagonal is a nonsingular M-matrix. *Conclusion.* $\zeta = 0$, hence $\delta\nu_\sigma = 0$ for every section with a trace row of $k$; every trace coordinate of $k$ is some $\zeta_J$ or $\pm$ some $\delta\nu_\sigma$ (Table 66: a $d_{c,k}$ or $b_{c,k}$ with zero limit value is the net flow of $k$ in a section where $\alpha_k$ is a trace mode), so $\delta t = 0$. $\mathbb{T}_k$ is square by Lemma 67. □

**Proposition 72** (the major block). Order the columns by decreasing length. The matrix $\mathbb{M}$ of derivatives at $v_\infty$ of the major rows with respect to the major coordinates is block lower triangular over the columns, with nonsingular diagonal blocks $\mathbb{M}_c$; hence $\mathbb{M}$ is nonsingular.

**Proof.** *Triangularity.* The major rows of $c$ are closed at $J_c$ (Table 48) and depend on $X_c$, $d_c$ and $f_c$, that is, on the coordinates of $c$ and of its ancestors.

*Non-binary $c = [i,m]$.* Let $\delta$ be a variation of the major coordinates of $c$ only, with $\mathbb{M}_c\delta = 0$. Then $\delta f_c = 0$ and $\delta d_{c,i} = \delta d_{c,m} = \delta d_{c,k} = 0$ ($k \notin [i,m]$), so $\delta d_c$ is supported on

$\{i+1,\dots,m-1\}$ with zero sum, and $\delta w_{\mathrm{T}_c} = \delta w_{\mathrm{B}_c} = \delta d_c$. For a root $\chi$ of $\Psi_{\mathrm{T}_c}$ or $\Psi_{\mathrm{B}_c}$, $\Psi'(\chi)\,\delta\chi = \mathcal{R}(\chi)$ with $\mathcal{R}(t) := \sum_p \alpha_p \delta d_p/(\alpha_p - t)$. Each major row is $G_{X_c}(\chi)/(\text{pivot})$ with $G_{X_c}(\chi) = 0$, so its variation is $[\ell_\chi \cdot \delta X_c + G'_{X_c}(\chi)\,\delta\chi]/(\text{pivot})$. The two rows of a middle root $\theta_j$ (in $\mathrm{T}_c$ and $\mathrm{B}_c$) give $G'_{X_c}(\theta_j)\,\mathcal{R}(\theta_j)\,(1/\Psi'_{\mathrm{T}_c}(\theta_j) - 1/\Psi'_{\mathrm{B}_c}(\theta_j)) = 0$, where $\Psi'_{\mathrm{T}_c} < 0 < \Psi'_{\mathrm{B}_c}$ (Lemma 24) and $G'_{X_c}(\theta_j) \neq 0$ (a simple zero of $G_{X_c}$, Proposition 35(e)); so $\mathcal{R}(\theta_j) = 0$ at the $m-i-2$ middle roots, and $\mathcal{R}(0) = \sum_p \delta d_p = 0$. Then $\mathcal{R}(t)\prod_{p=i+1}^{m-1}(\alpha_p - t)$ is a polynomial of degree at most $m-i-2$ with $m-i-1$ distinct zeros, hence zero, and $\delta d_c = 0$. (This is the invertibility of the Cauchy-type matrix $(\alpha_p/(\alpha_p - \theta_j))$ on $\{\sum \delta d = 0\}$.) Then $\delta\chi = 0$ for every mode, and the rows of $r_c$, the $\theta_j$ and $s_c$ say $\ell_\chi \cdot \delta X_c = 0$ at the $m-i$ interlacing points, with $\sum \delta X_c = 0$ and $\delta X_c$ supported on $[i,m]$: $\delta X_c = 0$ by Proposition 35(e).

*Binary c*, $\pi_p = \mathrm{B}$. The only major coordinate is $X_{c,p+1}$ (with $\delta X_{c,p} = -\delta X_{c,p+1}$); the net flows and the modes do not move, and the row of $\psi''$ varies by $(\alpha_{p+1}/(\alpha_{p+1} - \psi'') - \alpha_p/(\alpha_p - \psi''))\delta X_{c,p+1}/(\text{pivot}) \neq 0$, since $\alpha_{p+1} < \psi'' < \alpha_p$. $\pi_p = \mathrm{T}$: the same with $\psi^{\mathrm{mid}} \in (\alpha_{p+1}, \alpha_p)$. □

**Proof** (Proof of Lemma 51 and Theorem 52)**.** By (H5) and (H8) every section has $N$ simple positive modes and nonzero pivots at $v_\infty$; Lemma 26, continuity and Lemma 30(c) give Lemma 51 (with $n = \min_\sigma n_\sigma$, since $\bar\rho^{\,n_\sigma+1} \le \bar\rho^{\,n+1}$), and $G_\infty(v_\infty) = 0$ is the set of limit relations of (H5) (Lemma 30(b)). The zero block and the block-diagonal form of $\mathbb{T}$ follow from Lemma 69; the diagonal blocks are square by Lemma 67 and nonsingular by Propositions 71 and 72. □

**Example 73** ($N = 3$, pattern $(\mathsf{B},\mathsf{T})$)**.** Components A, B, C with volatilities $a > b > c$; columns $\mathrm{P} = [1,3]$ (prefractionator), $\mathrm{U} = [1,2]$, $\mathrm{W} = [2,3]$; A is sent down $\mathrm{B}_\mathrm{U}$ into the side product and C up $\mathrm{T}_\mathrm{W}$. The twelve coordinates are: for P, the trace flows $\varepsilon_\mathrm{C} := d_{\mathrm{P},C}$ and $\varepsilon_\mathrm{A} := b_{\mathrm{P},A}$ and the major contents $X_{\mathrm{P},B}, X_{\mathrm{P},C}$; for U, $d_{\mathrm{U},B}$, $d_{\mathrm{U},C}$ (traces), $X_{\mathrm{U},B}$ (major), $X_{\mathrm{U},C}$ (trace); for W, $d_{\mathrm{W},A}$, $b_{\mathrm{W},B}$, $X_{\mathrm{W},A}$ (traces), $X_{\mathrm{W},C}$ (major). The rows, by section (modes in decreasing order, pinch marked $*$): $\mathrm{T}_\mathrm{P}$ $(r_\mathrm{P}, u^*_\mathrm{P}, c)$: $r_\mathrm{P}$ at $J_\mathrm{P}$ (major) and $c$ at $J_\mathrm{U}$ (trace C); $\mathrm{B}_\mathrm{P}$ $(a, v^*_\mathrm{P}, s_\mathrm{P})$: $a$ at $J_\mathrm{W}$ (trace A) and $s_\mathrm{P}$ at $J_\mathrm{P}$ (major); $\mathrm{T}_\mathrm{U}$ $(\psi^{\mathrm{T}*}, b, c)$: $b$, $c$ at the condenser (traces B, C); $\mathrm{B}_\mathrm{U}$ $(\psi'^*, \psi'', c)$: $\psi''$ at $J_\mathrm{U}$ (major) and $c$ at $J_\mathrm{U}$ (trace C); $\mathrm{T}_\mathrm{W}$ $(a, \psi^{\mathrm{mid}}, \psi^{\mathrm{low}*})$: $a$ at $J_\mathrm{W}$ (trace A) and $\psi^{\mathrm{mid}}$ at $J_\mathrm{W}$ (major); $\mathrm{B}_\mathrm{W}$ $(a, b, \psi^{\mathrm{B}*})$: $a$, $b$ at the reboiler (traces A, B). No major coordinate enters a net flow ($m-i-2 = 0$ for P), so $\mathbb{M} = \mathrm{diag}(\mathbb{M}_\mathrm{P}, \mathbb{M}_\mathrm{U}, \mathbb{M}_\mathrm{W})$, where $\det \mathbb{M}_\mathrm{P}$ is, up to the pivots, the Underwood interpolation determinant $\det[\mathbf{1}^\top;\ (\alpha_k/(\alpha_k - r_\mathrm{P}))_k;\ (\alpha_k/(\alpha_k - s_\mathrm{P}))_k] \neq 0$, and $\mathbb{M}_\mathrm{U}$, $\mathbb{M}_\mathrm{W}$ are the nonzero scalars of Proposition 72. The trace blocks reduce to

$$A_\mathrm{A} = \begin{pmatrix} \gamma_{\mathrm{B_P}}^{-1} + \gamma_{\mathrm{T_W}}^{-1} & -\gamma_{\mathrm{B_W}}^{-1} \\ 0 & P_3 + \gamma_{\mathrm{B_W}}^{-1} \end{pmatrix}, \qquad A_\mathrm{C} = \begin{pmatrix} \gamma_{\mathrm{T_P}}^{-1} + \gamma_{\mathrm{B_U}}^{-1} & -\gamma_{\mathrm{T_U}}^{-1} \\ 0 & P_1 + \gamma_{\mathrm{T_U}}^{-1} \end{pmatrix},$$
$$A_\mathrm{B} = \mathrm{diag}(P_1 + \gamma_{\mathrm{T_U}}^{-1},\ P_3 + \gamma_{\mathrm{B_W}}^{-1})$$

(nodes $J_\mathrm{W}$, reboiler for A; condenser, reboiler for B; $J_\mathrm{U}$, condenser for C; each $\gamma$ is the coefficient of the component of its block). So $\det DG_\infty(v_\infty)$ is a product of explicit nonzero factors; the triangular solution starts at the product nodes and moves to the junctions. ⌟

# E From the limit system to steady states: proofs

**Lemma 74** (quantitative inverse function step)**.** Let $E$ be a finite-dimensional normed space, $\mathcal{G}_\infty, \mathcal{G} \in C^1(\bar{B}; E)$ on $\bar{B} := \bar{B}(v_\infty, r_0)$, $\mathcal{G}_\infty(v_\infty) = 0$, $A := D\mathcal{G}_\infty(v_\infty)$ invertible, $\Omega(r) := \sup_{|v-v_\infty|\le r} \|D\mathcal{G}_\infty(v) - A\|$, and $r_1 \in (0, r_0]$ with $\|A^{-1}\|\,\Omega(r_1) \le 1/4$. If $\sup_{\bar{B}} \|D\mathcal{G} - D\mathcal{G}_\infty\| \le 1/(4\|A^{-1}\|)$ and $\|A^{-1}\|\,|\mathcal{G}(v_\infty)| \le r_1/2$, then $\mathcal{G}$ has exactly one zero $v^*$ in $\bar{B}(v_\infty, r_1)$, and $|v^* - v_\infty| \le 2\|A^{-1}\|\,|\mathcal{G}(v_\infty)|$.

**Proof.** $\Phi(v) := v - A^{-1}\mathcal{G}(v)$ has $\|D\Phi(v)\| \le \|A^{-1}\|(\|A - D\mathcal{G}_\infty(v)\| + \|D\mathcal{G}_\infty(v) - D\mathcal{G}(v)\|) \le 1/2$ on the convex ball $\bar{B}(v_\infty, r_1)$, which $\Phi$ maps into itself: $|\Phi(v) - v_\infty| \le |\Phi(v) - \Phi(v_\infty)| + \|A^{-1}\|\,|\mathcal{G}(v_\infty)| \le r_1/2 + r_1/2$. Banach's fixed-point theorem gives the unique zero, and $|v^* - v_\infty| \le |v^* - v_\infty|/2 + \|A^{-1}\|\,|\mathcal{G}(v_\infty)|$. □

**Proof** (Proof of Theorem 53)**.** On $\mathcal{W}$ (Lemma 51), $\sup \|DG_n - DG_\infty\| \le C_1 \max_\sigma(n_\sigma + 2)\bar{\rho}^{n_\sigma+1} \le C_1(n+2)\bar{\rho}^{n+1}$ for $n \ge 1/\log(1/\bar{\rho})$ (where $x \mapsto (x+2)\bar{\rho}^{x+1}$ decreases), and $|G_n(v_\infty)| = |(G_n - G_\infty)(v_\infty)| \le C_2\rho_\infty^{n+1}$ by Lemma 30(c) at $w = w^\infty$ (every ratio $\mu$ at $v_\infty$ is at most $\rho_\infty$). $G_\infty$ is analytic, so $\Omega(r) \le C_3 r$. Take $r_0$ with $\bar{B}(v_\infty, r_0) \subset \mathcal{W}$, $r_1 := \min(r_0, 1/(4C_3\|A^{-1}\|))$, and $n_0$ with $C_1(n_0+2)\bar{\rho}^{n_0+1} \le 1/(4\|A^{-1}\|)$ and $\|A^{-1}\|C_2\rho_\infty^{n_0+1} \le r_1/2$, and apply Lemma 74 to $\mathcal{G} = G_n$. At $v_n$ every section satisfies $x_{\mathrm{top}} = c_\sigma M_\sigma^{n_\sigma+1} x_{\mathrm{bot}}$ with $c_\sigma \ne 0$ (Lemma 30(a)). □

**Lemma 75** (exact composite profile)**.** At $v_n$, take for a section $\sigma$ the eigen-data at $w_\sigma(v_n)$ with the right eigenvectors normalized by $\mathbf{1}^\top\hat{\xi}_r = 1$ (possible near $v_\infty$, since $\mathbf{1}^\top\hat{\xi}_r^\infty = 1$) and the left eigenvectors rescaled so that $\ell_r \cdot \hat{\xi}_r = 1$ (so $\ell_r \cdot x = [x]_r$ and the pivots are $[x_{\mathrm{bot}}]_{r_*}$, $[x_{\mathrm{top}}]_{r_*}$); put $a_s := \ell_s \cdot x_{\mathrm{bot}}/\varpi_{\mathrm{bot}}$ $(s \in D)$ and $b_r := \ell_r \cdot x_{\mathrm{top}}/\varpi_{\mathrm{top}}$ $(r \in U)$. For $0 \le k \le n_\sigma + 1$,

$$u_k := \frac{M^k x_{\mathrm{bot}}}{\varpi_{\mathrm{bot}}\lambda_{r_*}^k} = \hat{\xi}_{r_*} + \sum_{s\in D} a_s \mu_s^k \hat{\xi}_s + \sum_{r\in U} b_r \mu_r^{n_\sigma+1-k}\hat{\xi}_r, \tag{76}$$

and the stage compositions of Proposition 22 are $x_k = u_k/\mathbf{1}^\top u_k$.

**Proof.** $M^k x_{\mathrm{bot}} = \sum_r (\ell_r \cdot x_{\mathrm{bot}})\lambda_r^k \hat{\xi}_r$, and $E_r = 0$ $(r \in U)$ gives $\ell_r \cdot x_{\mathrm{bot}} = \mu_r^{n_\sigma+1} b_r \varpi_{\mathrm{bot}}$. □

The profile is a bottom half-profile decaying upward plus a top half-profile decaying downward, glued at the pinch; at $v_\infty$ these are the $v_k^{\mathrm{bot}}$ and $v_k^{\mathrm{top}}$ of (H6).

**Lemma 77** ($S > 0$ at every stage)**.** Let $c_0 := \min_\sigma \min(\inf_k \mathbf{1}^\top v_k^{\mathrm{bot}}, \inf_k \mathbf{1}^\top v_k^{\mathrm{top}}) > 0$ (H6). There are $n_1 \ge n_0$ and $C$ such that for $\min_\sigma n_\sigma = n \ge n_1$, every $\sigma$ and $0 \le k \le n_\sigma + 1$,

$$|u_k - v_k^{\mathrm{bot}}|_\infty \le C(\rho_\infty^{n+1} + \bar{\rho}^{n_\sigma+1-k}), \qquad |u_k - v_{n_\sigma+1-k}^{\mathrm{top}}|_\infty \le C(\rho_\infty^{n+1} + \bar{\rho}^{k}),$$
$$\mathbf{1}^\top u_k \ge c_0/2, \qquad S(x_k) \ge \frac{\alpha_N c_0}{2\,\mathbf{1}^\top u_k} > 0.$$

**Proof.** The coefficients $\hat{\xi}_r, \mu_r, a_s, b_r$ of (76) are analytic in $v$ on $\bar{B}(v_\infty, r_1)$ and equal at $v_\infty$ to those of (H6), so they differ from them by at most $C|v_n - v_\infty| \le C\rho_\infty^{n+1}$; since all $\mu \le \bar{\rho}$, $|\mu^k - (\mu^\infty)^k| \le (\sup_j j\bar{\rho}^{j-1})|\mu - \mu^\infty|$ uniformly in $k$. Comparing (76) termwise with $v_k^{\mathrm{bot}}$,

the $\hat{\xi}_{r_*}$- and $D$-terms differ by at most $C\rho_\infty^{n+1}$, and the $U$-terms are at most $C\bar{\rho}^{n_\sigma+1-k}$ ($b_r$ is bounded because $|\varpi_{\mathrm{top}}| \geq \varpi_0$); with $v^{\mathrm{top}}_{n_\sigma+1-k}$, the $U$-terms differ by at most $C\rho_\infty^{n+1}$ and the $D$-terms are at most $C\bar{\rho}^{k}$. Since $\min(\bar{\rho}^k, \bar{\rho}^{n_\sigma+1-k}) \leq \bar{\rho}^{(n+1)/2}$, for every $k$ one of the two half-profiles, $v \geq 0$ with $\mathbf{1}^\top v \geq c_0$, satisfies $|u_k - v|_\infty \leq \varepsilon_n := C(\rho_\infty^{n+1} + \bar{\rho}^{(n+1)/2})$. Then $\mathbf{1}^\top u_k \geq c_0 - N\varepsilon_n$ and $\alpha^\top u_k \geq \alpha_N c_0 - (\sum_i \alpha_i)\varepsilon_n$; take $n_1$ with $N\varepsilon_n \leq c_0/2$ and $(\sum_i \alpha_i)\varepsilon_n \leq \alpha_N c_0/2$, and note $S(x_k) = \alpha^\top u_k / \mathbf{1}^\top u_k$. □

**Proof** (Proof of Theorem 54). Theorem 53 gives $v_n$; Lemma 77 gives $S > 0$ at every stage and $\mathbf{1}^\top u_k \neq 0$, so the iterates of Proposition 22(b) are defined; $S > 0$ at the node compositions because they are affine in $v$ and within $C\rho_\infty^{n+1}$ of points of $\mathcal{X}$, where $S \geq \alpha_N$ (enlarge $n_1$ so that $S \geq \alpha_N/2$ there). Proposition 22(b) gives a steady state with (i) and (iv) (Lemma 41); (ii) follows from Theorem 53 and (H8), since the product compositions are affine in $v$; (iii) from Lemma 77; (v) from Theorem 53 and Proposition 22(a). □

## F Computations and reproducibility

No proof in this paper depends on a computation. The scripts below check identities, re-derive the constructions separately, and test the statements numerically; paths are relative to the problem folder `problems/halvorsen-skogestad-petlyuk-minimum-vapor/` of the repository. Environment: Linux (Fedora under WSL), Python 3.12 with `numpy`, `scipy`, `sympy`, `mpmath`, `python-flint`; one thread; runtimes from seconds to a few minutes unless stated. Each certificate script prints PASS or FAIL for each claim.

*Lower bound.*

- `total/network/scripts/unit_identities_exact.py` (exact rationals, `sympy`): Lemma 9 symbolically for $N = 2, \ldots, 6$; the summed node balance (the cut identity for an arbitrary set of units) on random unit graphs with recycles, arbitrary subsets and rational $\theta$.
- `total/network/scripts/section_identity_exact.py` (exact rationals): identity (14) and the single change of side per section, on 400 random stage-to-stage profiles with net flows of both signs.
- `total/foundations/scripts/unitnet.py`, `total/sections/scripts/check_stage_cut.py`, `total/network/scripts/cut_bound_check.py`, `total/simulation/scripts/test_stage_bound.py` and `test_section_sign.py`, `total/verification-early/scripts/vexact.py` and `vtests.py`: four separate steady-state solvers; Theorem 10 and its identity on thousands of random networks and on columns and Petlyuk arrangements; no violation.
- `total/network/scripts/permissive_sign_path.py`: the light-to-heavy sign path of the point of Remark 15 (40 digits); `verification/krawczyk_certificate.py`: interval certificate of that point.

*Attainment.*

- `total/sections/scripts/check_binary_attainment.py`: the quadratic of Appendix A (`sympy`).
- `attainment/design/scripts/check_design.py`: (H1)–(H8) and Propositions 35–44 on 240 random instances, $N = 3, \ldots, 7$, random patterns (high precision).

- `attainment/limit/scripts/check_combinatorics.py`: the case analysis of Lemmas 67–70 and Proposition 71 (rows, coordinates, closed-to-open moves, triangularity), exhaustively for $N \leq 12$ and all $2^{N-1}$ patterns (exact, integer arithmetic).
- `attainment/limit/scripts/check_limit_jacobian.py` and `check_unforced.py`: the block structure of Theorem 52 on random instances, $N = 3, \ldots, 6$; singularity without forcing (Remark 55).
- `attainment/analysis/scripts/check_analysis.py`: zeros of $G_n$ with generic numerical eigen-data, every balance re-checked with the actual streams, $N = 3, 4, 5$.
- `attainment/computation/scripts/paper_tables.py` (Tables 61–63, about 5 minutes) and `run_batch.sh`, `run_case.py`, `spurious_zeros.py`, `redteam_design.py`, `forcing_study.py`, `eta_dependence.py`, `unequal_lengths.py` (the 62-case campaign of Section 5).

*Verification of the attainment proof.* `attainment/verification-DL/scripts/v1_independent.py` (own construction of the design and of the limit system with a general eigen-solver, finite-difference Jacobians, finite-stage steady states with every balance re-checked; 23 cases) and `mutate_combinatorics.py` (ten deliberately broken models, all detected); `attainment/verification-A/scripts/v2_perturbation.py` (explicit Underwood and trace eigen-data, constants of Lemma 51, the inequalities of Lemma 74, contraction, physical re-shooting) and `v2_binary_prop10.py` (Appendix A: identities in `sympy` and shooting for six data sets).

*Underwood model.* `theory/scripts/run_theory_checks.sh` and `computation/scripts/run_certificates.sh` (the Underwood-model statements, the enumeration of regular-column configurations and the multistart minimizations mentioned in the introduction).

## References

[1] F. B. Petlyuk, V. M. Platonov and D. M. Slavinskii. Thermodynamically Optimal Method for Separating Multicomponent Mixtures. *International Chemical Engineering*, **1965**. 1

[2] Gerd Kaibel. Distillation Columns with Vertical Partitions. *Chemical Engineering & Technology*, **1987**.
DOI: 10.1002/ceat.270100112 1

[3] Ivar J. Halvorsen and Sigurd Skogestad. Minimum Energy Consumption in Multicomponent Distillation. 1. $V_{\mathrm{min}}$ Diagram for a Two-Product Column. *Industrial & Engineering Chemistry Research*, **2003**.
DOI: 10.1021/ie010863g 2, 5

[4] Ivar J. Halvorsen and Sigurd Skogestad. Minimum Energy Consumption in Multicomponent Distillation. 2. Three-Product Petlyuk Arrangements. *Industrial & Engineering Chemistry Research*, **2003**.
DOI: 10.1021/ie0108649 2, 15, 18

[5] Ivar J. Halvorsen and Sigurd Skogestad. Minimum Energy Consumption in Multicomponent Distillation. 3. More Than Three Products and Generalized Petlyuk Arrangements. *Industrial & Engineering Chemistry Research*, **2003**.
DOI: 10.1021/ie0108651 2, 3, 4, 6, 7, 9, 11, 12, 15, 23

[6] Ivar J. Halvorsen. Minimum Energy Requirements in Complex Distillation Arrangements. Dr. ing. thesis 2001:43. PhD thesis. Norwegian University of Science and Technology (NTNU), **2001**.
URL: https://skoge.folk.ntnu.no/publications/thesis/2001_halvorsen/ThesisHalvorsen2001-www.pdf 2, 5, 6, 9, 10, 27

[7] Ivar J. Halvorsen and Sigurd Skogestad. Minimum Energy and Entropy Requirements in Multicomponent Distillation. In: *Distillation and Absorption 2002*. Paper 6-27. **2002**.
URL:
https://skoge.folk.ntnu.no/prost/proceedings/distillation02/dokument/6-27.pdf 2, 7, 27

[8] Zbigniew Fidkowski and Lechosław Królikowski. Minimum Energy Requirements of Thermally Coupled Distillation Systems. *AIChE Journal*, **1987**.
DOI: 10.1002/aic.690330412 2

[9] Vishesh H. Shah and Rakesh Agrawal. A Matrix Method for Multicomponent Distillation Sequences. *AIChE Journal*, **2010**.
DOI: 10.1002/aic.12118 2, 3, 6

[10] Ulaganathan Nallasivam, Vishesh H. Shah, Anirudh A. Shenvi, Joshua Huff, Mohit Tawarmalani and Rakesh Agrawal. Global Optimization of Multicomponent Distillation Configurations: 2. Enumeration Based Global Minimization Algorithm. *AIChE Journal*, **2016**.
DOI: 10.1002/aic.15204 2, 7

[11] Radhakrishna Tumbalam Gooty, Rakesh Agrawal and Mohit Tawarmalani. Advances in MINLP to Identify Energy-Efficient Distillation Configurations. *Operations Research*, **2024**.
DOI: 10.1287/opre.2022.2340 arXiv: 2010.12113 2, 6, 7, 10

[12] Zbigniew T. Fidkowski and Rakesh Agrawal. Multicomponent Thermally Coupled Systems of Distillation Columns at Minimum Reflux. *AIChE Journal*, **2001**.
DOI: 10.1002/aic.690471211 2

[13] Zbigniew T. Fidkowski. Distillation Configurations and Their Energy Requirements. *AIChE Journal*, **2006**.
DOI: 10.1002/aic.10803 2, 6

[14] Tony Joseph Mathew. Advances in Modelling and Optimization of Multicomponent Distillation Configurations. PhD thesis. Purdue University, **2021**. 2

[15] Ivar J. Halvorsen and Sigurd Skogestad. Energy Efficient Distillation. *Journal of Natural Gas Science and Engineering*, **2011**.
DOI: 10.1016/j.jngse.2011.06.002 2

[16] Miguel A. Navarro-Amorós, Rubén Ruiz-Femenia and José A. Caballero. A New

Technique for Recovering Energy in Thermally Coupled Distillation Using Vapor Recompression Cycles. *AIChE Journal*, **2013**.
DOI: 10.1002/aic.14137 2

[17] José A. Caballero and Ignacio E. Grossmann. Optimal Synthesis of Thermally Coupled Distillation Sequences Using a Novel MILP Approach. *Computers & Chemical Engineering*, **2014**.
DOI: 10.1016/j.compchemeng.2013.10.015 2

[18] A. J. V. Underwood. Fractional Distillation of Multicomponent Mixtures. *Chemical Engineering Progress*, **1948**. 2

[19] N. L. Franklin and J. S. Forsyth. The Interpretation of Minimum Reflux Conditions in Multi-Component Distillation. *Transactions of the Institution of Chemical Engineers*, **1953**. Reprinted in Chem. Eng. Res. Des. 75 (1997) S56–S81, doi:10.1016/S0263-8762(97)80005-6. 2

[20] Zheyu Jiang. Multicomponent Distillation – Mathematical Modeling, Global Optimization, and Process Intensification. PhD thesis. Purdue University, **2019**. 2

[21] Zheyu Jiang, Mohit Tawarmalani and Rakesh Agrawal. Minimum Reflux Calculation for Multicomponent Distillation in Multi-Feed, Multi-Product Columns: Mathematical Model. *AIChE Journal*, **2022**.
DOI: 10.1002/aic.17929 2

[22] Ivar J. Halvorsen and Sigurd Skogestad. Optimal Operation of Petlyuk Distillation: Steady-State Behavior. *Journal of Process Control*, **1999**.
DOI: 10.1016/S0959-1524(99)00009-8 2

[23] Ilkka Malinen and Juha Tanskanen. A Rigorous Minimum Energy Calculation Method for a Fully Thermally Coupled Distillation System. *Chemical Engineering Research and Design*, **2007**.
DOI: 10.1205/cherd06086 2

[24] Andreas Acrivos and Neal R. Amundson. On the Steady State Fractionation of Multicomponent and Complex Mixtures in an Ideal Cascade. *Chemical Engineering Science*, **1955**. First of a series of six papers in the same volume.
DOI: 10.1016/0009-2509(55)85004-X 2

[25] Felix B. Petlyuk. Distillation Theory and its Application to Optimal Design of Separation Units. Cambridge University Press, **2004**.
DOI: 10.1017/CBO9780511547102 2

[26] Rakesh Agrawal. Synthesis of Multicomponent Distillation Column Configurations. *AIChE Journal*, **2003**.
DOI: 10.1002/aic.690490210 3, 6

[27] Zheyu Jiang, Gautham Madenoor Ramapriya, Mohit Tawarmalani and Rakesh Agrawal. Minimum Energy of Multicomponent Distillation Systems Using Minimum Additional Heat and Mass Integration Sections. *AIChE Journal*, **2018**.
DOI: 10.1002/aic.16189 3, 27

[28] Atle C. Christiansen, Sigurd Skogestad and Kristian Lien. Complex Distillation Arrangements: Extending the Petlyuk Ideas. *Computers & Chemical Engineering*, **1997**.
DOI: 10.1016/S0098-1354(97)87508-4 6

[29] Tony Joseph Mathew, Radhakrishna Tumbalam Gooty, Mohit Tawarmalani and Rakesh Agrawal. 110th Anniversary: Thermal Coupling via Heat Transfer: A Potential Route to Simple Distillation Configurations with Lower Heat Duty. *Industrial & Engineering Chemistry Research*, **2019**.
DOI: 10.1021/acs.iecr.9b04689 10

[30] Nils A. Carlberg and Arthur W. Westerberg. Temperature-Heat Diagrams for Complex Columns. 3. Underwood's Method for the Petlyuk Configuration. *Industrial & Engineering Chemistry Research*, **1989**.
DOI: 10.1021/ie00093a018 18

[31] Tony Joseph Mathew, Mohit Tawarmalani and Rakesh Agrawal. Relaxing the Constant Molar Overflow Assumption in Distillation Optimization. *AIChE Journal*, **2023**.
DOI: 10.1002/aic.18125 26

[32] Elling W. Jacobsen and Sigurd Skogestad. Multiple Steady States in Ideal Two-Product Distillation. *AIChE Journal*, **1991**.
DOI: 10.1002/aic.690370404 27

[33] Michael F. Doherty and John D. Perkins. On the Dynamics of Distillation Processes. IV: Uniqueness and Stability of the Steady State in Homogeneous Continuous Distillations. *Chemical Engineering Science*, **1982**.
DOI: 10.1016/0009-2509(82)80090-0 27

[34] P. N. Shivakumar and Kim Ho Chew. A Sufficient Condition for Nonvanishing of Determinants. *Proceedings of the American Mathematical Society*, **1974**.
DOI: 10.1090/S0002-9939-1974-0332820-0 30

[35] Abraham Berman and Robert J. Plemmons. Nonnegative Matrices in the Mathematical Sciences. SIAM, **1994**.
DOI: 10.1137/1.9781611971262 30